# The snug Balius battery case and the bare power board

Three-cell fit, mounting loads and vibration response

Started 9 November 2026 | Updated 12 November 2026 | SolidWorks native export and ANSYS MAPDL 2026 R1

## What fits at 1.25 mm

The three-pocket Balius SolidWorks part follows the spacing of the archived cells and clears all four outer washers at a nominal 1.250 mm gap between the mounting tabs and the underside of the PCB. The middle cell has only 0.003 mm of clearance. Print dimensions, battery sleeves and assembly alignment can therefore determine whether a real cell starts in contact.

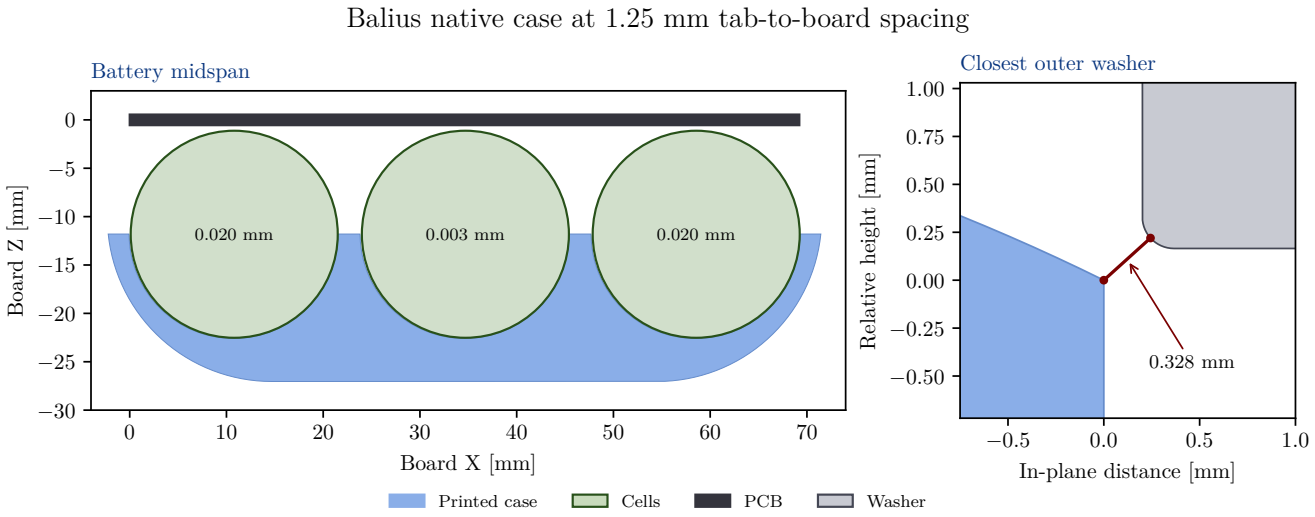



Figure 1: The left panel cuts through the battery midspan. Each cell label gives its shortest three-dimensional distance to the case. The right panel cuts through the two closest points on the outer washer and case, using a separate magnified scale.

| Assembly quantity | Bare board | Snug Balius geometry |
|---|---|---|
| Battery cells / nickel-strip attachments | 3 / 6 | 3 / 6 |
| Original battery-fastener preload | 6 × 750 N | 6 × 750 N |
| Added mounting screws | 0 | 4 (nylon or aluminium) |
| Printed solid volume | 0 | 62.980 $cm^3$ |
| Mass of board plus printed shell* | 251.82 g | 325.51 g |

*The bare value is the archived model mass. A dense PLA shell at 1.17 $g/cm^3$ adds 73.69 g, or 29.3%. The estimate excludes the four added screws and spacer material. Actual print mass has not been measured.

## Mounting and contact assumptions

The native export has 6.350 mm thick tabs at the four case screw bearings. The nominal distance from the PCB underside to each modeled screw-bearing plane is 7.600 mm. Both variants use the same hole registration and air around the cells. The added screw shanks have nominal M4 geometry. The nylon scenario uses a 3.0 GPa shank modulus and McMaster-Carr 93640A145 as representative M4 hardware. The aluminium shank modulus is 69 GPa. The 1.25 mm spacers are represented by the same assumed printed-PLA compression law. The four added screws have no specified installation torque.

In the unloaded CAD assembly, shifting the case 0.0030 mm in negative PCB X brings it into contact with the middle cell. In positive X the first limit comes from the mounting holes at 0.0969 mm. Along PCB Y the holes allow only 0.0071 mm in either direction. These one-axis limits assume a fixed 1.250 mm stand-off, zero rotation and four straight 4.00 mm shanks free to centre within their holes.

## What changed with the case?

The case reduced tightening deflection by **39% with nylon screws and 51% with aluminium**. At 8.5 N, local sideways travel was about **0.03 mm cased versus 0.20 mm bare** with the same nominal nickel law. It adds 74 g and can worsen vibration stresses (page 13). Contact changes limit those vibration predictions; neither screw option has a verified failure load. The finite campaign is closed (page 36).

## Comparison baseline and native rerun

Both screw-material comparisons use the October bare board, its three cells and six preloaded fasteners. The three native base-excitation runs use the eight original mounting faces, the 20–2000 Hz spectrum and the nine-second seeded records. The side-pull fixtures retain the original clamp preload and idealized weld footprints.

| Verified no-case result | Value |
|---|---|
| Peak PCB displacement during bolt tightening | 47.46 $\mu$m |
| First connected prestressed natural frequency | 485.96 Hz |
| Peak PCB Z displacement, base Z | 52.67 $\mu$m RMS |
| Peak PCB equivalent stress, base X | 90.33 MPa RMS |
| Bilinear nickel, force at +0.500 / −0.500 mm cell travel | +8.54 / −8.51 N |

Static displacement, RMS response and prescribed-travel reactions describe different load cases.

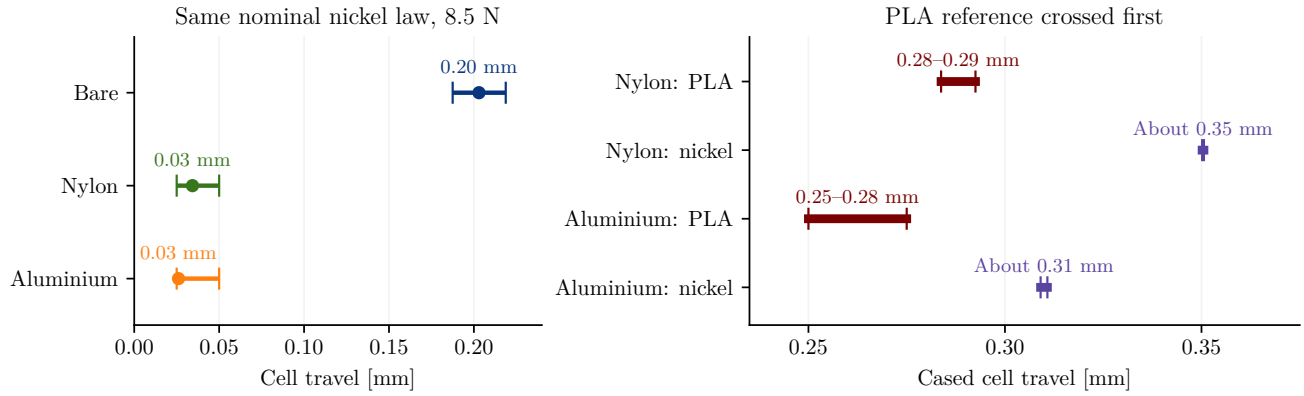
## Resolving the initial contact

The nominal 2 mm case mesh has 45,549 quadratic tetrahedra and 77,166 nodes. Its integrated volume differs from CAD by 0.00089%. The 3 and 1.5 mm meshes have 18,051 and 99,818 tetrahedra. All three have positive minimum scaled Jacobians. Both finer meshes cleared all three undeformed cells at both surface sampling levels. The 3 mm mesh showed an apparent middle-cell intersection at the coarser sampling level. At the higher level the middle cell first touched at 1.2412–1.2414 mm stand-off on the 2 mm mesh and 1.2406–1.2408 mm on the 1.5 mm mesh, compared with 1.2396–1.2400 mm in CAD. The native minimum initial gaps on the two finer meshes were 2.969 and 2.984 $\mu$m. Page 4 compares their actual loaded responses.

In the local side-pull fixture the left cell moves along PCB X. Replaying the bare cell's measured translation and rotation gave first contact at **0.0204 mm and 1.14 N toward the middle cell**, or **0.1653 mm and 6.7 N away**. On the native nylon forward pull the pocket was clear at 0.01750 mm and carrying pressure at 0.02875 mm (pages 19 and 21). ANSYS rejected the initial 2 mm local mesh before its first solution step. The replacement 1.5 mm mesh passed the native shape checks.

## Nickel damage under a sideways pull

With the same nominal multilinear nickel law, **8.5 N gives about 0.03 mm travel with either screw material versus 0.20 mm bare**. Both cased saved-state brackets are 0.025–0.050 mm, so their difference is unresolved. The bare bracket is 0.188–0.219 mm; the 0.50 mm result above uses the bilinear law. Bare nominal damage first appears at 3.25–3.28 mm, about 43 N, and reaches 2.78% locally at the last saved state, with force still rising.




Bars span adjacent saved native states; left dots interpolate. PLA denotes the first parent-mesh coupon crossing: through-layer stress for nylon screws, strain for aluminium. Nickel denotes section damage $D > 10^{-4}$. These are numerical reference crossings, not observed failures.

Nylon stopped on nonconvergence near **0.35 mm**; aluminium stopped on excessive shell-thickness change near **0.31 mm**. All 28 and 23 converged states pass the force checks; diagnostic states are excluded. PLA crosses its reference first at **0.28–0.29 mm for nylon and 0.25–0.28 mm for aluminium**. Before those crossings, nickel damage remains zero and initiation demand reaches about 50% and 35%. Later nickel damage **does not establish a tearing load**; neither PLA crossing is a converged failure limit.

Both mounts reduced PCB displacement at bolt lock (page 3) and raised base-Z PCB stress (page 12). Pages 23–24 give the refined PLA limit. Page 16 links the paired movies, and Appendix A preserves the earlier fit.

## Both snug mounts reduce PCB deflection during tightening

We tightened the six original battery fasteners to 750 N each, then locked their adjustments. The largest PCB displacement fell from 47.46 $\mu$m bare to 28.91 $\mu$m with nylon case screws and 23.09 $\mu$m with aluminium, reductions of 39.1% and 51.4%. With a 1.5 mm case mesh, the peaks were 28.97 $\mu$m for nylon and 23.10 $\mu$m for aluminium. The bare and nylon peaks occurred at the same board corner. With aluminium, the maximum moved 9.77 mm along the same edge. The comparison used the same 217,704 PCB nodes in all three assemblies.

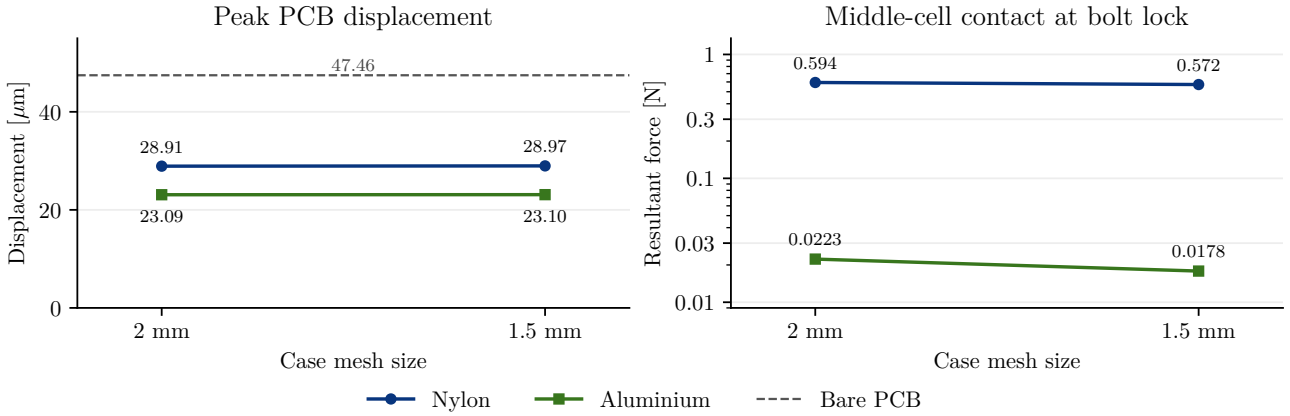



Figure 2: Native results at bolt lock with the original force criterion on both case meshes. The bare board gives one displacement reference. The right panel uses a logarithmic scale. The resultant at the four mounting tabs agreed with each plotted lock force within 5%.

## The middle pocket carries load before vibration begins

All loaded case faces belonged to the cylindrical wall of the middle pocket. The first active state occurred between 0.40 and 0.70 of the tightening step with nylon and between 0.70 and 1.00 with aluminium. Reconstructing the deformed case, three cells and six washers independently at two surface sampling levels also identified contact with the middle cell alone at lock. The closest sampled washer gaps were 0.321 mm with nylon and 0.324 mm with aluminium at the finer level.

| Nominal 2 mm case mesh | Nylon screws | Aluminium screws |
|---|---|---|
| Active contact faces after bolt lock | 5 | 1 |
| Highest contact pressure during tightening [MPa] | 1.080 | 0.163 |
| Highest contact pressure after lock [MPa] | 0.345 | 0.041 |
| Largest loaded-corner penetration over both steps [$\mu$m] | 0.222 | 0.034 |
| Peak PLA tension in-plane / across layers [MPa] | 0.334 / 0.344 | 1.052 / 0.760 |
| Peak screw extreme-fibre tension after lock [MPa] | 1.25 | 2.40 |

Each model converged at eight saved states. All six locked bolt forces were within 0.015 N of the 750 N target. The highest through-layer PLA tension reached 0.85% of the printed-coupon mean with nylon and 1.88% with aluminium. Page 4 gives the tighter nylon pressure result. The original contact pressures come from a small patch on the 2 mm case mesh. The mesh check on page 4 quantifies their sensitivity. Vibration and side pull are separate load cases.

## Mesh sensitivity of board displacement and pocket pressure

Three nylon-case meshes retained the same bare-board mesh, printed geometry, screw axes, 7.60 mm screw grip and PCB bearing nodes. All eight tightening and locking stations matched. The case mesh changes both the surface triangles and the nodes inside each tab's bearing footprint.

| Original force criterion, nylon mount | 3 mm | 2 mm | 1.5 mm |
|---|---|---|---|
| Quadratic case tetrahedra | 18,051 | 45,549 | 99,818 |
| Case-side bearing nodes per screw | 86–89 | 103–104 | 106–108 |
| Native minimum initial gap [$\mu$m] | 2.930 | 2.969 | 2.984 |
| Peak PCB displacement at lock [$\mu$m] | 28.919 | 28.915 | 28.970 |
| First saved state carrying contact, step 1 | 0.70 | 0.70 | 0.70 |
| Active case faces at lock | 4 | 5 | 7 |
| Peak contact pressure during tightening [MPa] | 1.025 | 1.080 | 1.178 |
| Peak contact pressure at lock [MPa] | 0.170 | 0.345 | 0.270 |
| Peak in-plane PLA tension [MPa] | 0.392 | 0.334 | 0.456 |
| Peak through-layer PLA tension [MPa] | 0.299 | 0.344 | 0.398 |

## Board displacement and pocket loading

All three meshes place the maximum displacement at the same PCB corner and the locked contact on the middle cell. The full spread in peak PCB displacement is 0.055 $\mu$m, or 0.19%. Each mesh predicts about 39% less displacement than the 47.46 $\mu$m bare-board result.

With the original stopping criterion, the 2 and 1.5 mm meshes reported peak contact forces of 8.323 and 8.571 N at 70% of tightening. At the same station the resultant reactions at the four PLA bearings were only 0.056 and 0.043 N. These contact peaks failed equilibrium on both meshes. At final bolt lock the 1.5 mm solution gave 0.5716 N independently at the pocket, bearings and cell.

## Both finer meshes with tighter force convergence

We repeated both installation steps with a fixed 0.01 N force-residual criterion. On the 2 and 1.5 mm meshes the middle pocket carried **0.12107 and 0.12014 N** at 70% tightening, then **0.56843 and 0.57158 N** after lock. All eight states on each mesh passed the independent force checks. The largest tab-versus-pocket vector discrepancy was 0.000011 N.

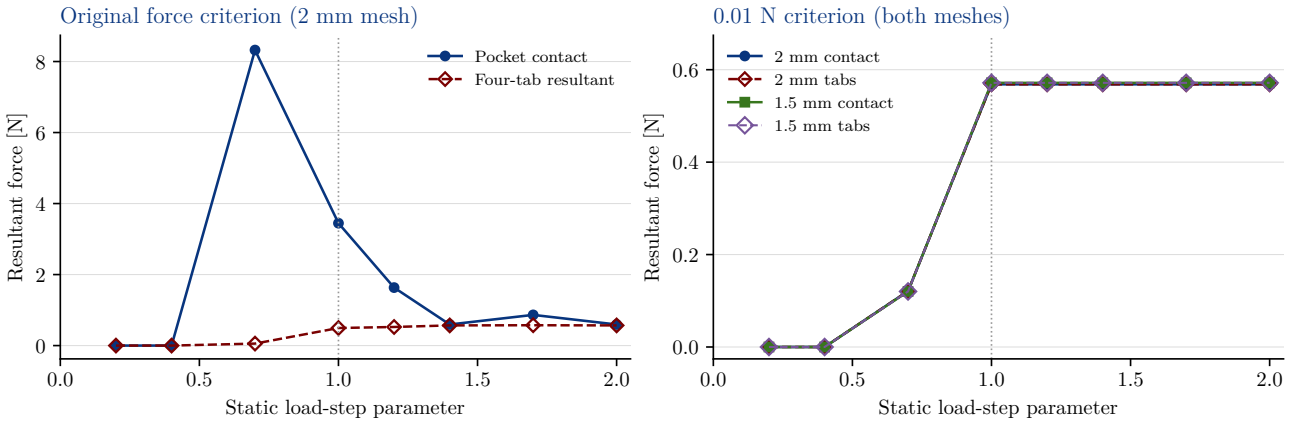



Figure 3: At parameter 1 the six original fasteners reached 750 N each. Step 2 locked their adjustment. Each panel uses its own vertical scale. The tab trace sums the four force vectors before taking their magnitude. The two tighter histories nearly coincide.

At lock, refinement changed the pocket force by **0.55%** and peak PCB displacement from 28.915 to 28.969 $\mu$m. Across all 217,704 common PCB nodes, the greatest vector difference was 0.0687 $\mu$m. Both meshes still reduced peak board deflection by about 39% from bare.

Local contact remained more sensitive. Five loaded points on the 2 mm mesh became ten on seven faces at 1.5 mm. The loaded quadrature area rose 12.9%, from 2.747 to 3.101 mm$^2$, while peak pressure fell **18.4%**, from 0.331 to 0.270 MPa. The area is a numerical integration measure. The tightly balanced total force alone does not establish local pressure convergence. The aluminium tolerance repeat is discussed on page 10. Vibration results retain their original installed states.

## The first two default-contact shapes persist with either mount

We repeated the October default-contact eigenanalysis on the refined board mesh before bolt tightening. The solver returned 20 positive frequencies for the bare board and for each snug mount. These controls reproduce the original mounting constraints and contact definitions. The operating-state modes require a separate extraction about the tightened assembly.

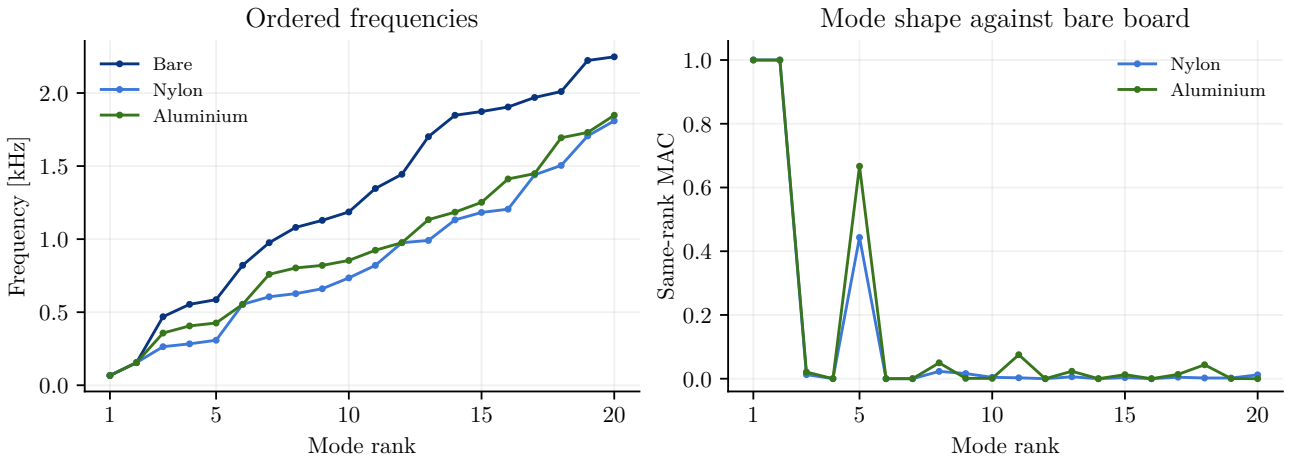



Figure 4: Left, the 20 lowest extracted frequencies for each assembly. Right, the modal assurance criterion (MAC) compares displacement vectors at equal rank on the 368,754 common structural nodes. A value of one indicates proportional shapes on those nodes.

Mode 1 was 66.611 Hz bare, 66.614 Hz with nylon and 66.615 Hz with aluminium. Both case-to-bare MAC values exceeded 0.99999. Mode 2 was also closely matched at about 154.8 Hz. At rank 3 the frequencies were 468.8, 263.9 and 357.2 Hz and the equal-rank shapes differed. The closest of the first 20 cased shapes to bare mode 3 occurred at rank 7 with nylon (MAC 0.714) and rank 4 with aluminium (MAC 0.823).

Mode 20 was 2247.6 Hz bare, 1809.5 Hz with nylon and 1847.7 Hz with aluminium. The same-rank MAC values were 0.012 and below 0.001, respectively. Even the closest shapes among the first 20 cased modes had MAC below 0.15 against bare mode 20. A frequency shift of a single physical mode cannot be inferred from these three rank-20 frequencies.

## Why the original mesh also lists a 1.69 Hz mode

The two coarse-board controls each recovered 20 frequencies with the Sturm check. Their 1.69 Hz mode is localized on the single solid element representing component L3. Every other solid reached less than 0.20% of its peak amplitude, and its largest directional effective mass was 4.44 parts per million of assembly mass. The first bare-board shape at 62.805 Hz reappeared at rank 2 with either screw, at 62.807 and 62.809 Hz (MAC above 0.999999 on 95,215 shared solid nodes).

## Paired 4K movies

The snug-fit movie catalogue links the bare and cased views for modes 1 and 20. The catalogue also includes the twentieth bolt-locked mode. Dedicated 20-mode extractions reproduced the wide-basis rank-20 frequencies of 1982.33 Hz with nylon and 2011.37 Hz with aluminium. Within each mount, the two extractions gave MAC above 0.9999 on the full solid mesh. Each modal film uses the same 6 mm peak display amplitude and camera pose. The nylon and aluminium panels identify the frequency, rank and shape match. The display amplitude is a visualization scale.

## Vibration of the tightened board and case

We restarted each assembly from its last converged bolt-locked equilibrium. The lowest frequency was 485.96 Hz bare, 294.94 Hz with nylon and 415.42 Hz with aluminium. Both mounts introduced shapes below the first bare frequency. Comparing the two screw materials, their lowest shapes had a full-solid MAC of 0.921 and a case-only MAC of 0.929. Aluminium raised the frequency of this matched shape by 40.8%. The cells' peak amplitudes were only 3.8% and 8.6% of the PCB peaks in the nylon and aluminium first modes, respectively. When the middle pocket is held open, the lowest frequencies are 266.38 and 363.65 Hz, respectively (page 25). The open nylon mode at 293.81 Hz most closely retains the seated first shape (mass-weighted MAC 0.946).

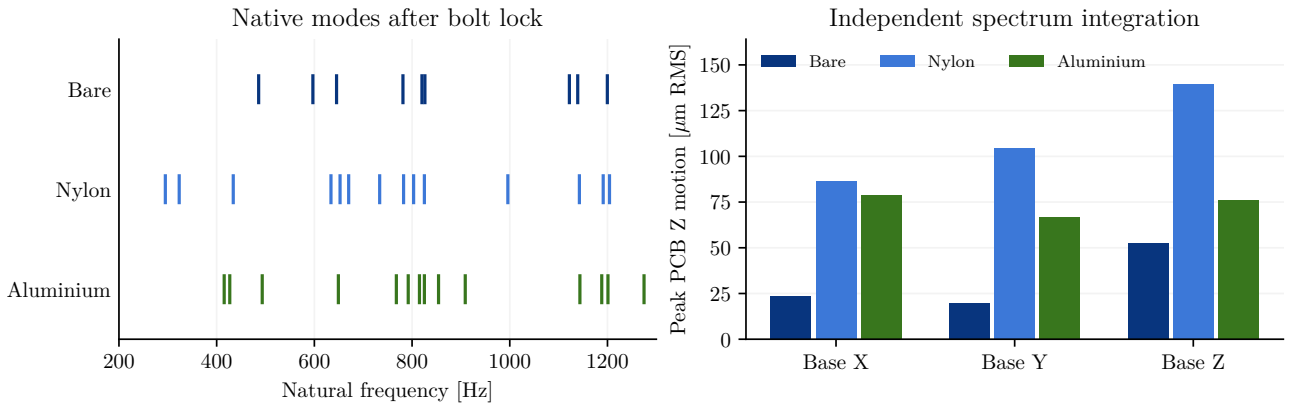



Figure 5: Left, each tick marks a native natural frequency through 1.3 kHz. Right, the maximum board-normal RMS displacement anywhere on the PCB, relative to the base. Each axis was excited separately with the October 20–2000 Hz spectrum and 2% modal damping. The independent calculation integrates the signed native modes and their participation factors.

## Which modes correspond to the bare board?

On the common original assembly, the best matches to the first bare shape occurred at 633.99 Hz with nylon (MAC 0.697) and 426.84 Hz with aluminium (0.694). The first bare mode therefore cannot be tracked reliably by its rank. The 826.30 Hz bare shape persisted at 825.20 and 825.21 Hz, with MAC values above 0.995 for both screws.

## Board response to the original base spectrum

| Peak PCB Z displacement [$\mu$m RMS] | Base X | Base Y | Base Z |
|---|---|---|---|
| Bare | 23.37 | 19.45 | 52.67 |
| Snug, nylon screws | 86.31 | 104.58 | 139.49 |
| Snug, aluminium screws | 78.96 | 66.78 | 75.86 |

Both mounts reduced PCB deflection during bolt tightening. Under base-Z vibration, the calculated RMS board motion increased to 2.65 times bare with nylon and 1.44 times bare with aluminium. The native bases retained 38 bare modes and 48 modes for each mount up to 4 kHz. Removing modes above 3.5 kHz changed all nine tabulated peaks by less than 0.003%. The independent bare calculation reproduced the 18 archived native displacement peaks within 0.013%. The six cased MAPDL spectra reproduced the independent peak PCB Z displacements within 0.002 $\mu$m. Pages 11–12 compare the native stresses across directions.

## Case-mesh qualification

After tightening, MAPDL flagged the same PLA tetrahedron in both mounts at its 179.9° edge-angle limit. On each locked mesh, an independent polynomial bound confirmed a positive Jacobian determinant throughout all 45,549 case elements. The flagged element had the smallest lower bound, 0.01286 mm$^3$ in both cases. The 1.5 mm nylon mesh passed the same orientation check for all 99,818 PLA tetrahedra. No PLA element triggered the native edge-angle error-limit warning on that mesh. Page 13 compares its vibration response.

## Outer cells can reach the case during vibration

The modal model holds the contact conditions present at bolt lock. To test this assumption, we applied the signed modal displacement at physical scale to the case, three cells and six washers. For each base direction we selected the frames with the lowest and highest input acceleration in the original 480-frame film interval. This gives six samples at identical excitation times for each screw material. We reconstructed the quadratic contact surfaces at two subdivision levels and first checked that the zero-motion geometry reproduced the native tightening result.

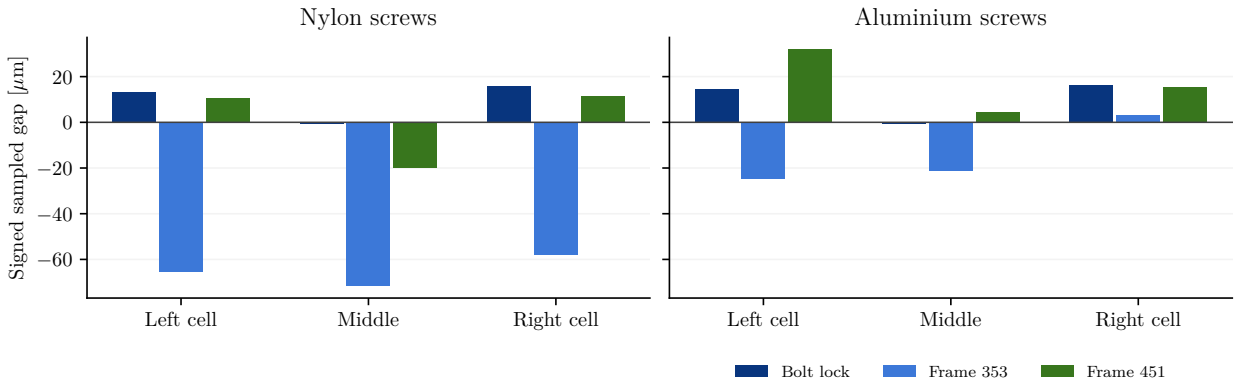



Figure 6: Closest sampled signed distance from each cell to the PLA at bolt lock and at the two selected base-Z frames. A negative distance places a sampled cell-surface vertex inside the case. Triangle-intersection tests also checked each complete surface. Left and right follow increasing PCB X.

At bolt lock the middle pocket carried load with both screw materials. At base-Z frame 353 (1.02155 s), all three cell surfaces intersected the nylon-mounted case. With aluminium screws, the left and middle cells intersected. At frame 451 (1.02753 s), both outer cells cleared both cases. The aluminium model also opened a 0.0046 mm sampled gap at the middle cell. Both spatial refinements agreed on every classification, and all six washers cleared at every sampled instant.

| Base axis | Film frame | Time [s] | Intersecting cells, nylon | Intersecting cells, aluminium |
|---|---|---|---|---|
| X | 75 | 1.00458 | Left, middle | Middle, right |
| X | 183 | 1.01117 | Middle | Middle |
| Y | 85 | 1.00519 | Left, middle | All three |
| Y | 28 | 1.00171 | Middle | Middle |
| Z | 353 | 1.02155 | All three | Left, middle |
| Z | 451 | 1.02753 | Middle | None |

Outer-pocket engagement also appeared during the 0.10 s startup ramp. We selected two earlier base-Z stations using left-cell motion relative to the case and checked all the surfaces again. At 82.14 ms all three cells intersected the nylon-mounted case, while only the middle cell intersected the aluminium-mounted case. At 83.83 ms the aluminium case also intersected the right cell. Both geometric refinements agreed. These instants precede the first calculated PLA strength-reference crossing on page 9.

The extra supports can redirect load once an outer cell engages. The spectrum values on page 6 quantify the response with the installed contact state held fixed. To check load transfer at frame 353, paired local nonlinear models will drive the same region of each mount with the signed base-Z displacement. Each has a contact-enabled and a free control at two cut sizes. The six samples above establish neither the first contact time nor its duration.

## Base-X vibration raises board travel and reduces peak stress

MAPDL applied the original 20–2000 Hz random acceleration along PCB X to the bare assembly and both installed cases. The three native runs used the same 2% modal damping. The table gives the largest one-sigma response within each named part. Each maximum can occur at a different node.

| Base X, native MAPDL response | Bare | Nylon screws | Aluminium screws |
|---|---|---|---|
| PCB Z displacement [$\mu$m RMS] | 23.37 | 86.31 | 78.96 |
| PCB absolute Z acceleration [$g$ RMS] | 52.29 | 95.37 | 78.87 |
| PCB equivalent stress [MPa RMS] | 90.32 | 82.68 | 79.35 |
| Cell-body equivalent stress [MPa RMS] | 21.02 | 18.03 | 20.50 |
| Nickel-strip equivalent stress [MPa RMS] | 291.90 | 267.55 | 259.29 |
| PLA through-layer normal stress [MPa RMS] | — | 4.97 | 11.87 |
| Screw longitudinal stress [MPa RMS] | — | 7.74 | 23.36 |

On the nominal case mesh, peak PCB equivalent stress fell by 8.5% with nylon screws and 12.1% with aluminium screws. The fine nylon model placed the peak within 1% of bare (page 13). The independent calculation on page 6 reproduced all three native PCB Z displacement maxima within 0.001 $\mu$m. The equivalent board material gives a comparison of assembly response. A service margin for the populated PCB requires the laminate data and package-specific board-strain limits.

## How the middle cell transfers vibration load

At bolt lock the middle cell touches the case and remains attached to its two nickel strips. We summed the signed native modal element forces separately at the two strip bonds and at the loaded pocket. The case-side resultant opposed the force recovered on the middle cell outside the bond footprints. Including all three interfaces reproduced the independently integrated modal cell inertia within a relative error of $6 \times 10^{-6}$ across the modes. The other two cells continued to transmit load through the strip bonds in this linearization.

With base-X excitation, independent spectrum integration gave middle-pocket force components of 17.53 N RMS along X and 7.75 N RMS along Z with nylon screws. Aluminium gave 15.83 and 6.07 N RMS. These dynamic components assume the middle contact stays engaged. The native base-X solid-element force records provided a separate check on the modal calculation. Page 7 shows where the fixed contact assumption breaks in the seeded motion.

## Nickel response and movie coverage

The global nickel strips are linear elastic. The archived local damage model uses an exploratory yield reference of 185 MPa. Including preload, the first base-X seeded stress crossing fell at 38.12–38.13 ms bare, 37.83–37.84 ms with nylon and 40.39–40.41 ms with aluminium. Establishing permanent strain or tearing requires the nonlinear strip model. The movie catalogue lists each reviewed pair by base direction and response component. The RMS camera orbits show a fixed statistical field on a shared colour scale.

## Printed-tab tension peaks beside the screw bearing

For each nine-second excitation we combined the locked stress tensor with the signed modal stresses using the native MAPDL coordinates at each time station and element corner. The search covered all 182,196 unaveraged PLA corners and all 589,825 time stations per base direction. Across the layers, the largest value reached 31.17 MPa under base Y with nylon and 84.20 MPa under base Z with aluminium. Both occurred 0.60 mm from the coupled bearing nodes of the tab near PCB $(X, Y) = (66, 4)$ mm.

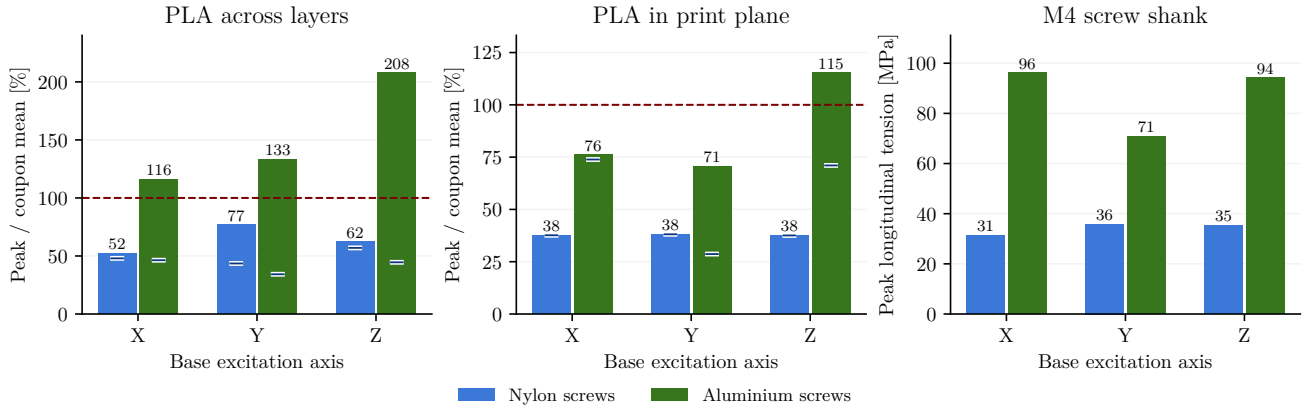



Figure 7: The first two panels divide the peak signed PLA tensile stress by the corresponding printed-coupon mean. The dashed line marks the coupon mean. Small horizontal marks on the bars show the highest result at least 2 mm from any tied bearing node. The right panel gives peak axial-plus-bending tensile stress at a modelled screw-section point. Each panel uses the same seeded input for a given base direction.

## What the percentages measure

The reference strengths are the reported means for uniaxial printed specimens, 40.5 MPa through layers and 52.3 MPa in plane. Nylon mounting reached 77% through layers and 38% in plane, both under base Y. Aluminium reached 208% through layers and 115% in plane under base Z. Across all three aluminium runs, excluding the 2 mm bearing band brought the highest layer-normal and in-plane results to 46% and 74% of their respective means. The rigid screw-bearing footprint and the actual print orientation matter to this local comparison. PolyLite PLA reference data.

For aluminium screws the calculated through-layer tension first crossed the coupon mean at 129.64–129.65 ms under Z, 190.57–190.58 ms under X and 517.99–518.01 ms under Y. The in-plane reference was crossed only under Z, at 845.18–845.20 ms. The exact oscillator integration reproduced all four crossing brackets at the original 1/65,536 s sample interval. The nylon histories remained below both coupon means throughout.

At both worst through-layer instants, the deformed surfaces of all three cells intersected the case in the linear reconstruction. The elastic continuation holds the initial contact tangent fixed even when a new pocket would engage. The aluminium peak is a demand beyond the coupon reference. Predicting cracking or a failure time would require changed contact and a calibrated print failure law.

## What the screw carries

The four circular M4 beam shanks were evaluated at 25 section points at each end. The highest tensile value was 35.90 MPa for nylon under base Y and 96.21 MPa for aluminium under base X. Bending contributed 35.00 and 91.73 MPa, respectively, at those exact instants. The nylon part has no verified tensile rating for this comparison, and the aluminium alloy and temper are unspecified. The native base-X one-sigma stresses agree with the independent spectrum integration over all 200 shank points to better than 0.004% in relative field norm.

An exact oscillator solution for the piecewise-linear acceleration changed the 24 PLA maxima across both screw materials, tensile directions, base axes and bearing-distance scopes by at most 0.43% from the average-acceleration integration. Both methods use the same installed stiffness and material law.

## The nylon middle pocket carries about 0.6 N at bolt lock

The six original fasteners were each tightened to 750 N. Only the middle pocket carried a case-to-cell load. The original 2 and 1.5 mm runs gave 0.594 and 0.572 N at lock over five and seven faces. The tighter 2 mm solve gave 0.568 N over the same five faces. The 8.323 and 8.571 N intermediate peaks on the two original meshes failed the four-tab equilibrium check (page 4).

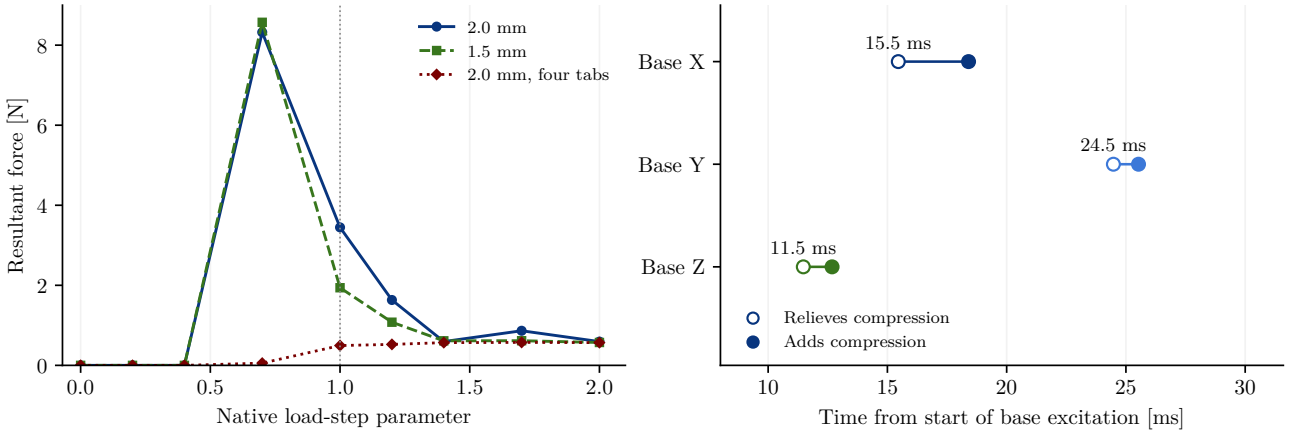



Figure 8: Left, native contact resultants on both meshes and the independently recovered resultant at the four 2 mm-mesh tab bearings. Right, the 2 mm model gives the times when the signed modal increment along the lock-force direction first passed +0.594 and −0.594 N during each separate base acceleration. The open marker and label show when unloading first exhausted the initial compression. The solid marker shows an increment of the same magnitude in compression.

## Why the fixed-contact force becomes inconsistent

The five active frictionless faces point towards the same side of the middle cell. Over their full curved surfaces, the cosine between the case normals and the resultant direction exceeded 0.993. A compressive force on any of those faces contributes with the same sign along this direction. Integrating the stresses in the deformed cell and shell established which sign relieves compression (page 26). Once the signed unloading increment exceeds the initial compression, the frozen patch would have to pull on the cell.

The nominal model reached that condition at 15.46 ms for base X, 24.48 ms for base Y and 11.47 ms for base Z. The 1.5 mm model gave 15.46, 20.19 and 11.58 ms. On both meshes, reducing the modal cutoff from 4 to 3 kHz preserved the three crossings at the original sample interval. Page 26 gives the corresponding aluminium estimates.

At both force crossings for each base direction, the outer cells remained clear on both meshes. On the finer mesh, an intersection within the seven-face installed patch confirmed middle-cell contact even where the whole-shell search lacked sufficient resolution. The force check detects a need to redistribute the installed pressure before those snapshots show an outer cell reaching its pocket. Complete lift-off remains unresolved. The nonlinear controls include an intact shell with all three cells, six nickel strips and six washers so both outer pockets can engage.

## How closely did the installed models balance?

At the rejected 2 mm peak, independent stress integration gave 2.441 N at the PLA pocket, 0.260 N on the cell and 0.056 N at the four tab bearings. With tighter convergence all three gave 0.1211 N at that station. At bolt lock all three gave 0.5684 N, and their moments about the centre of the four bearing pilots agreed within 0.0004 N mm. Individual nylon tab loads remained larger than their net resultant because their directions differed. On the original aluminium mesh at lock the four tab magnitudes ranged from 0.724 to 0.868 N, while their resultant was 0.02229 N against 0.02230 N at the pocket. The tighter aluminium repeat passed the mounting-force check at every station and retained 0.02230 N at lock. The finer nylon repeat also passed all eight states. Its 0.57158 N lock force is 0.0052% below the earlier fine result. At 70% tightening it carried 0.12014 N, replacing the rejected 8.571 N output. Direct barrel reconstruction reproduced every loaded face and the locked resultant within 0.22 $\mu$N. Page 4 compares both meshes at the tighter criterion.

## Under base Y, stress rises across most of the PCB

The peak equivalent stress under Y excitation increased from 17.85 MPa RMS bare to 50.21 MPa with nylon screws and 48.40 MPa with aluminium. The bare peak lies at $(10.0, 16.0)$ mm in the PCB plane. The nylon peak shifts to $(9.5, 118.2)$ mm, and the aluminium peak to $(62.6, 5.5)$ mm.

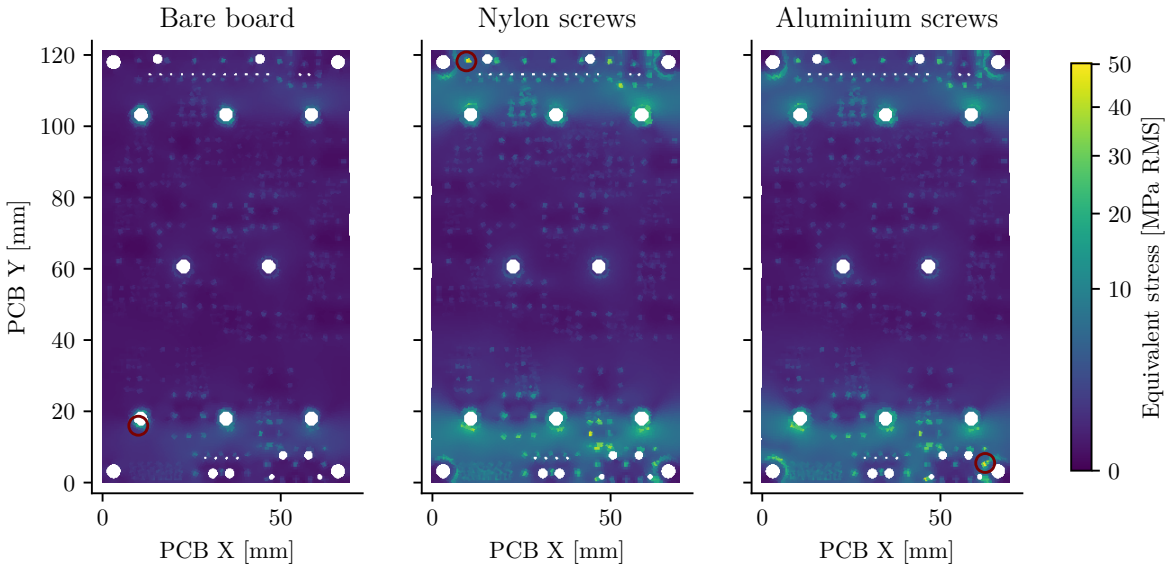



Figure 9: Native base-Y stress fields in the original board coordinates. A ring marks the peak in each assembly. The three maps share one square-root colour scale, labelled in MPa RMS. Each mesh footprint shows the largest stored equivalent stress among all corners through the board thickness.

## How much of the board changes?

We matched the two thickness layers by their common mid-plane nodes, giving 19,507 columns across 39,014 quadratic PCB elements. The footprint areas sum to 8,150.067 mm$^2$. Dividing the native element volume at the first bare preload station by nominal thickness gave 8,150.066 mm$^2$. Comparing each column to the same place on the bare board, the base-Y stress rose by more than 10% over 98.0% of the footprint with nylon and 97.6% with aluminium.

| Base-Y equivalent stress [MPa RMS] | Bare | Nylon | Aluminium |
|---|---|---|---|
| Area-weighted median | 0.264 | 0.695 | 0.622 |
| Area-weighted 95th percentile | 1.85 | 8.84 | 7.22 |
| Highest element-corner result | 17.85 | 50.21 | 48.40 |

Under base X the highest stress fell with both mounts, but the area-weighted 95th percentile rose from 3.54 MPa bare to 9.97 MPa with nylon and 10.06 MPa with aluminium. A lower peak at one corner can therefore coexist with higher stress across a substantial part of the board.

## Board-normal vibration separates screw stiffness from peak stress

Under base Z, aluminium screws limited peak board-normal motion to 75.86 $\mu$m RMS, compared with 139.49 $\mu$m with nylon. The aluminium assembly nevertheless developed the larger peak PCB stress, 98.65 versus 69.37 MPa RMS. The bare values were 52.67 $\mu$m and 32.48 MPa. The nine native spectra used the original load level and damping and held each assembly's contact state fixed at bolt lock.

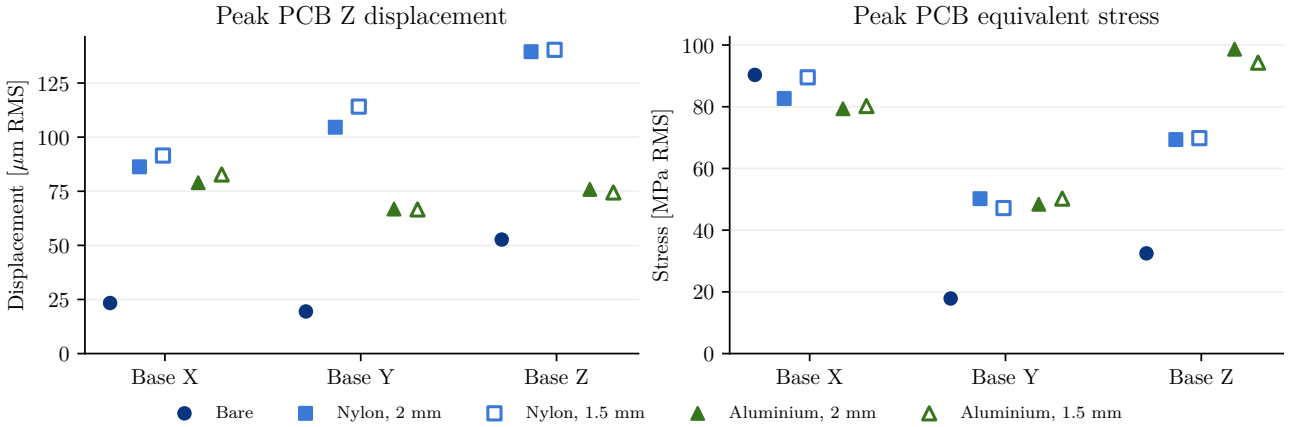



Figure 10: Separate X, Y and Z base excitations. Filled markers show native MAPDL spectra for the bare board and 2 mm case meshes. Open markers show independent integration on each native 1.5 mm basis. Each maximum is taken over the whole PCB. The two mesh sizes use their own installed contact state.

## Which excitation frequencies drive the peak stress?

At each assembly's own governing PCB corner, we integrated the signed modal stresses over disjoint forcing-frequency bands, retaining all cross-correlations between modes. The bands sum to the full mean square. Under base X, the 500–800 Hz band accounted for 95–97% of the mean square in all three assemblies. Under base Z, the strongest band changed with the mount.

| Base-Z assembly | Strongest forcing band | Share of stress mean square |
|---|---:|---:|
| Bare | 800–1200 Hz | 74.9% |
| Snug, nylon screws | 200–350 Hz | 94.7% |
| Snug, aluminium screws | 350–500 Hz | 92.1% |

The full signed-tensor calculation reproduced all nine native peak PCB stresses within 0.1%. Reducing the modal cutoff from 4 to 3.5 kHz changed those peaks by at most 0.20%. This checks modal truncation within the 2 mm case model. Page 13 compares separately tightened 1.5 mm meshes for both screw materials. The largest aluminium PCB stress changed by +1.1%, +3.8% and −4.4% under X, Y and Z. The finer base-Z peak remained 2.90 times bare.

Peak nickel-strip equivalent stress under base Z also increased, from 38.13 MPa RMS bare to 122.82 MPa with nylon and 138.21 MPa with aluminium. The signed linear startup histories first crossed the local model's assumed 185 MPa yield reference at 70.30–70.31 ms and 61.25–61.26 ms, respectively. The bare history stayed below it over the first 100 ms. These conditional elastic crossings locate a demand for nonlinear analysis. Permanent strain and tearing require the local material and contact histories.

## How much does the case mesh change the vibration result?

We repeated tightening and the prestressed eigensolve with 99,818 quadratic PLA elements. The first nylon frequency moved from 294.94 to 293.37 Hz and aluminium from 415.42 to 413.00 Hz. First-mode correlations on 369,584 original assembly nodes were 0.993 and 0.996. Of the 48 nominal modes, 31 nylon and 45 aluminium modes matched above 0.95. The finer bases contained 49 and 48 modes through 4 kHz.

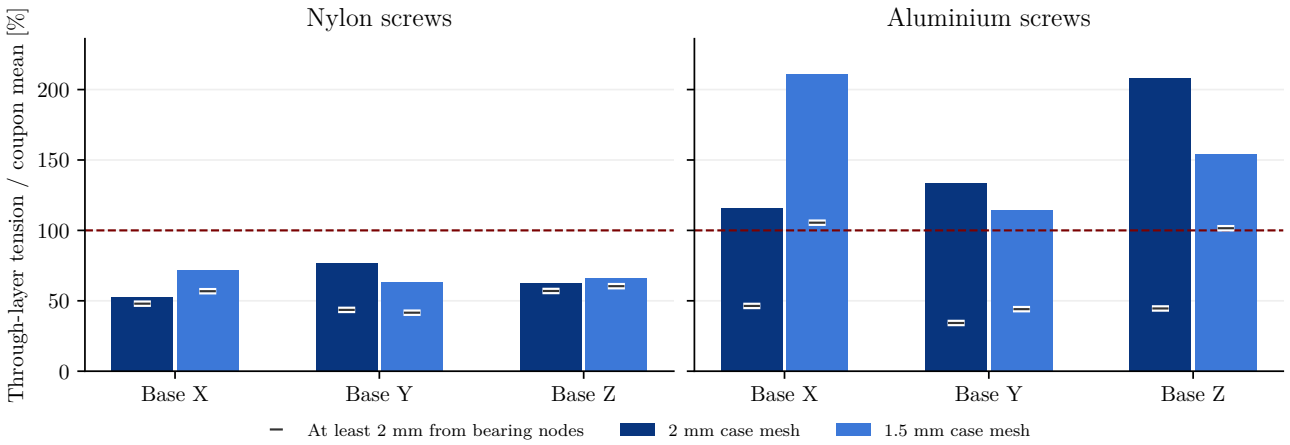



Figure 11: Peak signed through-layer PLA tension during each nine-second input, including preload. Bars show all corners and horizontal marks exclude points within 2 mm of a bearing node. Independent integration uses native modes. The reference is the 40.5 MPa printed-coupon mean.

## Board stress depends on the excitation axis

For nylon under base X, the 2 mm model gave a peak board stress 8.5% below bare, while the 1.5 mm calculation came within 0.9% of bare. The base-X 95th percentile across the board footprint was 9.97 and 9.78 MPa on the two case meshes, compared with 3.54 MPa bare. Under base Z, the board-normal RMS motion changed by 0.62% on refinement and peak stress changed by 0.66%. Under base Y, the same quantities changed by +9.1% and −6.1%.

| Peak PCB equivalent stress [MPa RMS] | Base X | Base Y | Base Z |
|---|---|---|---|
| Bare, native MAPDL spectrum | 90.32 | 17.85 | 32.48 |
| Nylon, 2 mm, native MAPDL spectrum | 82.68 | 50.21 | 69.37 |
| Nylon, 1.5 mm, integrated from native modes | 89.54 | 47.16 | 69.83 |
| Aluminium, 2 mm, native MAPDL spectrum | 79.35 | 48.40 | 98.65 |
| Aluminium, 1.5 mm, integrated from native modes | 80.23 | 50.22 | 94.29 |

## Printed tab and middle-pocket contact

The largest nylon layer-normal tension changed from 31.17 to 28.94 MPa (77.0% to 71.5% of the coupon mean). Aluminium changed from 84.20 MPa under Z to 85.38 MPa under X (207.9% to 210.8%). Both fine maxima lay about 0.6–0.7 mm from constrained bearing nodes. Two time integrations agreed within 0.37% on all fine tensile maxima.

Away from the bearings, finer aluminium reached 42.72 MPa under X and 41.12 MPa under Z. These maxima occurred at one node of the locked middle-pocket face, 39.52 mm from a bearing node. Two incident tetrahedral corner records crossed the layer reference under X and one under Z. Within that region, no tetrahedron had all four corners above it at those instants.

The force-balance bound requires the finer aluminium pocket to redistribute pressure within 6.00, 7.22 and 5.65 ms under X, Y and Z. Its first layer-reference crossings occurred much later, at 183.11, 459.70 and 486.34 ms; the two integrations agreed to one sample. These local elastic demands use the retained contact state. Actual cracking requires a model of the evolving contact and printed material.

Peak M4 longitudinal tension reached 37.4 MPa with finer nylon and 96.2 MPa with finer aluminium, compared with 35.9 and 96.2 MPa on the nominal meshes. At the finer aluminium maximum, bending contributed 91.7 MPa and direct tension 4.53 MPa at the same section point. Exact piecewise-linear input integration reproduced all 24 refined screw peaks within 0.302%.

## Board-normal vibration shifts the highest bond load to the middle cell

Under base-Z excitation the largest dynamic force at a cell-to-nickel attachment rose from 30.22 N bare to 40.09 N with nylon case screws and 50.50 N with aluminium, increases of 33% and 67%. The right cell's low-Y bond governed on the bare board. The middle cell's high-Y bond governed in both snug mounts. Under base X the largest bond load increased with nylon screws and changed by less than 1% with aluminium.

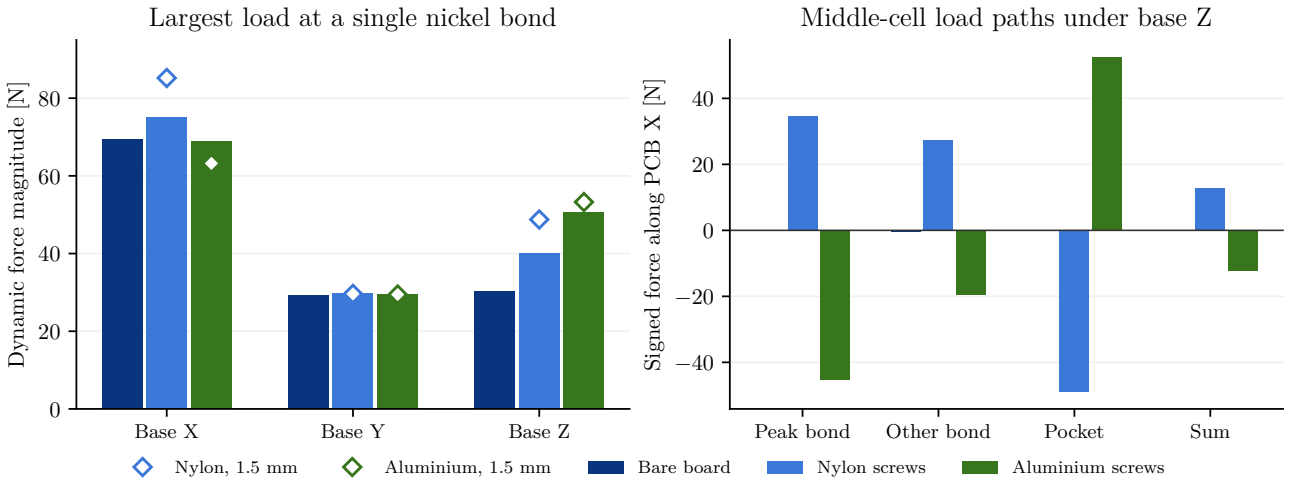



Figure 12: Left, the maximum simultaneous three-component force magnitude at any of the six bonds in each nine-second record. Diamonds give both finer case meshes. Right, nominal-mesh signed PCB-X force on the middle cell under base Z, evaluated separately for each assembly at its own largest middle-cell bond load. Opposing bond and pocket forces partially cancel in the total.

## How much does the case mesh matter?

Refining the nylon mesh from 2 to 1.5 mm raised the governing bond force under base X from 74.95 to 85.20 N (+13.7%) and under base Z from 40.09 to 48.77 N (+21.6%). Under Y it stayed near 29.6 N, while the pocket's maximum force increased from 17.98 to 28.94 N. The same bond governed each direction on both meshes. On finer aluminium, the peaks changed from 68.95 to 63.22 N under X (−8.3%), 29.52 to 29.45 N under Y (−0.2%), and 50.50 to 53.25 N under Z (+5.5%). The middle cell governed X and Z on both aluminium meshes.

## Where the load goes

At the middle cell's governing instant under base Z, its two bonds carried +34.6 and +27.3 N along PCB X with nylon screws. The pocket contributed −49.0 N, leaving +12.9 N across the three interfaces. For aluminium, the corresponding components were −45.3, −19.5 and +52.6 N, giving −12.2 N. The modest whole-cell resultant therefore hides substantial demands on each attachment.

## Checks and interpretation

The reconstruction uses the signed native modal forces on each complete bonded area and the nominal prestressed mode shapes. For all three assemblies we checked native MAPDL coordinates against independent integration through all 589,825 stations on each axis. A second integration, exact for a linearly interpolated input, changed the nine largest bond loads by at most 1.18%. A 3.5 kHz mode cutoff changed them by at most 0.92%. The chart uses native MAPDL histories for the nominal meshes and bare board, with independent integration for both finer case meshes. The two integrations agreed within 0.88% for the aluminium bond peaks. At every station we summed both bonds and the installed middle pocket and checked the result against the cell's independently mass-weighted modal force operator. The largest relative difference over a whole history was below $2.4 \times 10^{-6}$.

These loads describe intact bonds and contact held at its installed state. Page 10 shows the early pressure redistribution required by the nylon model. Joint strength, actual contact forces after redistribution, and damage in a weld require separate evidence.

## Base-X time histories give higher peak PCB stress in both snug mounts

Across the 480 original movie frames, the maximum dynamic PCB equivalent stress reached 154.41 MPa bare, 177.87 MPa with nylon case screws and 170.75 MPa with aluminium. These increases of 15.2% and 10.6% apply to the displayed 29.24 ms of the seeded base-X record. The largest bare and nylon stresses occurred at the same element corner. The aluminium maximum moved to another corner and occurred at a different time.

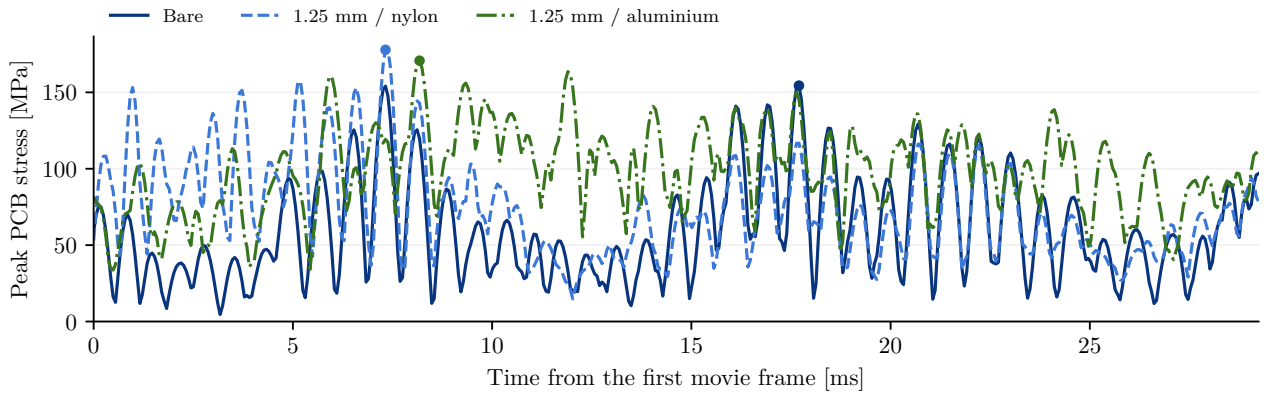



Figure 13: Maximum dynamic von Mises stress across 312,112 unaveraged PCB corners at each matched instant. The solver supplied the signed modal coordinates. We summed all six stress components before forming the equivalent stress. The dots mark the maximum for each assembly.

## Comparing the boards at the same instant

| Governing assembly | Time from start [s] | Bare [MPa] | Nylon [MPa] | Aluminium [MPa] |
|---|---|---|---|---|
| Nylon | 1.007385 | 154.09 | 177.87 | 119.21 |
| Aluminium | 1.008240 | 119.63 | 143.00 | 170.75 |
| Bare | 1.017761 | 154.41 | 116.70 | 145.56 |

At the instant of the nylon maximum, the aluminium board stress was lower than bare. At its own maximum, aluminium exceeded both other boards. A single comparison frame would miss this change in ordering. The plotted values are linear dynamic increments about bolt lock. The RMS spectra on pages 8 and 12 characterize the response to the input spectrum over its frequency band. Neither quantity provides a failure threshold for the populated PCB.

## Watching the base-X comparisons

The movie catalogue includes the assembly's X displacement, the board's Z displacement and its dynamic equivalent stress. Each subject pairs the bare board with both snug mounts at matching times and on one physical scale. The views magnify displacement 50× and slow playback by about 273×. The native simulations hold the installed contact state fixed throughout the original nine-second records.

## The snug case increases PCB travel in all three motion clips

With the original nine-second acceleration records, ANSYS produced 589,825 modal-coordinate samples per axis for the bare board and both snug mounts. The plotted maxima cover the 480 displayed instants from 1.000061 to 1.029297 s. Under base Z the bare PCB moved up to 0.168 mm normal to the board. The nylon mount reached 0.390 mm and the aluminium mount 0.248 mm. Peak motion under base Y was about 5.8 times bare with nylon screws and 3.4 times bare with aluminium.

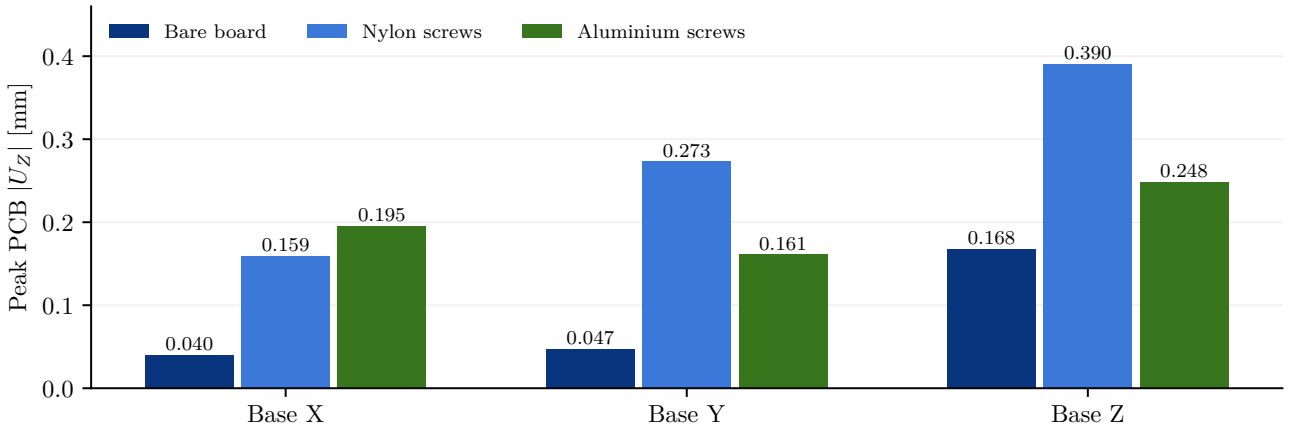



Figure 14: Largest absolute board-normal displacement on the PCB movie surface. Each bar uses the same 480 instants, with the structure moving relative to its mounting base. Page 12 gives the corresponding one-sigma spectrum results.

## Which movie shows which response?

The complete movie catalogue has 22 pairs. The three mode subjects use a common 6 mm display normalization. In the twelve spectrum subjects the camera moves around a fixed one-sigma displacement field. The seven seeded subjects replay signed motion or dynamic PCB equivalent stress at successive times. Each comparison has a bare movie and a companion showing both screw materials.

## How the movie fields were checked

Every bare and cased video decoded at $3840 \times 2160$ and 60 frames/s. The corresponding panels share the response quantity, physical interval and scale. Independent integration recovered the scalar field actually painted on each of the 21 seeded panels to within 0.0026% in relative field norm over the whole clip. We also inspected three decoded frames from each of the 44 files. The animation magnifies motion 50×. The values in the graph use the physical displacement.

## Which outer cell reaches its pocket first?

Both outer cells clear the case when the bolts lock. Under the same seeded base-Z vibration, the nylon assembly's right cell reaches its pocket first, at about 31.1 ms. With aluminium screws the left cell reaches the case first, at about 42.6 ms. These times describe motion along the nominal 2 mm model's prestressed modes. The middle cell already carries load when vibration starts.

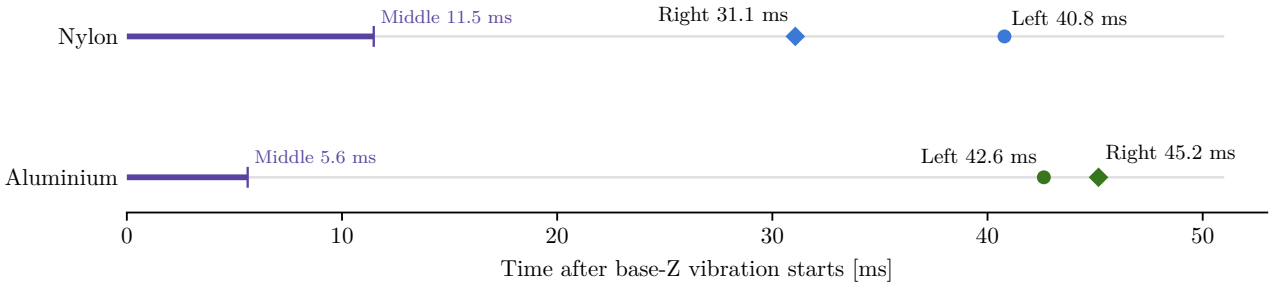



Figure 15: Outer-cell markers show the first sampled surface crossing bracketed from the native MAPDL motion history. Purple marks locate the loss of net middle-pocket compression on the nominal fixed-contact trajectories, at 11.47 ms with nylon and 5.62 ms with aluminium. The aluminium preload comes from independent stress integration (page 26). Once that pressure changes, the real assembly can follow a different path.

| | | First sampled crossing interval [ms] | | |
|---|---|---|---|---|
| Case screws | Outer cell | 2 mm native | 2 mm independent | 1.5 mm independent |
| Nylon | Right | 31.051–31.083 | 31.051–31.083 | 31.036–31.067 |
| Nylon | Left | 40.771–40.803 | 40.771–40.787 | 40.756–40.772 |
| Aluminium | Left | 42.602–42.634 | 42.602–42.634 | Pending |
| Aluminium | Right | 45.150–45.167 | 45.150–45.167 | Pending |

Bounds rounded outwards. Each lower endpoint is the last sample in a sequence with both surfaces proven separate. At the upper endpoint the two curved faces share an interior point. A sample spans 0.01526 ms.

## How the crossings were checked

At every sampled station from rest, we searched the entire 16,596-face printed shell against the quadratic faces of each outer cell. Subdivided Bézier control hulls establish separation. At a crossing we evaluated the original six-node interpolations and found the same point on both surfaces to within $10^{-9}$ mm, with nonparallel normals. This avoids the premature overlap that can appear on faceted renderings of the curved cells. Independent integration of the same modes and acceleration, exact for linearly interpolated input, reproduced the order for both screw materials.

## Does a finer case mesh change the order?

For nylon screws, both independent brackets moved one sampling interval earlier when the case mesh was refined from 2 to 1.5 mm. The right cell still reached its pocket about 9.7 ms before the left. Allowing for the width of both brackets, the largest timing difference compatible with these samples is 0.046 ms. The largest base-Z nickel-bond load increased by 21.6% on the same refinement (page 14). The timing comparison alone cannot establish the impact load.

## What the local fixtures can recover

The 15 mm left-side fixture includes both faces at the verified left-cell crossings for nylon and aluminium. The 9 mm fixture includes the native aluminium left crossing but misses the native nylon left crossing. Both radii omit the right cell and every PLA face that carried middle-cell load at bolt lock (five with nylon, one with aluminium). Their local reactions consequently cover only part of the assembly. The intact-shell controls include all three cells and the original hardware. A nonlinear vibration solution is still needed to determine impact force, separation and rebound.

## How far does the left cell move relative to its clamps?

The lateral pull fixture drives the left cell along PCB X. In the nine-second base-X vibration record, the peak displacement along that axis relative to the two nickel-strip clamps was 48.2 $\mu$m bare, 34.3 $\mu$m with nylon screws and 42.6 $\mu$m with aluminium screws. The snug case reduced the peak by 28.8% and 11.7%, respectively. Under base Y and base Z, both mounts increased this cross-axis motion.

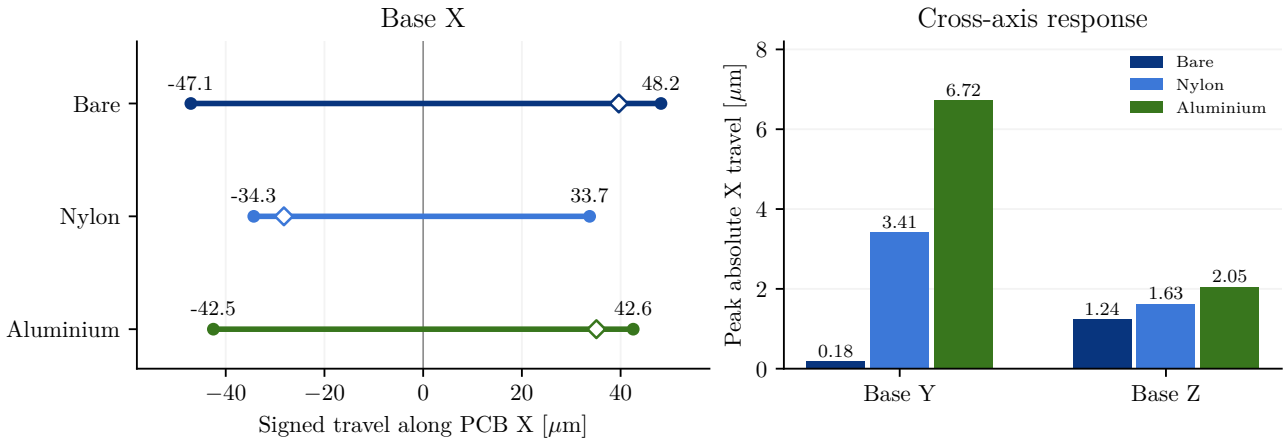



Figure 16: Left, signed extremes of lateral travel from bolt lock under base X. Positive points toward the middle cell. The open diamond shows the contribution from rotation at the larger absolute peak. Right, peak absolute X travel under base Y and Z. Each assembly receives the same drive for each base axis.

## Where the displacement is measured

The cell's mass centroid lies within 0.002 mm of the bare fixture's drive point. We fitted the motion of the two washer faces and subtracted it at the cell centre. Using a volume-weighted rotation of the cell, we also followed a virtual point 10.625 mm above its centre, level with the flat nickel legs. All values are increments from bolt lock.

At the largest absolute base-X excursion, rotation of the cell relative to its clamps accounted for **39.6 of 48.2 $\mu$m bare**, **28.2 of 34.3 with nylon** and **35.1 of 42.6 with aluminium**. The remaining travel at strip height was 8.5, 6.1 and 7.5 $\mu$m. This virtual measurement describes the motion of the cell as a body. Measuring slip at a weld would require displacements on both sides of that weld.

## Numerical checks and interpretation

All three assemblies retained the same washer-face nodes and quadratic connectivity. Fits directly to the reconstructed displacements at each signed extreme reproduced the modal projection to within $10^{-7}$ mm. Independent integration of the same modes with linearly interpolated acceleration changed the 18 signed extrema by at most 1.07%. Refining the nylon case from 2 to 1.5 mm reduced its independently calculated base-X peak by 4.1%, to 33.1 $\mu$m. The finer base-Z value increased by 11.6%, to 1.81 $\mu$m.

## When do the outer cells reach the case under base X and Y?

On the nominal 2 mm native fixed-contact trajectory the left cell reaches the case first for both screw materials. With nylon it reaches the case at 38.28–38.30 ms, about 0.46 ms after the first nickel-strip corner crosses the assumed 185 MPa yield reference. With aluminium the left cell reaches the case at 29.09–29.12 ms, roughly 11.3 ms before that nickel reference is crossed. The right pocket follows at about 40.4 ms in both models. For nylon the middle pocket already needs to redistribute its installed pressure by about 15.46 ms under base X.

The quasistatic fixture prescribes one cell displacement while the global model excites the board, the clamps and all three cells. The case itself also vibrates, so clamp-relative cell travel cannot by itself establish pocket closure, nickel damage or pull force. The results above use the fixed contact state at bolt lock. Under base Y the left cell crosses first at 38.0 ms with nylon and 48.1 ms with aluminium, about 34.2 and 42.3 ms before the respective nickel reference crossings. Independent integration reproduced the outer-cell order for both mounts under all three base axes. The largest change between integration methods was one sample (0.0153 ms). Page 20 compares the event order on two nylon case meshes.

## The snug case restrains the edge cell in a sideways pull

At about **0.107 mm** of cell travel, the local fixture required **47 N with nylon screws** and **116 N with aluminium**, versus about **5.4 N bare**. This is the last common saved station below all tested corner references, not a safe-load rating. Refinement moves the aluminium PLA reference crossing to **0.107–0.132 mm** (page 24). The cased decks differ only in screw properties.

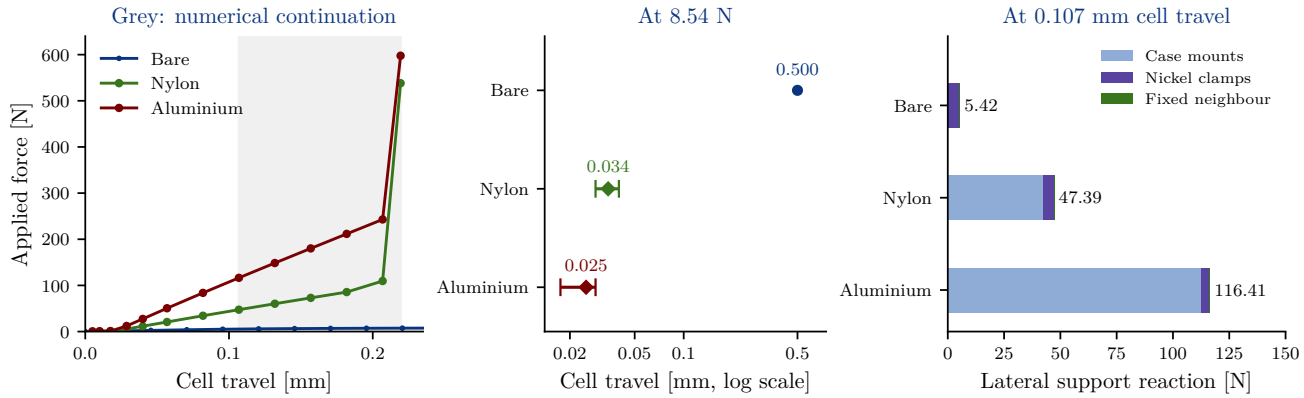



Figure 17: Left, native forces; grey marks continuation beyond the last common reference-screened point. Centre, travel at 8.54 N; bars span saved states and diamonds interpolate. Right, reactions at 0.107 mm. Bare forces interpolate between saved states.

## Equal load and equal travel answer different questions

At 8.54 N, bare travel was 0.500 mm. Nylon brackets this load at 0.02875–0.04000 mm, with a linear estimate of **0.03436 mm**. Aluminium brackets it at 0.01750–0.02875 mm, with an estimate of **0.02515 mm**. The aluminium interval crosses first contact; the bracket remains essential. Both mounts first carried pocket pressure in that interval.

## Higher restraint transfers more load into the mounts

In the numerical continuation at 0.207 mm, the four screw supports carried **90.31 N with nylon** and **239.03 N with aluminium**. The nylon case also loaded the fixed right cell by 12.69 N; aluminium left all neighbouring hardware unloaded. Nickel clamps carried 6.54 and 3.88 N, respectively. Bearing forces and moments agree with beam ends, spacers and separate DPF support reactions at all saved stations.

| At 0.206875 mm imposed cell travel | Nylon screws | Aluminium screws |
|---|---|---|
| Peak PLA in-plane tension [MPa] | 17.94 | 43.21 |
| Peak PLA through-layer tension [MPa] | 17.97 | 23.75 |
| Peak screw-shank tension [MPa] | 27.64 | 108.88 |
| Driven-cell mean contact pressure [MPa] | 1.74 | 3.84 |
| Driven-cell peak contact pressure [MPa] | 16.12 | 37.22 |
| Driven-cell loaded quadrature area [mm$^2$] | 59.57 | 62.65 |

The screw peaks occur at the fixed PCB ends. Bending supplies 26.68 MPa of the nylon peak and 106.88 MPa of the aluminium peak. Face forces and deformed areas independently reproduce the pressures. These peaks need spatial convergence; screw, thread, cell-wall and sleeve limits remain unknown. These parent-mesh PLA peaks understate the refined sharp-corner demand; the aluminium reference has already been crossed.

## The mounting assumption matters

The nylon tabs moved 0.0817–0.0837 mm sideways, while aluminium tabs moved 0.0176–0.0184 mm. Tied tab and screw-tip rotations require opposing normal loads within the 3.75 mm bearing radius: at least 6.33–8.11 N per nylon tab and 5.80–7.28 N per aluminium tab. Rotation-release controls are deferred. The complete-board case (page 34) permits PCB and neighbouring-cell motion.

## Verified scope and closed campaign

All 24 aluminium checkpoint equilibria pass their original balance screens. Later material refinement restricts interpretation: the roughly 538 N nylon and 598 N aluminium results at 0.219 mm (page 21) are numerical continuation. Both reverse pulls are audited (page 35). Both nominal damage cases are closed with numerical limits (page 2). Both increment controls passed all 42 saved states per case (page 28). The final target-resolution check also passed (page 36). The finite campaign is closed; broad sweeps remain deferred.

## Refining the case mesh changes the base-X event order

For the snug nylon-screw assembly, the 2 mm case mesh puts the first nickel stress crossing about 0.46 ms before the left cell reaches the case. On the 1.5 mm mesh, the left cell reaches the case **9.64–9.68 ms before** the nickel crossing. Both models predict that the left cell reaches its pocket first under base X. The relative timing of pocket contact and nickel loading remains sensitive to mesh size.

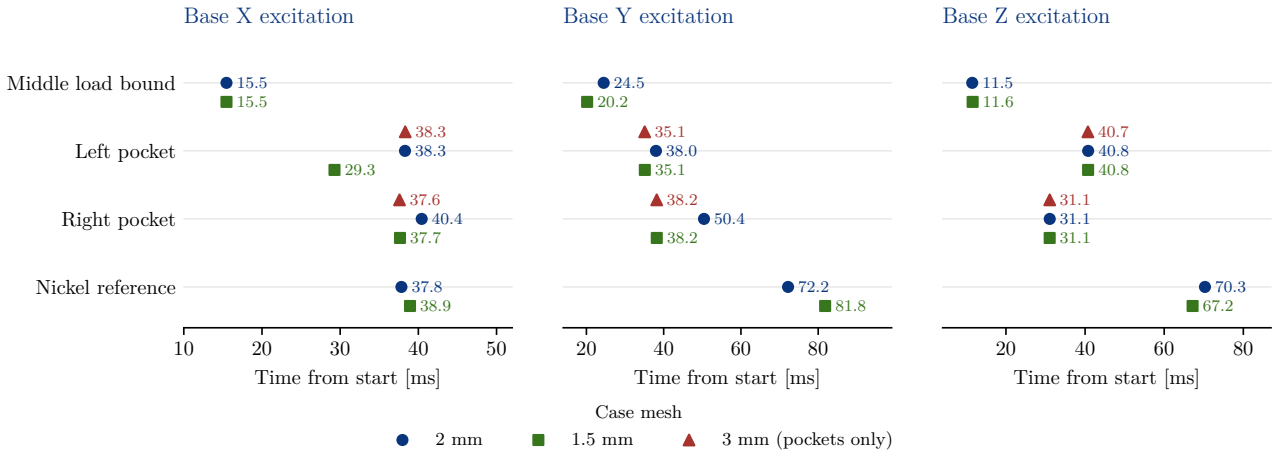



Figure 18: Times from the start of the matched base acceleration. Pocket markers give the midpoint of the first confirmed surface-intersection interval. The middle-load marker bounds persistence of the initial pressure distribution. The nickel marker gives the first sampled 185 MPa crossing in the strips. The 3 mm basis supplies outer-pocket displacements only. Each mesh uses its own installed state and mode shapes.

## Which prediction changes?

Under base X, the fine mesh puts the left-pocket crossing at 29.27 ms and the right at 37.65 ms. These are 9.02 and 2.78 ms earlier than on the nominal mesh. The nickel reference moves later, from 37.83 to 38.93 ms, and a different strip governs. The initial sampled gap at the left outer cell is slightly *larger* on the finer mesh, 13.56 versus 13.14 $\mu$m. Tighter initial clearance alone cannot explain the earlier crossing.

Under base Y the left cell reaches the case first at 35.09, 38.00 and 35.12 ms on the 3, 2 and 1.5 mm meshes. The right-cell crossings are 38.20, 50.44 and 38.22 ms. The nonmonotonic sequence prevents claiming convergence. The nickel reference follows the first outer crossing on the two finer meshes by about 34 and 47 ms.

Under base Z, the right cell reaches the case first on all three meshes, near 31.1 ms. The 3 mm mesh gives 31.052–31.067 ms on the right and 40.710–40.726 ms on the left. Refining from 2 to 1.5 mm shifts each pocket bracket by one sample (0.0153 ms). The nickel reference moves from 70.30 to 67.18 ms and follows both outer crossings.

## What the calculation establishes

We integrated the same piecewise-linear acceleration record at 65,536 Hz on the two native MAPDL modal bases. For each sample the signed modal stresses were added to the locked preload before evaluating the equivalent stress at all 11,348 nickel element corners. The two meshes have identical original nickel elements, corner nodes and coordinates. Retaining modes only through 3.5 kHz shifted the first nickel crossing by at most one sample across all six mesh/axis combinations. Under base X, the 3.5 and 4 kHz calculations also gave identical sampled crossing intervals for both outer pockets on each mesh. The native nominal histories confirmed the nickel crossing intervals.

In both finer base-X calculations, the middle pocket already requires a change in pressure by about 15.5 ms. The later times depend on continuing along the locked-contact modes. On the 3 mm mesh, the right pocket crosses at 37.582–37.598 ms and the left at 38.315–38.330 ms, reversing the order on the two finer meshes. Its initial faceted candidate at 29.33 ms was rejected: deeper convex hulls proved separation. Both later crossings have independent quadratic-surface witnesses. The three meshes do not establish converged outer-contact timing or order. These fixed-contact trajectories identify where a contact-changing solution is needed.

## Pocket pressure rises as the fixed middle cell takes load

In the numerical continuation beyond the refined aluminium PLA reference, both mounts first load the fixed middle cell between **0.207 and 0.219 mm**. At the upper station the nylon actuator supplies **538.13 N**, of which the middle cell takes **424.29 N** (78.8%). Aluminium requires **597.68 N**, with **348.67 N** (58.3%) at the middle cell. These neighbours are rigidly restrained in the local fixture; on the full board they move.

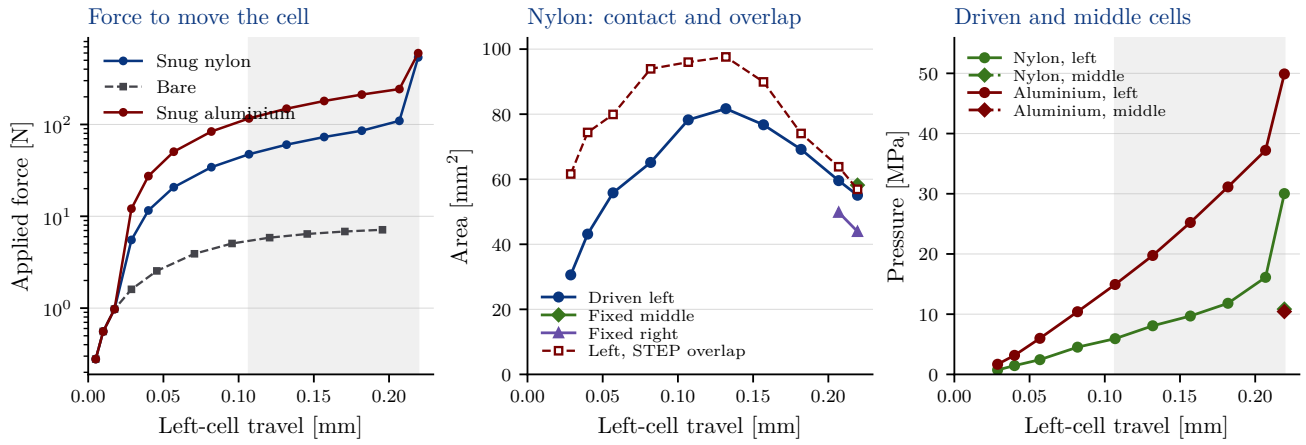



Figure 19: Left, native forces with bilinear nickel. Centre, nylon loaded area and STEP overlap. Right, cell pressures. Grey marks numerical continuation beyond the last common reference-screened point.

## Which cell sees the highest pressure?

With nylon, driven-cell loaded area fell from 59.57 to 55.08 mm$^2$ across this interval; mean pressure rose from 1.74 to 9.68 MPa and the peak from **16.12 to 30.03 MPa**. Aluminium gave **49.90 MPa peak** and 10.44 MPa mean over 56.96 mm$^2$ at the upper station. Both middle cells averaged 7.29 MPa, but over 58.20 mm$^2$ with nylon and 47.84 mm$^2$ with aluminium. Their peaks were 10.85 and 10.43 MPa. Nylon also loaded the right cell (3.31 MPa peak); aluminium left it clear.

## How much does the faceted cell surface matter?

At 5.53 N, 112 of 18,372 pocket points carried native pressure, covering 30.58 mm$^2$. Holding the deformed case and rigid cell pose fixed, successively finer target skins put 201 and 224 points inside the cell. The trimmed STEP solid gave 227 points (61.61 mm$^2$). At 11.57 N the finest skin and STEP agreed on the count of 278, compared with 164 loaded natively. The completed native control changed matched forces by less than 1%. At 0.040 mm, loaded area rose about 12% and peak pressure about 6% (page 36).

Independent force, area and target-ownership checks reproduce the native pressure fields. Rigid cells provide no sleeve or can strength limit. Nylon stopped after about 16 hours at 0.470 mm travel. Two later states failed the Z-force balance; the first 63 passed. The tighter material interpretation is separate: the paired aluminium reference bracket is now 0.107–0.132 mm (page 24).

## How much does the installed case restrain each cell?

At bolt lock, the middle cell moved **9.9% less with nylon screws** and **13.9% less with aluminium** under the same small sideways force. Refining the nylon case from 2 to 1.5 mm gave a 9.4% reduction. The change from bare at either outer cell was below 0.36%. We solved each full stiffness matrix independently, applying a unit force along PCB X through each cell's mass centre in turn. Contacts retained their installed state.

| X travel per unit force on the named cell [$\mu$m/N] | Bare | Nylon | Aluminium |
|---|---|---|---|
| Left cell relative to board supports | 0.5618 | 0.5607 | 0.5600 |
| Middle cell relative to board supports | 0.5235 | 0.4717 | 0.4509 |
| Right cell relative to board supports | 0.7969 | 0.7954 | 0.7941 |
| Left cell relative to its own strip clamps | 0.2726 | 0.2726 | 0.2727 |

Cutting off the modes at 4 kHz lowered the last row by less than 0.8%. Relative cell rotation contributed 0.213 $\mu$m/N on each board and 17.81 of 17.95 $\mu$m/N in the local bare and nylon fixtures over 5–10 $\mu$m of travel. Page 25 traces the initial force into the strips and PLA. The bare nonlinear check gives 2.327 $\mu$m relative to the clamps at 8.542 N (page 34).

## How much closer do neighbouring cells move?

During the nine-second base-X history, the snug case reduced the left cell's motion relative to its own strip clamps (page 18). Its peak approach toward the middle cell nevertheless increased from **19.1 $\mu$m bare** to **85.5 $\mu$m with nylon screws** and **70.8 $\mu$m with aluminium screws**. The middle cell moves in the whole-board model. In the local side-pull fixture on page 21 it is held fixed.

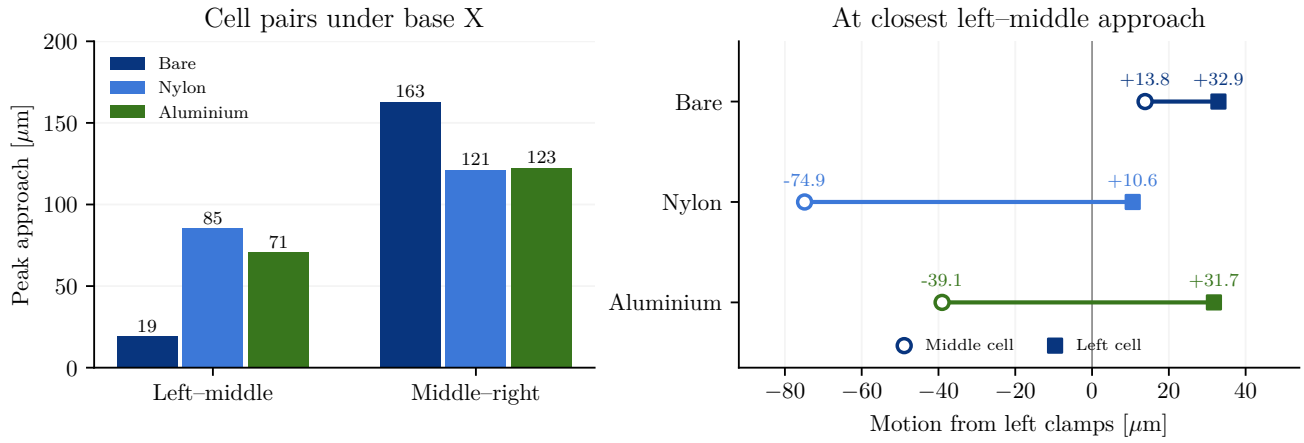



Figure 20: Left, greatest decrease in each pair's centre spacing along its installed line of centres. Each pair can peak at a different time. Right, both cell motions at the instant of peak left–middle approach in each assembly, measured relative to the same two left strip clamps. Positive points along the row towards the right cell.

## Which cell supplies the relative motion?

At the bare left–middle peak, both cells moved in the same direction relative to the left clamps, by +32.9 and +13.8 $\mu$m. With nylon mounts the left cell moved +10.6 $\mu$m while the middle cell moved −74.9 $\mu$m. With aluminium mounts the corresponding motions were +31.7 and −39.1 $\mu$m. The other pair approached by 162.9 $\mu$m bare, 121.1 $\mu$m with nylon and 122.5 $\mu$m with aluminium.

| Base drive | Bare | Nylon | Aluminium | Fine nylon |
|---|---|---|---|---|
| | Peak left–middle approach [$\mu$m] | | | |
| X | 19.10 | 85.46 | 70.80 | 84.98 |
| Y | 0.53 | 15.63 | 29.57 | 19.80 |
| Z | 3.41 | 33.75 | 32.45 | 26.52 |

The first three columns use native MAPDL coordinates. The last uses independent integration on the 1.5 mm nylon modes. Relative to independent integration of the nominal 2 mm nylon modes, mesh refinement changed the left–middle peak by −0.1% under X, +26% under Y and −21% under Z.

## Measurement and scope

At each of the 589,825 samples we subtracted simultaneous mass-centre translations. We checked the extrema against the nodal displacement fields of the three cells. Using the full three-dimensional centre distance changed the result by less than 0.001 $\mu$m at every sample. Across both pairs and all drives, independent integration reproduced the nominal approach and opening peaks within 0.63%. Reducing the modal cutoff from 4 to 3.5 kHz changed those peaks by less than 0.69%.

Cell-centre spacing and cell-to-pocket clearance measure different motions. For example, the nominal nylon left cell first reached its pocket at 38.30 ms under X, when the left and middle centres had approached by only 3.85 $\mu$m. The nine-second comparisons continue along the modes at bolt lock. Predicting subsequent contact forces and the assembled board's resistance to a sideways pull requires the corresponding nonlinear loading and restraints.

## The first PLA tensile crossing occurs at two mounting-tab roots

On the 1.5 mm case mesh, peak tension across the printed layers during the nylon forward pull rose from **38.79 to 40.79 MPa** between **0.2950 and 0.3039 mm** of left-cell travel. The 40.5 MPa reference comes from a printed PLA tensile coupon. This was the first of the three printed-material references to be reached. It occurred at the two tab roots on the pulled-cell side of the case.

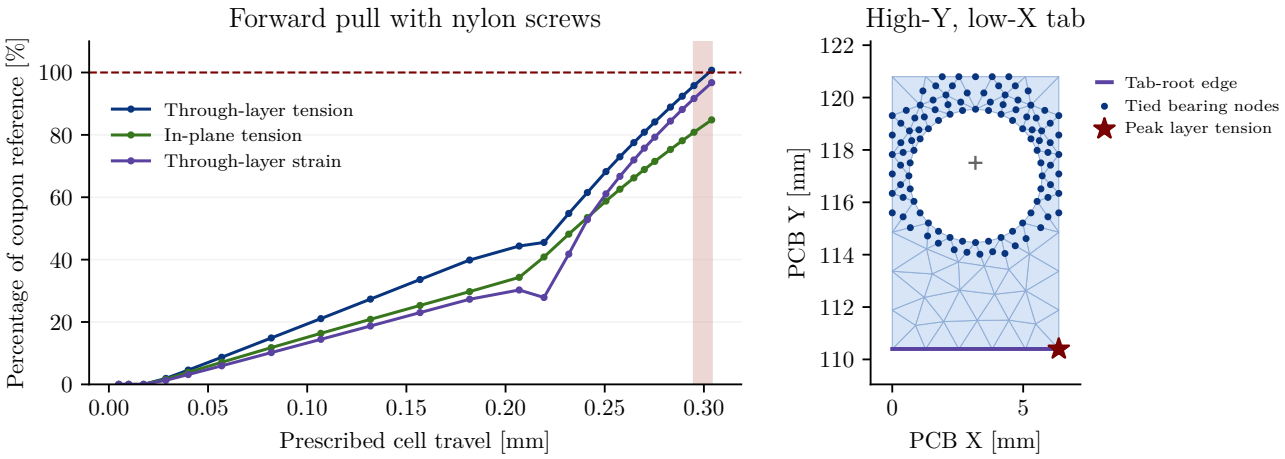



Figure 21: Left, the most highly loaded PLA corner for each material quantity at every saved forward-pull station through the first crossing. Shading brackets the crossing. Right, the native bearing-face mesh on the high-Y tab, viewed from the PCB. The cross marks the screw axis. The star lies at the junction of three planar STEP faces. The purple line marks the 6.36 mm tab-root edge.

## How local is the peak?

Only two of the 99,818 PLA tetrahedra had a corner above 40.5 MPa. They occurred at corresponding ends of the mounting tabs, 4.19 mm from the nearest nodes tied to either screw head. The low-Y tab reached 40.68 MPa. At the high-Y tab, nine elements met at the same corner and their unaveraged layer stresses ranged from 6.76 to 40.79 MPa. No tetrahedron had all four corners above the coupon value. Averaging the signed through-layer stress along each 6.36 mm root edge gave 16.51 MPa at low Y and 17.02 MPa at high Y. Even choosing the highest incident-element line average on each segment gave only 29.39 and 30.38 MPa. These results locate the crossing at the corners. Page 24 tests how its travel and peak value change with local mesh resolution.

## What happened to the screws and strips?

At the upper station the highest tensile stress in a nylon screw shank was 59.01 MPa. The installed screw and its threads have no verified strength limit in this study. The PLA in-plane tension reached 44.35 MPa (84.8% of its separate coupon mean), while the highest through-layer strain reached 96.7% of its reference at the left pocket. Maximum loaded-corner penetrations were 0.0191 mm at the moving cell and 0.0084 mm at the fixed hardware, both below the 0.05 mm numerical screen.

At the last station before the PLA crossing, 0.2950 mm, the fixture reaction was 2,138 N versus 7.78 N interpolated bare. Peak nickel equivalent plastic strain reached 17.14% on the flat strip next to the idealized cell attachment. At least one exterior shell face reached 10% across 4.68 $mm^2$ of the two strips' 924.64 $mm^2$ reference area. Both bare states bracketing the same travel stayed below 0.65% everywhere. These are bilinear strip strains with intact welds. The fixed-neighbour fixture and the localized PLA crossing do not establish an assembled-pack failure load.

## Sharp PLA corners prevent a converged strength prediction

Refining the forward-pull model moves the aluminium case's first tested layer-strength reference crossing to **0.107–0.132 mm** of cell travel. Nylon's finest tested bracket is **0.219–0.232 mm**. Neither crossing is mesh-converged. These are comparisons with the **40.5 MPa printed-coupon mean**, not predicted cracking loads.

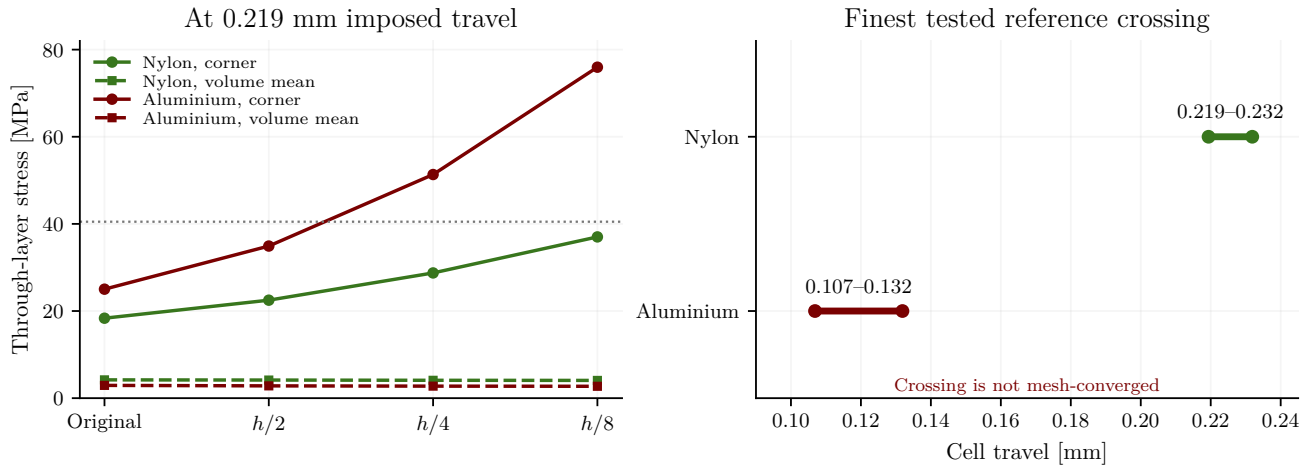



Figure 22: Independent elastic patches driven by native ANSYS displacements. Left, corner peaks and averages over fixed material volumes at the same travel. Aluminium is already beyond its refined reference there. Right, brackets at the finest tested local resolution. The two mounts have different governing corners.

## What changed with aluminium screws?

Its governing corner lies on the low-Y tab root, about 11 mm from the nearest tied screw node. At 0.232 mm travel, successive element halvings raise the layer tension from about **26 to 37, 54 and 80 MPa**. The mean over the same **4.45 mm**$^3$ of PLA changes only from 3.08 to 2.87 MPa. A larger cut region gives 81 MPa at the finest resolution and the same crossing bracket. The high-Y corner that governed nylon stays below 40 MPa throughout this aluminium checkpoint, even at the finest tested resolution.

## How the check was made

Each patch retains the original orthotropic stiffness and quadratic cut-face displacements. Rigid motion is removed, while physical free surfaces stay free of active contact and screw ties. There are 16 combinations of local resolution, cut size, subdivision diagonal and quadrature at each aluminium corner. All pass equilibrium and fixed-volume checks. Unrefined patches reproduce native corner stresses within 0.2 MPa; increased quadrature reproduces the refined stress. Changing cut size or subdivision diagonal preserves the crossing classification. The earlier nylon controls show the same unresolved sharp-corner sensitivity.

## How this changes the comparison

At the last common saved point below every tested corner reference, about **0.107 mm**, the native forces are **47 N nylon and 116 N aluminium**, versus approximately **5.4 N bare**. The earlier 0.207 and 0.219 mm force, pressure, work and nickel comparisons remain useful numerical diagnostics, but no longer sit inside the revised paired material screen. Their equilibrium audits and numerical values are unchanged. The low-load 8.54 N comparison precedes this reference crossing.

The actual printed corner radius and local print quality are unmeasured. A small, mesh-sensitive peak cannot establish a physical failure load, and a low volume average cannot dismiss that peak. This independent local check does not redistribute loads through the assembly or simulate cracking. The main study reports this limitation without adding another refinement campaign.

## Where does a sideways force on the middle cell go?

On the bare board the two nickel strips shared the middle cell's lateral load almost equally. After tightening the snug case, the PLA pocket carried **10.4% of an additional small load with nylon screws** and **14.5% with aluminium screws**. Both nickel strips carried less than on the bare board. The high-Y strip showed the larger reduction with either screw material.

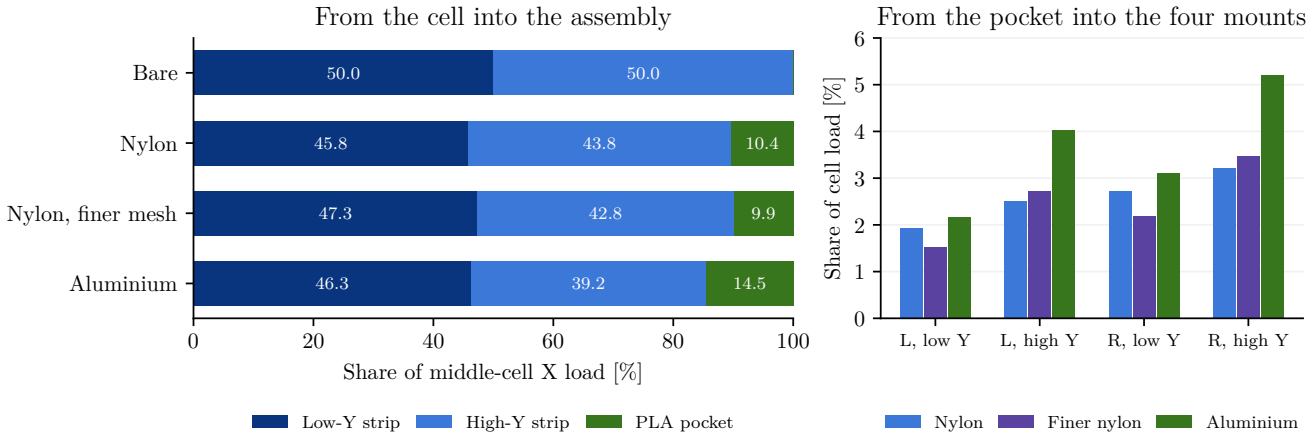



Figure 23: Independent whole-board calculation at bolt lock. The left panel divides the additional PCB-X force among the two ends of the middle cell and its case pocket. The right panel resolves the same pocket force at the four screw-bearing rings. Both panels use the total applied cell force as the denominator. L and R denote the low- and high-X mounting tabs.

## What changed with the screws and the mesh?

For each newton applied along PCB X, the low- and high-Y strips carried 0.458 and 0.438 N with nylon screws. With aluminium they carried 0.463 and 0.392 N. The largest share at a single mount occurred at the high-X, high-Y tab in all three cased calculations. It carried 3.2% of the applied load with nylon and 5.2% with aluminium on the 2 mm meshes. Refining nylon to 1.5 mm changed the total pocket share from 10.37% to 9.90% and this tab's share from 3.21% to 3.47%.

## Why the load location matters

We applied a separate load through the mass centre of each cell. With either outer cell loaded, both outer pockets remained open in the installed configuration, and the loaded cell transferred its full additional X force to its two strips. The board flexed enough to change the reaction at the already seated middle pocket. Per newton on the left cell, its signed X increment was −0.0066 N with nylon and −0.0068 N with aluminium. Under load on the middle cell, the corresponding increments were +0.1037 and +0.1452 N.

We recovered the forces from the native element matrices. Each cell balanced within $10^{-9}$ N/N in all three axes. The four tab reactions balanced the pocket within $10^{-11}$ N/N. Across 38 bare, 48 nominal and 49 fine-nylon modes, the largest relative difference from native bond and pocket force fields was below $4.1 \times 10^{-7}$.

## When the middle pocket is open

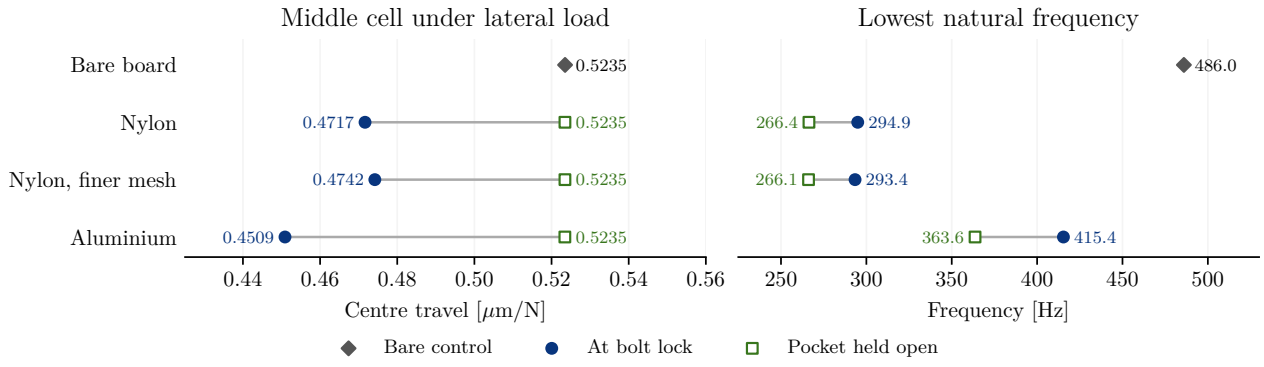



Figure 24: All case mounts remain attached. Removing the middle contact from the installed stiffness matrix brings the middle cell's lateral travel within 0.011% of bare. The lowest frequency also falls. Each frequency is the lowest in its own model. Mode shapes can change order.

## How much force keeps the middle cell seated?

The nominal ANSYS models gave **0.594 N of middle-pocket compression with nylon screws** and **0.0223 N with aluminium** after bolt lock. The aluminium repeat directly recovered 0.022296 N from the contact elements. It reproduced all eight original displacement fields, bolt loads and contact histories. Integrating the PLA pocket stresses independently gave the force vector within $3.5 \times 10^{-7}$ N. Its single loaded triangle carried much less initial compression, even though the pocket took a larger fraction of a small additional load into it (page 25).

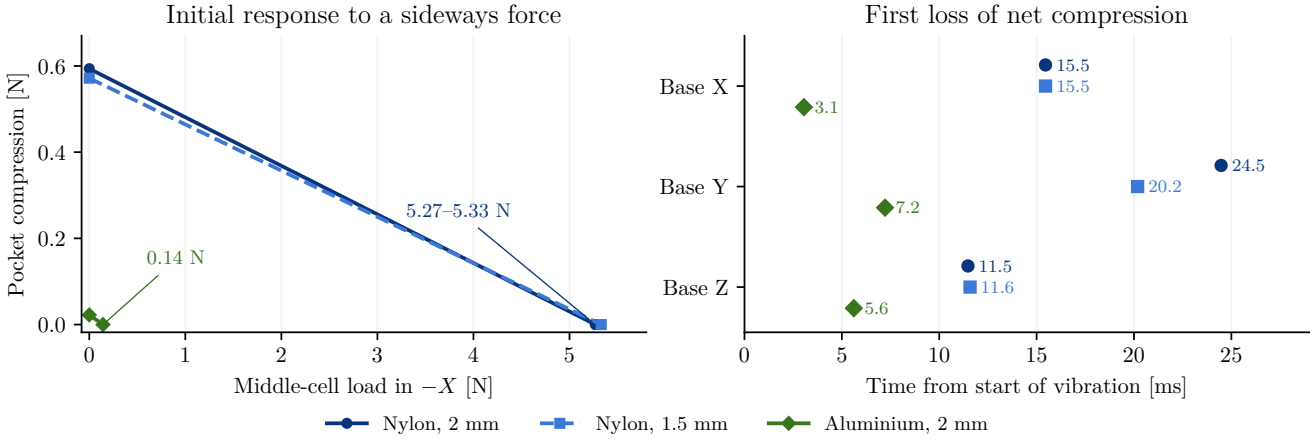



Figure 25: Left, the initial whole-board force slope projected along the installed compressive reaction. Each line ends when that projection reaches zero. Right, the first sample on the fixed-contact trajectory at which the old patch would have to pull on the cell. The nominal cases use native modal coordinates. The finer nylon model uses independent integration. Mesh sizes refer to the printed shell.

## An applied force can relieve contact

A force along negative PCB X on the middle cell relieves the seated pocket. Extrapolating the initial slope gives zero net compression at 5.27 N with nylon and 5.33 N on its finer mesh. The aluminium estimate is 0.14 N. Loading the *left* cell in positive X also relieves the middle pocket, through motion of the board. That intercept is 82.2 N with nylon and 3.07 N with aluminium. The whole-board pulls pass through 1.00 and 8.54 N on the left cell before unloading. New contact or a redistribution over the original patch can occur before any of these intercepts.

## When vibration uses up that compression

With aluminium screws the net projection first changed sign at 3.05 ms for base X, 7.22 ms for base Y and 5.62 ms for base Z. With nylon it changed sign at 15.46, 24.48 and 11.47 ms, respectively. Independent Newmark and piecewise-linear-input integrations reproduced all six nominal native crossings within the original sample. Reducing the mode cutoff from 4 to 3 kHz preserved all nine plotted crossings. Refining nylon moved the base-Y crossing from 24.48 to 20.19 ms. At the plotted instants all original patch normals still lay within the compressive half-space. The times mark a limit of the frozen-contact calculation. Complete separation and subsequent impact require a changing-contact solution.

## Checks on magnitude and direction

We integrated stress over the deformed quadratic solids on both sides of the interface. For aluminium, the PLA pocket, the four tab reactions and the middle cell outside its two nickel bonds gave 0.022296, 0.022290 and 0.022290 N. On the fine nylon mesh those three values and the native contact output agreed within 0.00002 N in magnitude. On nominal nylon the solid values were 0.568–0.571 N against 0.594 N from the contact elements. The tab, cell and native contact forces were put on the same side of the interface before comparison. Refining aluminium from 2 to 1.5 mm reduced the native lock force from 0.02230 to 0.01785 N (20.0%). The four tabs, PLA pocket and middle cell each gave 0.01784–0.01785 N by stress integration on the finer mesh. Both meshes loaded one barrel point. Its quadrature area fell from 0.550 to 0.314 mm$^2$, while the local lock pressure rose from 0.0406 to 0.0568 MPa. The 0.003 mm nominal middle-cell clearance makes this comparison sensitive to the as-printed fit.

## Does zero contact force mean the middle cell has cleared the case?

In the independent calculation the initial aluminium contact unloaded at 3.07 N. Another case triangle still crossed the curved middle cell at 5.807 N. The surfaces separated by 5.850 N. In the native pull (page 34), the original face still carried pressure at 2.886 N. The force was zero at 4.771 N even though the curved surfaces still intersected. During unloading the original face resumed carrying pressure.

## Does the middle cell lift off during vibration?

With aluminium screws and excitation normal to the board, the independently calculated force at the initially loaded contact fell to zero at **5.608 ms**. That contact stayed open for **0.543 ms** and its spring opened by at most **44 nm**. At peak opening, whether the original PLA face clears the cell depends on how its curvature is represented. A second face still overlaps the cell in both geometry checks.

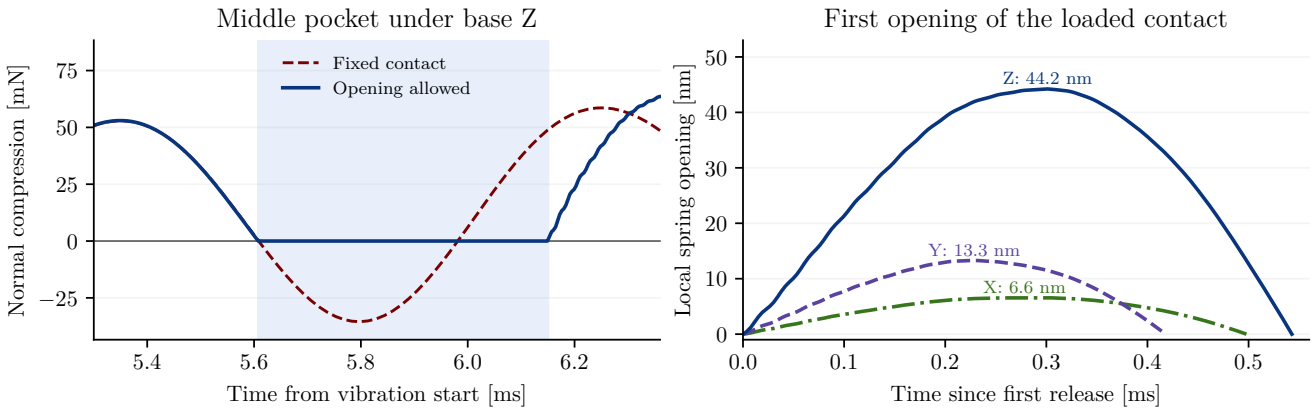



Figure 26: Base-Z compression from the native fixed-contact history and the independent spring model. Both use 0.0223 N at lock. Shading marks release to reclosure. At right, each trace starts at its own release time.

Under base X the first release lasted from 3.048 to 3.546 ms, and under base Y from 7.214 to 7.632 ms. We compared the local gap on two PLA faces about 10 mm apart along the cell. At bolt lock ANSYS reported 0.0406 MPa peak pressure on the first face and zero on the second.

## Keeping the displacements and restoring the STEP curvature

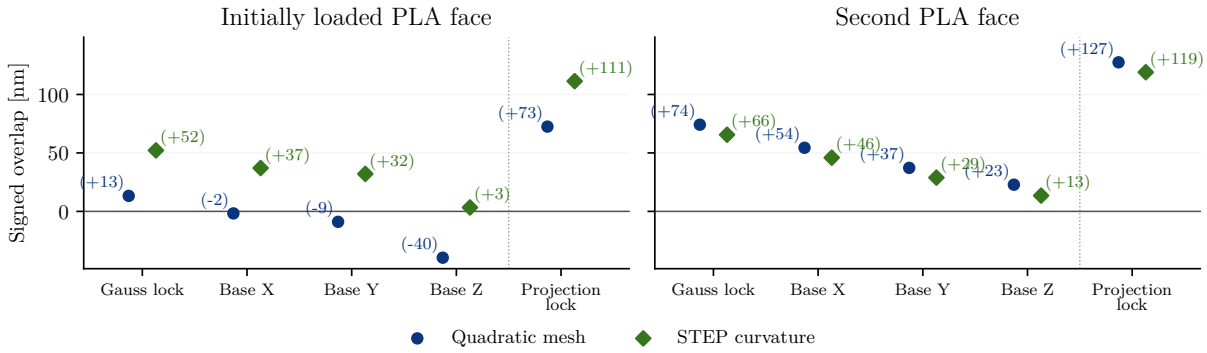



Figure 27: Positive values indicate radial overlap. The first four poses retain the nominal Gauss-detection preload and its independently calculated peak openings. The final pose uses the separate native projection preload at lock. STEP curvature restores the 10.795 mm pocket and 10.700 mm cell radii at each pose's own displacements. Values are rounded to nanometres.

At the base-Z peak the original mesh face cleared the middle cell by 39–40 nm. Restoring curvature gave 3–4 nm of overlap; on the second face it reduced overlap from about 23 to 13 nm. The sign at the original face also changed under base X and Y. Each maximum was bounded to 0.5 nm, with witnesses on the trimmed STEP surfaces. This reconstruction retains the existing equilibrium and supplies no new pressure.

The projection preload passed force balance but reported zero force despite up to 0.128 $\mu$m of overlap. The cylindrical correction initially failed balance by 22%. Its tighter repeat (0.0001 N criterion) passed all eight states: **0.044 N at lock**, with **0.03% imbalance**, versus **0.022 N nominal**. Local contact remains formulation-sensitive; these controls do not replace the nominal vibration bases.

## How accurately is the local opening calculated?

Adding 16 static displacement vectors to the 48 installed modes recovered the static contact flexibility. Increasing from 12 to 16 vectors changed peak openings by at most 1.3%; time-step refinement changed force-increment histories by less than 0.25% in relative norm. Other contacts and 2% damping stayed fixed. Both whole-shell controls stopped at native mesh-shape checks before solving, blocking their contact and transient dependents. Physical clearance and recontact force remain unresolved.

## How much work does it take to move the edge cell?

At the bare fixture's terminal force of **8.54 N**, lateral actuator work was about **0.0846 mJ with nylon screws**, **0.0449 mJ with aluminium** and **3.30 mJ bare**. The cased values interpolate native load intervals, including first contact for aluminium. At the same imposed travel of **0.206875 mm**, the actuator supplied **9.55, 23.14 and 0.962 mJ**, respectively. The local fixture fixes the four screw supports and neighbouring cells. Values at 0.207 mm are numerical continuation beyond the refined aluminium PLA reference.

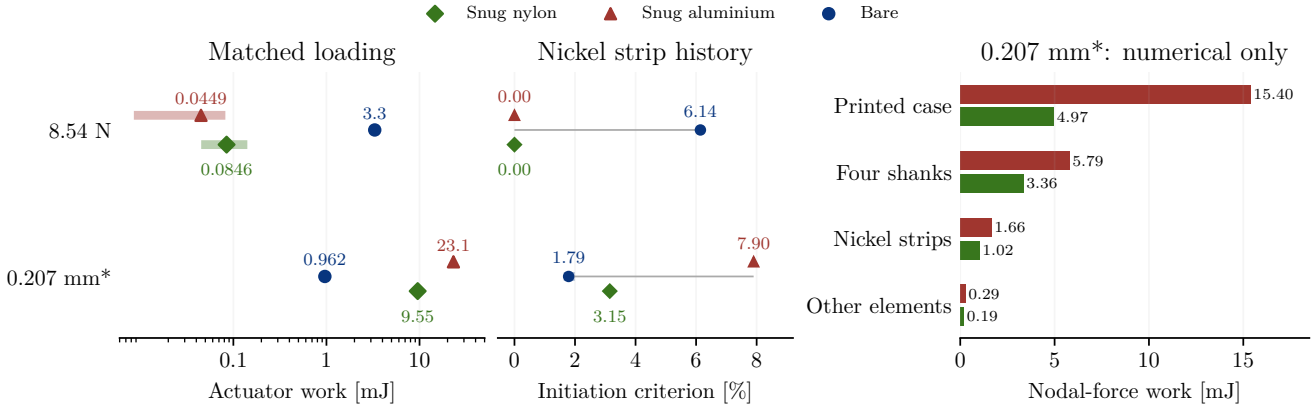



Figure 28: Left, work at matched force or travel. Native endpoint bands at 8.54 N span 0.045–0.141 mJ for nylon and 0.0085–0.0821 mJ for aluminium. Centre, independent nickel initiation replay. Right, component nodal-force work. Asterisks mark post-reference numerical continuation.

## Where the work goes

At 0.206875 mm the PLA and four shanks stored 4.974 and 3.365 mJ with nylon screws, together 87.3% of actuator work. Aluminium raised these elastic energies to 15.400 and 5.794 mJ, or 91.6%. Independent nodal-force integration reproduced the elastic storage. Nickel work was 1.022 and 1.662 mJ, respectively. Summed nodal work matched integrated actuator reactions within 0.0009 mJ across nylon's checkpoint and 0.000021 mJ across aluminium's.

## Smaller increments preserve both force results

Both selected controls completed **42 equilibria through 0.219 mm**; all passed the fixture checks. With the same meshes and bilinear nickel law, reaction changed by **less than 1%** at all 13 matched stations for each screw material. Nylon's endpoint force changed from **538 to 539 N** and work from **13.6 to 13.2 mJ**, about 3%. Aluminium's force stayed about **598 N**, while work fell from **28.4 to 27.9 mJ**, about 2%. At 8.5 N, travel remained about **0.034 mm** with nylon; aluminium's estimate shifted from **0.025 to 0.026 mm**. Early contact work is more step-sensitive and is not an impact-energy rating. Both controls stay below their own parent-mesh PLA references; earlier corner-refinement bounds are not transferred to them. These checks establish neither full convergence nor a failure load. The final target-resolution result is on page 36.

## How close did the nickel get to damage initiation?

Following the ANSYS ductile-damage law, each plastic-strain increment is divided by its critical strain at the current stress triaxiality. A sum of one marks initiation. Independent replay agreed with the native criterion within $2.4 \times 10^{-6}$ on the bare damage-law pull and three controls. On aluminium's bilinear history, computing triaxiality from the full stress tensor changed the peak criterion by less than $6.4 \times 10^{-9}$.

At 8.54 N the bare fixture reached 6.14% of the nominal criterion; both snug fixtures remained elastic at both bracketing loads. At 0.2069 mm travel, demand was **3.15% with nylon and 7.90% with aluminium**. At 0.2194 mm it rose to **6.70% and 14.44%**, against 1.63–1.99% at the surrounding bare stations. Aluminium's maximum was in the top section of a flat strip element. The weakest assumed curve gave 20.1% and 43.3% at the same endpoint. These are fractions of an initiation criterion, not allowable-stress utilization or observed damage. They retain the bilinear deformation history; the damage law has no feedback. The two nominal native damage comparisons are reported on page 2; the parameter grid is deferred.

## Limit of the comparison

The refined aluminium corner now bounds the paired comparison at the 0.107–0.132 mm interval (page 24). The 0.207 and 0.219 mm quantities above describe numerical continuation, not an intact-case strength or energy rating. Neither mount has a converged sharp-corner failure threshold.

## When does vibration change the installed contact state?

After the six original battery bolts locked, **one of the four spacers carried compression with nylon case screws, and two with aluminium**. In these models the four M4 case screws began at their stress-free length. The 750 N installation preload applies to each original battery bolt. The installed M4 preload on the hardware is unknown.

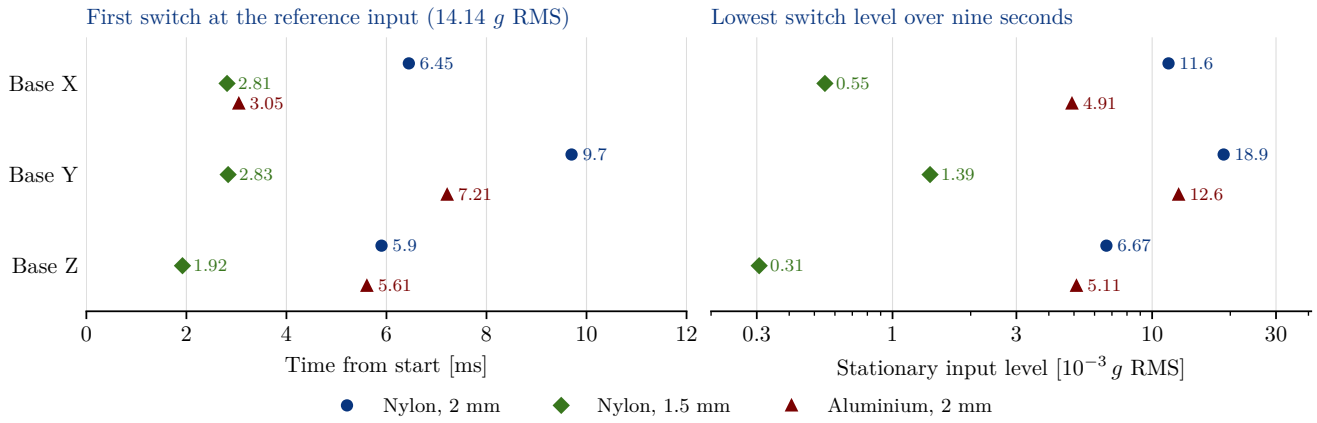



Figure 29: Each axis uses its own nine-second waveform. The left panel gives the first contact change at full input. The right panel gives the lowest input level for a change at any middle-pocket integration point or tab spacer. The markers show the attained values. Nominal cases use native modal histories. Finer nylon uses independent integration of its native modes.

## Installed forces at the four tabs

The nylon mount compressed the high-X, low-Y spacer by 4.07 nm and loaded it to 0.0586 N. On the finer nylon mesh the same spacer carried 0.0633 N. With aluminium screws, the low-X, high-Y and high-X, low-Y spacers carried 0.0154 and 0.0151 N. The combined shank and spacer forces agreed with independent integration at each PLA bearing within 0.00012 N.

## How much vibration changes a contact?

Under base Z, a previously clear point on the nominal nylon pocket touched the middle cell at **0.00659–0.00667** $g$ **RMS**, before the spacer would seat at 0.0131 $g$ RMS. Its gap at bolt lock was 9.67 nm. At full input the same point touched at 5.889–5.902 ms. We checked every initially clear pocket point against all 308 quadratic faces of the middle cell. Earlier spacer or pocket-unloading levels held in the other eight model and axis combinations. On finer nylon the weakest point unloaded at **0.000306** $g$ **RMS**, leaving **0.571 N** of pocket compression. The measured print and fastener preload would be needed to resolve which transition governs the hardware.

At each model's first-change level we searched every PLA corner over all nine seconds, retaining the original tightening stress. Peak through-layer tension ranged from **0.86–0.98%** of the printed-coupon mean with nylon and **1.91–1.99%** with aluminium. At the same 0.000306 $g$ base-Z level, peak dynamic bond-force increments were 0.655 mN bare, 0.869 mN with nominal nylon, 1.057 mN on finer nylon and 1.095 mN with aluminium. These small loads illustrate the sensitivity of lightly seated contacts. Page 9 gives the printed stresses at the full reference input.

## Effect on the lowest natural frequency

Keeping the middle pocket and the rest of the assembly at their locked stiffness, we changed the four axial spacer stiffnesses and re-solved the whole-board eigenproblem. With nylon screws the lowest frequencies were **294.25, 294.94 and 295.93 Hz** for all spacers open, the installed state and all four bearing. With aluminium they were **415.40, 415.42 and 415.44 Hz**. Both spans are small compared with the difference from the bare board at 485.96 Hz. The installed calculation reproduced the first four native frequencies for each screw material.

## Does tightening the M4 screws squeeze the middle cell?

We shortened the four shanks equally to an assumed **1.00 N mean tension per screw**. The required shortening was 0.354 $\mu$m with nylon screws on either mesh and 0.082 $\mu$m with aluminium. All four spacers bore 0.86–1.05 N each. The middle pocket gained only **8–9 mN** of compression in all three models. One initially loaded point released on finer nylon. None of the initially clear middle-pocket points touched on any mesh. The change in middle-cell displacement per unit sideways force stayed below 0.033%. Actual screw tension remains unmeasured.

## How much does the case restrain a cell as the pocket unloads?

With an 8.54 N load on the middle cell toward negative PCB X, the bare cell centre moved **4.47 $\mu$m**. An independent whole-board calculation that allowed the loaded pocket points and tab spacers to change contact gave **4.20 $\mu$m with nylon screws** and **4.46 $\mu$m with aluminium**. The reductions were 6.1% and 0.24% for this direction and load. These are incremental motions from bolt lock with the original board supports. Page 19 treats the separate local fixture with fixed neighbours.

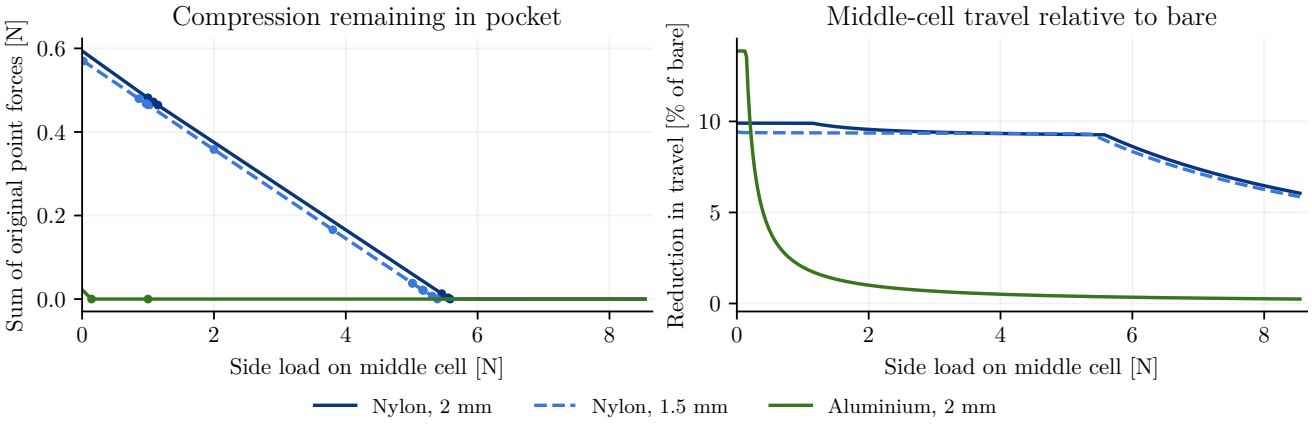



Figure 30: Both panels use a sideways force applied at the middle cell's centre of mass. Left, the compressive forces at the original pocket points are summed by magnitude. Dots indicate saved loads, including changes in bearing. Right, travel reduction compares each cased solution with the bare board at the same force. Initially open case faces cannot acquire load in this calculation.

## How the original pocket points release

On the nominal nylon mesh, five points carried compression at lock. Under middle-cell load the second spacer engaged at 1.00 N. The first pocket point unloaded at 1.08 N and the last at 5.58 N. On the finer nylon mesh ten points carried compression over seven triangles. The weakest unloaded at only 0.018 N and the last at 5.39 N. The finer mesh predicted 4.21 $\mu$m of middle-cell travel at 8.54 N. With aluminium the only loaded point unloaded at 0.143 N.

Loading the *left* cell toward positive PCB X gave a different result. At 8.54 N its centre moved 4.799 $\mu$m bare, 4.789 $\mu$m with nylon screws and 4.785 $\mu$m with aluminium. This independent calculation retained 0.532 N on the nominal nylon middle pocket. The native control on page 34 retained 0.506 N after the tighter zero-load hold. The aluminium middle pocket lost its original compressive force at 3.07 N, although the applied force was on the adjacent cell.

## Checks and limits

The contact elements retained the same installed nodal forces across every prestressed mode. For both nylon meshes we checked the forces on each triangle against the separate static ANSYS output. For aluminium the inherited resultant was 0.0222964 N. The difference from the independent solid-stress integration was 0.00000034 N in vector norm. This use of stored base contact data follows the ANSYS Element Reference, section 5.9.2.

At the 8.54 N middle-cell endpoint, the deformed surfaces of all three cells cleared the case for both screw materials and both nylon mesh sizes. Loading the left cell instead left a middle-cell surface intersection on both nylon meshes. The aluminium mesh cleared at the endpoint.

Under the left-cell load we followed every initially unloaded integration point on the middle pocket through each interval of constant bearing. We bounded the distance to the complete quadratic cell surface, including the ends, throughout each interval. The closest approach was 7.4–7.6 nm with nylon screws on the 2 mm case mesh, 81.3–81.8 nm with the same screws on the 1.5 mm mesh and 18.5–18.7 nm with aluminium screws on the 2 mm mesh. No additional point touched. The margins depend strongly on the mesh. In the native pulls, all five nylon faces retained pressure. The single aluminium face released and re-engaged. No new face carried pressure with either mount (page 34).

## Does the middle battery stay seated during vibration?

Under the original board-normal vibration record, the middle pocket repeatedly unloaded and carried compression again in all three cased models. On the nominal 2 mm mesh the first interval with zero force at every originally loaded pocket point was **5.61–6.15 ms with aluminium screws** and **11.54–11.78 ms with nylon**. The first such interval on the finer nylon mesh was shorter, 11.64–11.67 ms. Both the pocket and the four tab spacers were free to open or close in this independent whole-board calculation.

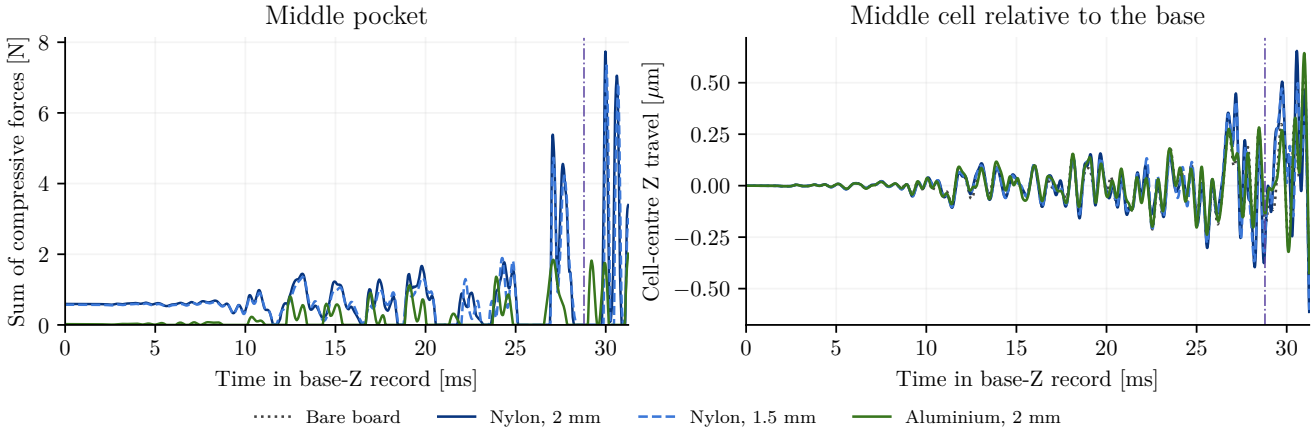



Figure 31: First 31.25 ms of the same base-Z excitation. The left panel sums the modeled compressive forces at the middle pocket. The right panel gives signed, mass-weighted middle-cell travel relative to the moving base. Bare uses the native MAPDL history. Each cased trace uses the independent switching-contact model. The vertical line identifies the common instant used to check all three case surfaces.

## One comparison time for each excitation

We selected the stored input sample at which the smallest opening among the originally loaded pocket points, across all three cased models, was greatest. We then checked the *entire* deformed quadratic surface of the middle cell against the case in each model at that same time. The values below bound the shortest distance between the two meshes. Each bound includes subdivisions of both curved surfaces. The marked times describe the selected snapshots; the smallest surface distance may reach its peak at a different time.

| Base | Time [ms] | Nylon, 2 mm [$\mu$m] | Nylon, 1.5 mm [$\mu$m] | Aluminium, 2 mm [$\mu$m] |
|---|---|---|---|---|
| X | 31.08 | 5.00–5.11 | 4.74–4.85 | 6.09–6.20 |
| Y | 24.70 | Overlap | Overlap | Overlap |
| Z | 28.79 | 3.90–4.01 | 2.44–2.55 | 1.73–1.84 |

Over the full 31.25 ms board-normal interval, the largest middle-cell Z travel was 0.56 $\mu$m bare, 0.65 $\mu$m with nylon screws (0.60 $\mu$m on finer nylon), and 0.64 $\mu$m with aluminium. These peaks occur at different times.

## What the startup calculation resolves

The calculation retains the native prestressed modes and adds five groups of static displacement fields driven by unit pocket forces, spacer forces and three base inertia loads. The original pocket contains five loaded integration points on nominal nylon, ten on finer nylon and one on aluminium. All three models retain 2% damping in their installed modes. Removing a group of displacement fields changed the pocket-force histories by at most 2.4% in relative RMS norm. An independent matrix-exponential solution located changes of contact within each input interval and checked the implicit time integration.

At the matched 24.70 ms base-Y sample, we identified an intersecting case triangle in each model. All three triangles had been unloaded at bolt lock. Page 32 evaluates the additional forces at the prescribed pose. The surface checks give the geometry at the calculated positions. Coupled reactions on newly engaged faces, the effects of actual print dimensions and any damage after recontact remain unresolved.

## Where does the moving battery reach a new contact point?

At 24.70 ms under base-Y excitation, the nylon models placed the newly reached contact points near the **high-Y end** of the middle battery. With aluminium screws the twelve new points lay **0.6–14.0 mm from the low-Y end**. The locations below are contact integration points on the cylindrical pocket wall. A triangle can intersect the cell between those points.

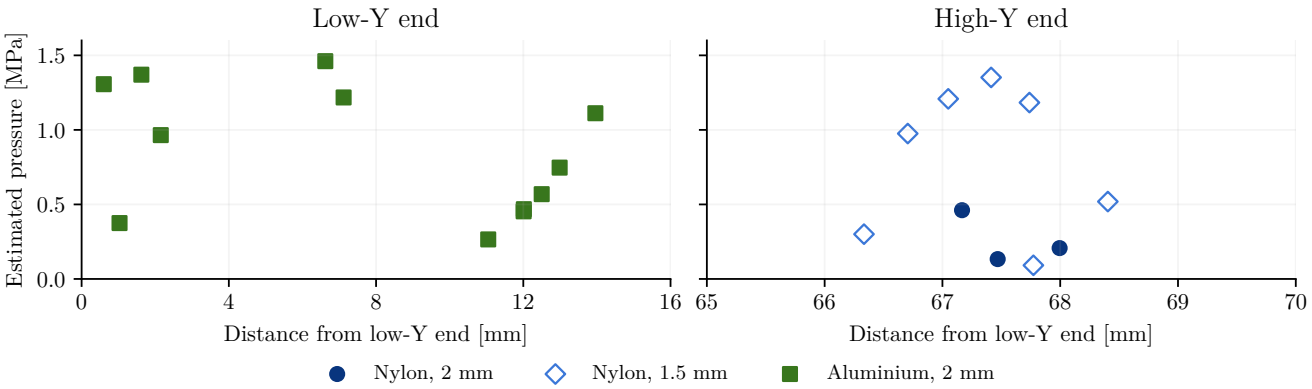



Figure 32: Each symbol is one newly reached integration point on the deformed FE surfaces at the common base-Y instant. Axial distance is measured from the low-Y end of the undeformed cylindrical cell wall. Pressure uses the ANSYS normal stiffness with the assembly held at the calculated pose.

## How much force would those points carry at that pose?

| | Nylon, 2 mm | Nylon, 1.5 mm | Aluminium, 2 mm |
|---|---|---|---|
| New FE points inside the cell | 3 | 7 | 12 |
| Largest radial penetration [$\mu$m] | 0.095 | 0.222 | 0.300 |
| Resultant on the case [N] | 0.354 | 1.495 | 5.775 |
| Points inside with STEP curvature | 39 | 68 | 80 |

The original seated points all had zero force at this instant in the independent vibration calculation. We held its displacements fixed and evaluated the force required by the penalty law on the new points. The aluminium run supplied its own normal stiffness at all twelve newly engaged points. Their stiffnesses matched those used in the earlier nylon-based estimate, reproducing 5.775 N at this prescribed pose. If the new forces act on the assembly, its displacements can change. The fourfold change in the nylon resultant with mesh refinement limits the accuracy of a force estimate.

## Geometry and force checks

For each of the three Gauss points on every middle-pocket triangle, we intersected a radial ray with the deformed quadratic cylindrical cell surface. Repeating the calculation with the original STEP curvatures and the same displacements increased the counts in all three models. The axial rim and flat cell ends are outside this radial calculation. On aluminium the triangle intersection identified on page 31 was at the low-Y rim. All three points of that case triangle lay beyond the cylindrical barrel; the twelve barrel contacts came from other triangles.

At bolt lock the FE calculation identified all five loaded points on nominal nylon and all ten on finer nylon. Multiplying penetration by the native normal contact stiffness and the area represented by each point reproduced the locked pocket resultants of 0.594 and 0.572 N to within 0.000002 N. The geometry matched all loaded faces in the tighter nylon run and the nominal aluminium run at each of eight stations. The aluminium face-force errors stayed below 0.000002 N, including the three stations that failed the separate mounting-force check. Agreement with the local pressure law can coexist with an unbalanced assembly. The ANSYS normal stiffness and force records are defined in the CONTA174 element output table.

Under base Z, the first previously unloaded cylindrical point entered the mesh at 5.91 ms with nominal nylon, 10.07 ms on finer nylon and 6.33 ms with aluminium. The later gaps on page 31 use a trajectory that omits the forces at those new points. They quantify the geometry of that trajectory; the actual battery motion through recontact remains open.

## How does M4 tension change the first pocket contact event?

A mean M4 shank tension of 1 N seated all four spacers. Under board-normal vibration, the first change still occurred at a middle-pocket point. We used the same nine-second input as on page 29 and changed only the assumed assembly state and the stiffness of the points bearing in that state. The screw tension in the hardware is unknown.

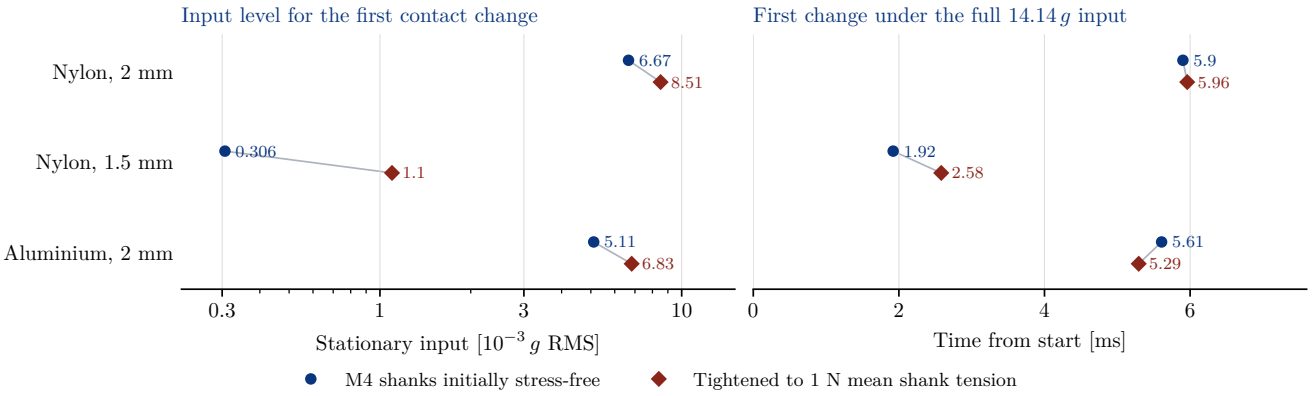



Figure 33: Left, lowest observed input level for a middle-pocket or spacer change. Right, first change in the ramp at the full reference level. Both panels describe the same board-normal waveform. The points show the attained upper ends of the numerical intervals.

## Where the contact changes

On the nominal nylon mesh, a previously clear point touched the middle cell at **0.00842–0.00852** $g$ **RMS** after tightening, compared with 0.00659–0.00667 $g$ before tightening. With aluminium screws a new point touched at **0.00676–0.00683** $g$ **RMS**. An originally loaded point had unloaded first at 0.00511 $g$ in the original assembly. The first frequency with all spacers bearing was 295.93 Hz with nylon and 415.44 Hz with aluminium on these meshes.

On finer nylon, one weakly loaded point opened during tightening. Its contact spring seated again at **0.00110** $g$ **RMS**. At that station the deformed quadratic mesh still gave 0.015 nm of clearance. At the same time sample, continuing the tightened linear response closed the geometric gap at 0.00112 $g$ RMS. This difference sets the precision of the predicted seating level for that point.

## Comparison with the bare board

Middle-cell sideways displacement per unit force was 0.5235 $\mu$m/N bare. At the assumed screw tension it was 0.4715 $\mu$m/N with nylon, 0.4743 $\mu$m/N with finer nylon and 0.4509 $\mu$m/N with aluminium. In the cased models tightening changed this static quantity by less than 0.033%, even as the lightly loaded contacts responded to different vibration levels.

## Checks on the tightened state

We linearized the equations about the independently equilibrated 1 N state using two-percent modal damping. Forcing and displacements were carried in the original native mode shapes plus additional stiffness responses. A separate full-matrix calculation checked the first four frequencies with all spacers bearing and the original pocket stiffness retained. Two and five blocks of additional responses, and a half sample interval, preserved the governing point on all three models. We swept each originally clear middle-pocket integration point against all 308 curved middle-cell faces. Other interfaces, locations between integration points and motion after a contact changes require separate checks.

## Does the snug case restrain the left cell on the complete board?

At 8.542 N the left cell moved **4.824 $\mu$m bare**, **4.816 $\mu$m with nylon screws** and **4.812 $\mu$m with aluminium screws**. Relative to the two strip clamps, travel was 2.3275, 2.3278 and 2.3279 $\mu$m, respectively. The nominal complete-board models give less than a 0.25% change in centre travel with either mount.

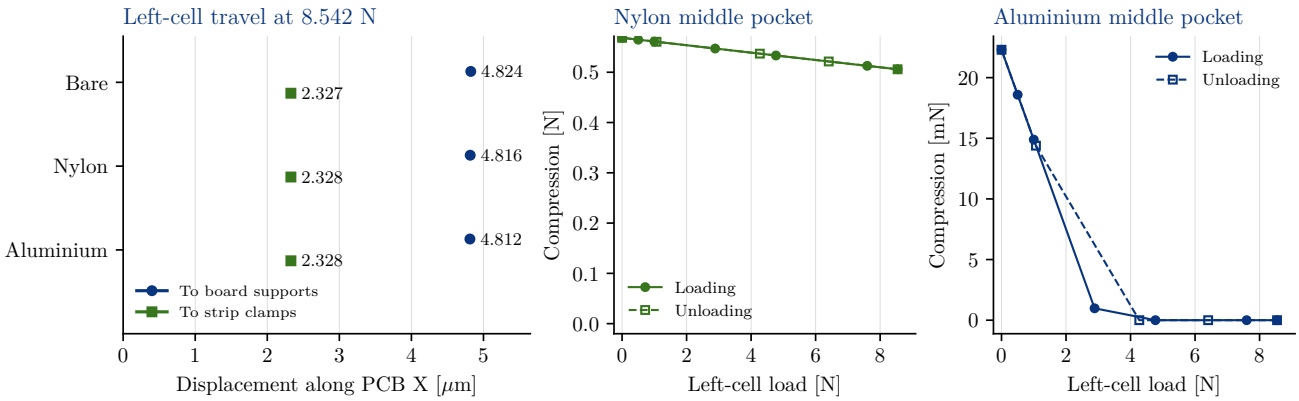



Figure 34: Cell-centre motion from the zero-added-load hold and native compression along each middle pocket's installed reaction direction. Markers show converged states. The two force panels use different vertical units.

## Why the local pull gives a different travel

The local fixture on page 19 prescribed rigid-cell travel with locally supported, bilinear nickel strips. There the left cell reached its pocket and travel at the same load fell from 0.500 mm bare to about 0.034 mm with nylon screws. On the complete boards, the stand-offs support the PCB and force acts on 1,353 nodes in proportion to their contribution to the cell mass. The elastic solid strips carried essentially the same lateral increments in all three models. The loaded left cell stayed clear of the case at every saved station.

| At 8.542 N on the left cell | Bare | Nylon | Aluminium |
|---|---|---|---|
| Lateral increment at the two left-cell nickel bonds [N] | 4.262 / 4.286 | 4.262 / 4.285 | 4.262 / 4.285 |
| Peak nickel equivalent stress including bolt preload [MPa] | 167.30 | 167.52 | 167.59 |
| Largest local increase in nickel equivalent stress [MPa] | 27.08 | 26.93 | 26.89 |
| Largest local increase in PCB principal tensile strain [$\mu\varepsilon$] | 367 | 366 | 364 |
| Work supplied by the distributed load [$\mu$J] | 20.597 | 20.565 | 20.549 |

Both cased nickel peaks reached 90.6% of the assumed 185 MPa yield reference, compared with 90.4% bare. Through-layer PLA tension reached 0.335 MPa with nylon screws and 0.789 MPa with aluminium (0.83% and 1.95% of the coupon mean). Peak shank fibre tension was 1.24 and 2.39 MPa. The PCB assembly, ideal welds and screws have no verified hardware allowables in this study.

## Which pockets still touch?

With nylon screws, the tighter hold relaxed the middle-pocket force to 0.5685 N. At maximum load it retained 0.5059 N on five faces. With aluminium the middle pocket carried 0.98 mN at 2.886 N and zero at 4.771 N. The curved meshes still intersected at 4.771 N and were separated at 7.599 N. At 8.542 N their minimum gap was 58.7–59.8 nm. As the load returned to zero the same face carried pressure again.

At maximum load, the smallest driven-cell gap was 9.878–9.889 $\mu$m with nylon and 11.018–11.029 $\mu$m with aluminium. All six modeled washers cleared the case by at least 0.311 mm with nylon and 0.314 mm with aluminium at the saved stations. These bounds use the nominal mesh and solved displacements. Print and sleeve tolerances remain unmeasured.

## Both snug mounts restrain reverse sideways travel

When the left cell is pulled away from its neighbours, **both screw variants bracket travel at 0.146–0.171 mm under 8.5 N**, compared with **0.500 mm bare**. Both linear estimates round to **0.16 mm**, or about two-thirds less travel. This interval spans first pocket contact. The saved steps do not resolve the small difference between the two interpolated estimates.

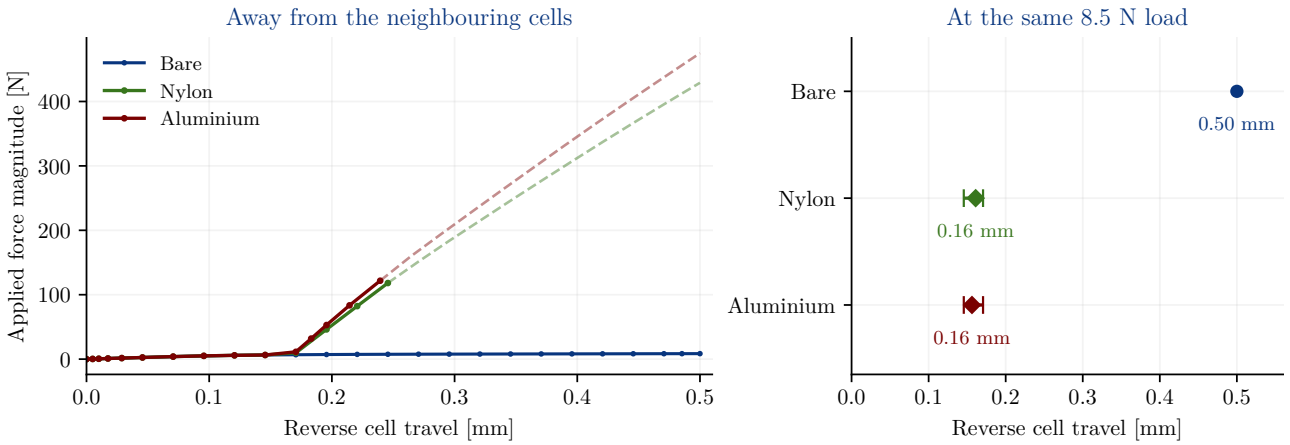



Figure 35: Native negative-X local pulls, shown as positive magnitudes. Each solid cased curve ends at its last saved station below the parent-mesh PLA and contact screens. Dashes retain later numerical continuation. Diamonds interpolate within the saved-state brackets shown by the horizontal bars.

## Compare the same travel in both mounts

At the common saved travel of about **0.196 mm**, the reaction is **46 N with nylon screws**, **53 N with aluminium** and **7.1 N bare**. Most of the cased reaction passes through the printed shell into the fixed middle cell. Aluminium carries more load through its case mounts. These are local fixtures with fixed neighbouring cells and screw ends, not the compliance of the complete board.

| At 0.196 mm reverse travel | Nylon screws | Aluminium screws |
|---|---|---|
| Reaction through case mounts [N] | 5.1 | 12.8 |
| Reaction through fixed middle cell [N] | 34 | 34 |
| Reaction through nickel clamps [N] | 6.5 | 6.1 |
| Peak PLA in-plane tension [MPa] | 14 | 16 |
| Peak PLA through-layer tension [MPa] | 3.6 | 4.0 |
| Peak screw-shank fibre tension [MPa] | 3.8 | 8.9 |

## What the validation establishes

Both native solves completed 0.500 mm: 33 saved equilibria for nylon and 34 for aluminium. All fixture force and moment, bearing, fixed-cell, pressure and work checks passed. The largest force-residual components were 0.025 and 0.046 N. The predicted first-contact travel of about 0.165 mm lies within both native clear/loaded brackets.

The **52.3 MPa in-plane PLA coupon reference** is first crossed at **0.246–0.271 mm with nylon** and **0.239–0.264 mm with aluminium**. These overlapping parent-mesh brackets do not establish a failure-order difference or a converged fracture load. The forward corner-refinement limit cannot be transferred to reverse loading. Terminal forces of 429 and 475 N are retained only as numerical continuation. Actual printed strength, screw allowables and battery-sleeve limits remain unverified.

## Final contact check and main-validation outcome

The last selected native control refined the driven battery's rigid target from **8,856 to 35,162 triangles**. The 1.5 mm PLA mesh, bilinear nickel law, fixed neighbours, four-core SMP runtime and prescribed increments stayed unchanged. All **20 saved equilibria through 0.040 mm** passed the fixture checks. The largest force-residual component was below 0.001 N.

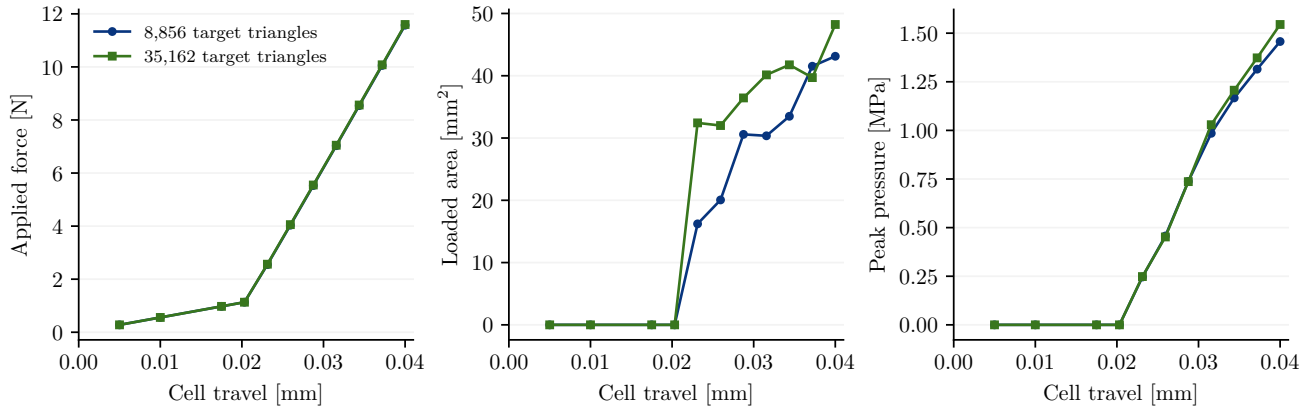



Figure 36: Native quantities at the same saved travel in the original and refined target controls. Loaded area is the contact quadrature measure. The curves describe the local fixed-support fixture, not free battery motion on the complete board.

## Force is stable; local contact area is less stable

Across the 11 common stations, applied force changed by **less than 1%**. Both models first carried pocket pressure between **0.020 and 0.023 mm**. At 8.5 N, interpolated cell travel remains about **0.034 mm**; each native bracket is retained in the audit. Contact pressure is compared only at actual shared travel, without interpolation across the force crossing.

At **0.040 mm**, force changed from 11.57 to 11.60 N, loaded area from **43 to 48 mm$^2$**, and peak pressure from **1.46 to 1.54 MPa**. The latter changes are about 12% and 6%. At first loading, the area approximately doubled despite the small force change. Thus the restraint conclusion survives this check, while local contact area is not converged. Rigid cell surfaces supply no physical sleeve or can strength limit.

## The selected computational work is closed

The finite scope contains **16 audited case records**, including reused prerequisites and documented partial solutions, **two preprocessing failures** and **six blocked dependents**. No selected case remains to dispatch. The two whole-shell controls failed before solving, so their contact and transient descendants provide no result. The 281 deferred cases are not resumed.

## What can be concluded

The case reduces tightening deflection and restrains sideways travel once a cell reaches its pocket. Its added mass and stiffness can increase board vibration stresses and attachment loads; aluminium generally transfers higher local loads than nylon. Fixed-contact vibration histories lose predictive scope when contact changes. Printed-coupon crossings, growing corner peaks and numerical stops do not establish fracture or tearing loads. The report preserves native ANSYS evidence, independent calculations and CAD-only findings separately. The prior fit remains in Appendix A, and all 22 shared-scale movie subjects remain available.

# Appendix A. Previous battery-case study

Assembly fit, bolt tightening, side pull and vibration response

9 November 2026 | Native structural analysis in ANSYS MAPDL 2026 R1.02 | 22 paired movies

## The mounting height determines whether the cells are already loaded

Placing the four tabs directly against the board creates approximately 1.18 mm of interference at the cell crowns. Each of the three cell solids shares about 1.70 $cm^3$ with the undeformed enclosure. A 3 mm tab stand-off clears the cells, nickel strips, washers and fasteners present in the 35-body mechanical STEP assembly. Thus a flush installation must displace or deform parts as the screws tighten. With the supplied Balius STEP, the outer washers touch at 2.397 mm stand-off. The Balius SolidWorks part has different washer clearance (page 29). Translating a rigid cell along the PCB X axis first closes a gap after 0.146–0.325 mm, depending on the cell and direction. Flexible relative motion determines whether contact occurs under load.

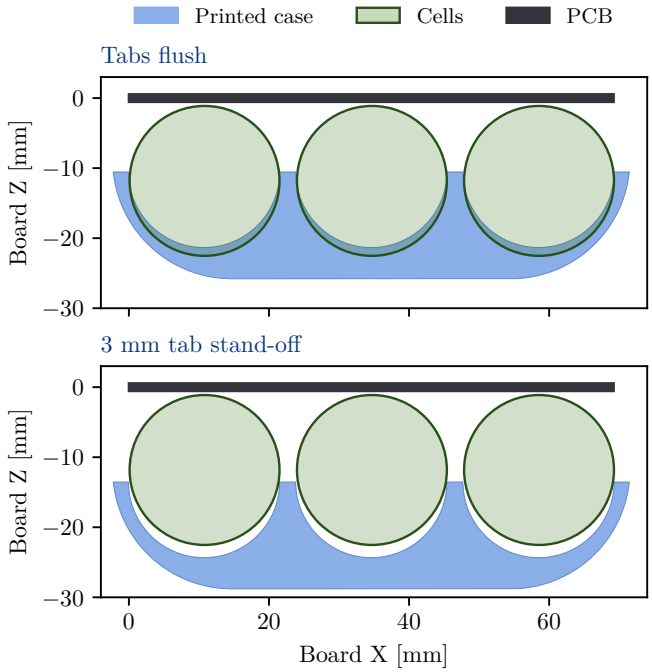



Figure A.1: Section through the middle of the cells, in PCB coordinates. The solid outlines show as-designed geometry before tightening. The mounting tabs lie beyond this section.

## What the raised case changes

Under X-axis vibration, the 3 mm case increased peak board-normal motion from 23.37 $\mu$m RMS bare to 145.55 with nylon screws and 120.00 with aluminium. Under base Z the peak PCB stress rose from 32.5 MPa RMS bare to 64.8 with nylon and 84.2 with aluminium (page 13). With the bilinear nickel law, an 8.54 N pull moved the edge cell 0.50 mm bare and about 0.20 mm with either raised case. With the multilinear law, a fixed 0.3375 mm travel required 10.15 N bare, 63.82 N with nylon and 125.98 N with aluminium screws (page 26). Nylon also gave the lowest cell-body X acceleration. Aluminium reduced board movement relative to nylon and increased the local stress across the PLA layers at a screw bearing. In the flush mounting trials, the printed-PLA tensile reference was exceeded before the tabs touched. At the last nylon equilibrium, only 4–5% of each tab lay within 0.05 mm of the board. With aluminium, that fraction was 18–19% even at the full 3 mm screw travel (page 19).

## Reference assembly

The October assembly has 208 bodies and a modeled mass of 251.82 g. Eight stand-off end faces carry the base motion. Two nickel strips attach each cell to the assembly. In the connected reference, six existing screws were loaded to 750 N apiece and locked. Its first mode was 485.964 Hz. Independent base excitation between 20 and 2000 Hz gave a maximum PCB directional displacement of 0.052674 mm RMS under Z excitation and a maximum PCB equivalent stress of 90.325 MPa RMS under X excitation. These archived quantities supply reference checks for the repeats.

## Registration and fastener selection

The four nonplated corner holes in PCB1 ODB++ have finished diameter 4.200 mm. The hole pattern measures $62.865 \times 114.427$ mm. The case bores have two diameters, 5.000 and 5.090 mm, on a $62.855 \times 113.241$ mm pattern. Centring the two patterns leaves about 0.593 mm of offset at each end. The sum of the radial clearances of a nominal 4 mm shank in the smaller case bore and in the board is 0.600 mm. The approximately 0.007 mm difference is the available nominal alignment margin there. Printed hole size, fastener major diameter and drilling tolerances therefore matter for installation.

| Tab stand-off [mm] | Cell overlap [mm$^3$/cell] | Washer bodies intersected | Other intersections |
|---|---|---|---|
| 0 | 1696–1698 | 4 | 6 nickel strips |
| 1.2715 | 0 | 4 | None found |
| 2 | 0 | 4 | None found |
| 3 | 0 | 0 | None found |

Table A.1: Solid intersections between the unstrained case and mechanical STEP bodies. The board's lower face defines the tab seating datum.

Starting from 3 mm of tab stand-off, the four outer washers meet the descending case after 0.603 mm. The middle cell first touches after 1.760 mm and the two outer cells after 1.790 mm. Once a washer touches, actual motion depends on deformation of the case and fasteners. A rigid assembly at the nominal registration therefore needs more than 2.397 mm of stand-off for clearance. Manufacturing tolerances and assembly misalignment would change that value.

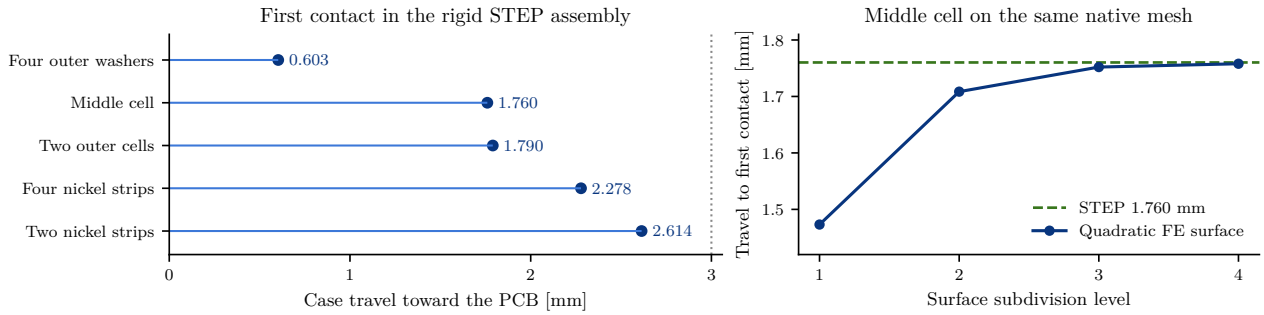



Figure A.2: Left, downward travel from the 3 mm position. Solid intersections 0.05 mm beyond each onset independently confirm contact. Right, subdividing the same curved finite-element surfaces moves the middle-cell contact toward the STEP solution. The underlying case mesh and the physical geometry are fixed.

## Materials and mounting representation

The supplied single-solid STEP has volume 63.069 cm$^3$. At a printed-PLA reference density of 1.17 g/cm$^3$, a dense case weighs 73.79 g, about 29.3% of the bare assembly mass. The material axes place the print layers parallel to the PCB for this scenario. PolyLite PLA coupon tensile moduli are 3.427 GPa in the print plane and 3.065 GPa through the layers. Shear moduli and Poisson ratios are engineering assumptions. Actual infill, orientation, print parameters and case mass require measurement. The published coupons used 100% infill. Polymaker technical data sheet.

The candidate model uses four M4 beam shanks with an effective tensile area of 8.78 mm$^2$. The comparison uses 69 GPa for aluminium and an exploratory 3.0 GPa for nylon. The nominal nylon hardware is McMaster-Carr 93640A145, M4 × 0.7 thread, 12 mm long. The represented bearing patches transfer loads between each end and its solid mesh. Compression-only members represent tab seating, with an assumed PLA annulus at the raised attachment points. The four case screws have no measured installation torque. The 750 N in the reference applies to the six original battery fasteners. McMaster-Carr metric socket-head screws.

## Numerical checks and remaining comparisons

The initial mesh has 16,108 curved quadratic tetrahedra. A 43,659-element refinement is available. Both have positive minimum scaled Jacobians. The native preparation check counted four screw beams, four compression seats and 8,038 candidate case contact faces. The raised cases passed the preload checks used for their vibration comparisons. In both flush cases the PLA exceeded its tensile reference before the tabs seated (page 19). The October bare-cell damage run and its repeat first registered damage around 43 N. A physical breaking load remains undetermined.

## The bare-board repeat reproduces the six locked bolt loads

The original battery attachment provides the initial stress field for the connected modal and vibration analyses. In the first nonlinear load step each of the six pretension elements receives 750 N. In the second step MAPDL holds the attained section adjustment fixed. The reaction at the pretension degree of freedom then measures how much load the locked joint retains. The four case screws are absent from this reference run.

| Existing bolt | Locked adjustment [$\mu$m] | Locked reaction [N] |
|---|---|---|
| 1 | 28.297 | 749.99935 |
| 2 | 22.205 | 749.99945 |
| 3 | 18.597 | 749.99908 |
| 4 | 30.457 | 749.99680 |
| 5 | 20.686 | 749.99942 |
| 6 | 18.772 | 749.99868 |

Table A.2: Fresh MAPDL extraction after length locking. Every row agrees with the October export at its stored precision. The largest departure from the applied 750 N is 0.00321 N.

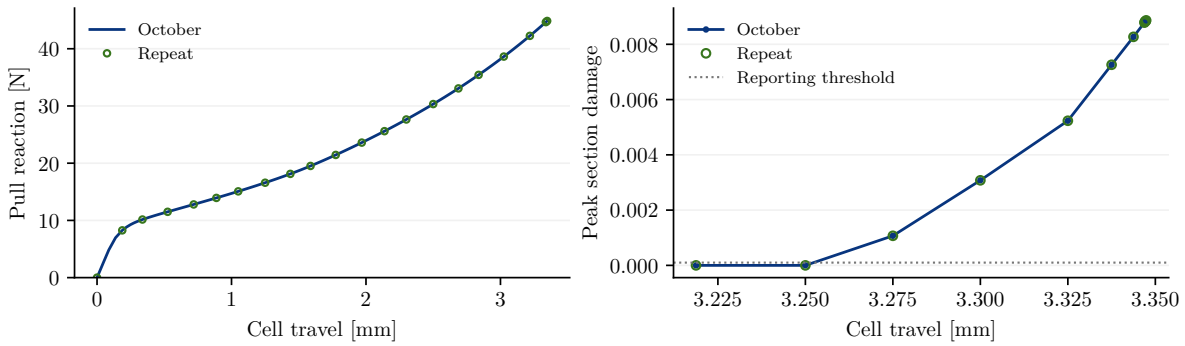
The solver stored eight equilibrium states and completed the two load steps in approximately 59 minutes on 12 distributed processes. The largest total displacement at lock was 0.0474595 mm. The repeated result differs from the October Mechanical export by $3.4 \times 10^{-8}$ in relative terms. The model produced 45 warnings, including overlapping contacts, automatically removed redundant constraints and the fixed orientation of pretension elements during large-displacement analysis. The warning text printed in the main solver listing matched the reference run. This comparison establishes repeatability for the existing attachment. Page 7 compares the locked bare and raised assemblies. The vibration response requires separate modal and random-response results.

## How the side-pull comparison measures load transfer

The local model drives the outer cell sideways through its rigid pilot while the two nickel strips bend. In the washer-clamped fixture, each strip is compressed with 750 N and the clamp height is then locked. During the pull, the cell can translate in the other two directions and rotate. The cased fixture adds a deformable printed enclosure, four screw shanks and compression-only tab seats. The neighboring cells and screw tops are fixed in this local approximation. The native model check counted 9,502 potential PLA contact faces. These contacts carry compression between PLA and the cell or washer surfaces. Direct PLA-to-nickel contact is omitted from this local fixture. That omission matters most in the flush geometry, where the CAD also intersects the strips.

The audit sums external forces and moments about the displaced cell centre at each converged increment. Both nominal nickel studies used the same bare mesh and 750 N clamp preloads. At 0.500 mm lateral travel, the bilinear model (148 MPa initial yield) required 8.54 N, and the multilinear damage model (185 MPa initial yield) required 11.33 N. At the same 8.54 N, the refined multilinear bare and raised-nylon histories both gave 0.20190 mm between saved states at 0.20166 and 0.20215 mm. The two loading schedules used different increments. Page 6 compares enclosures using the bilinear law throughout.

With the continuous nickel-damage law, the October run and its fresh repeat reached the same 110 equilibria. Both first crossed the $10^{-4}$ damage threshold between 3.250 and 3.275 mm, at 42.87–43.37 N. The maximum force difference over the two curves was $2.5 \times 10^{-8}$ N. At their last converged state (3.347 mm, 44.86 N), four flat shell elements exceeded the threshold on the bottom face, two per strip, over 1.50 mm$^2$ of original shell area. The force was still rising when convergence stopped. The rupture load remains undetermined.




Native bare edge-cell pull through the last converged equilibrium. The right panel enlarges the onset interval. Damage is the largest reported shell-section value in the two strips.

## Both default-contact meshes reproduce the October modal spectra

Each unpreloaded model used the archived mounting constraints and searched for 20 undamped natural frequencies between 1 and 10,000 Hz. The original mesh returned modes from 62.805 to 2205.547 Hz and the refined mesh from 66.611 to 2247.595 Hz. The maximum relative differences from the corresponding October exports were $1.78 \times 10^{-8}$ and $1.74 \times 10^{-9}$. Repeating a given discretization thus produced the same frequency spectrum to the stated precision.

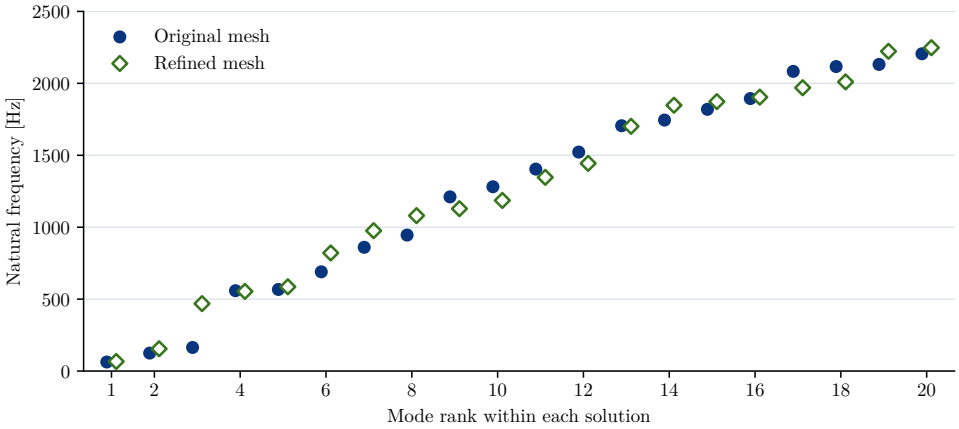



Figure A.3: Fresh ANSYS results with the original default contacts. Rank orders the frequencies within each mesh. A shared rank can contain different mode shapes, particularly where modes exchange order.

The first original-mesh run returned only four modes and exited normally. The repeat kept the same input geometry, material properties, supports and search range. It changed the Block Lanczos algorithm to its alternate formulation and requested the Sturm sequence check. This recovered the complete spectrum. The default models retain the detached nine-body group in the earlier connection diagnosis. The bearing, locking and contact states of the six-bolt model provide the starting point for the connected vibration comparison.

## PCB stress over the 480 frames of the seeded motion movie

Each of the three 9 s October base records has 589,825 samples at 65,536 Hz. After a 0.1 s ramp, the periodic 20–2000 Hz input reaches 14.136 g RMS. The native histories used 38 prestressed modes for the bare board, 49 with raised nylon screws and 48 with raised aluminium screws. All use 2% damping. Each clip shows 480 signed states from 1.000061 to 1.029297 s at 60 frames per second.

The six signed dynamic stress components were summed at each of 312,112 PCB element corners before calculating von Mises stress. The fresh native X history reproduced all 480 October bare-board frame maxima within 0.000028%; the independent integration agreed within 0.00182%. The largest native value in this clip was 154.41 MPa bare, 169.95 MPa with nylon screws and 151.03 MPa with aluminium screws. The nylon peak occurred about 3.85 ms earlier and at a different board node from the bare peak.

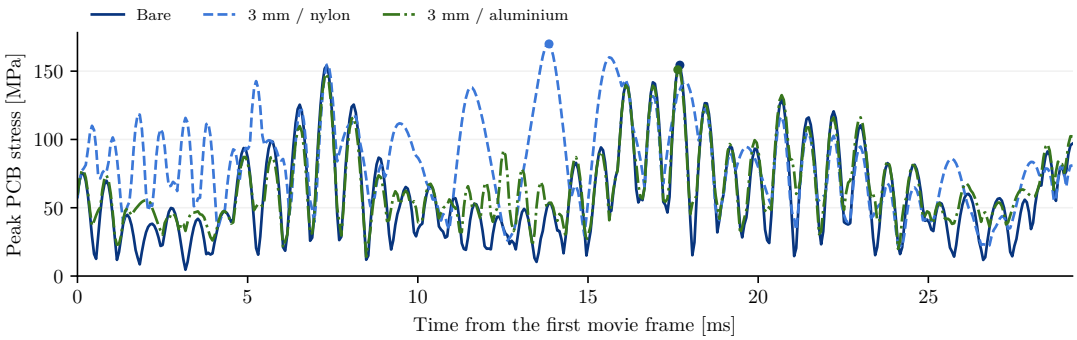



Figure A.4: Maximum dynamic PCB stress at each of the original 480 movie times. The three curves use signed native MAPDL coordinates on the corresponding tightened modal bases. Each dot marks a clip maximum.

Over the same frames, native peak board-normal displacements for base X/Y/Z were 0.040/0.047/0.168 mm bare, 0.244/0.154/0.290 mm with nylon and 0.114/0.125/0.314 mm with aluminium. All 22 pairs use native fields. The seven time-history pairs display motion at 50× scale and 273× slow motion. Paired movies.

## Does the contact model start with the gap measured in CAD?

An enclosure that presses on the cells at the start of a raised-case calculation will appear to stiffen the assembly before the real parts would touch. We checked the undeformed contact pairs separately from the load solution. The local fixture uses one target for the moving edge cell and another for the fixed neighboring cells and washers. The global assembly uses deformable targets. The search radius tells MAPDL which nearby facets to examine; it does not set the physical clearance.

| At 3 mm tab stand-off | Default search | 0.6 mm search | 2 mm search |
|---|---|---|---|
| Local: moving edge cell | +0.212 | +0.212 | +0.212 |
| Local: fixed cells and washers | −7.685 | +0.200 | +0.200 |
| Global: enclosure to assembly | +0.199 | +0.199 | +0.199 |

Table A.3: Initial native contact check in millimetres. Positive values are gaps; negative values are overlaps. Default search radii ranged from 10.6 to 11.1 mm. The 1 mm search reproduced the 0.6 and 2 mm columns.

The CAD solids do not intersect at this stand-off. We rejected the first raised local calculation because its 7.685 mm overlap contradicted that independent check. It did not reach a converged load increment. The replacement uses a 1 mm search radius for the two added contact pairs and leaves the nickel clamp contacts unchanged. The comparison calls for the cell travel at first closure, contact pressure, penetration, and load transferred to the case mounts.

The shortest unloaded distance from any cell to the STEP case is 0.144–0.161 mm. Translating the edge cell rigidly along X first closes the gap at 0.163 mm in positive X and 0.308 mm in negative X. In the loaded fixture the cell also rotates and moves vertically.

## The cell reaches the STEP case before the native contact pair carries load

We recovered each cell pose from the motion of 784 nickel nodes rigidly attached to the cell. In both raised nominal damage fixtures, those motions fitted one rigid transformation to within $10^{-5}$ mm and reproduced the independently exported cell-pilot translations. A separate displacement readback covered all 29,610 printed-case nodes. At the stations below, their maximum movement was below $10^{-9}$ mm.

| Cell travel | STEP solids | Meshed contact faces | Native contact reaction |
|---|---|---|---|
| 0.1375 mm | 0.02583 mm gap | Disjoint | 0 N |
| 0.1875 mm | 0.1183 mm$^3$ overlap | Intersecting | 0 N |
| 0.211914 mm | 0.4864 mm$^3$ overlap | Intersecting | 0 N |

Both screw materials gave all three rows. The final row is the last open state of each refined pull. Independent intersection tests used the actual cell target facets and only the PLA faces selected for contact. Both surface refinements gave the same classification.

At 0.1875 mm the cell was 23.4–23.6 $\mu$m past first STEP contact along X with its rotation and transverse translations held at their measured values. At the final open station for either screw material this excess travel reached 47.8–48.0 $\mu$m. Testing only the target vertices missed intersections between faces at 0.1875 mm. The finer load steps on page 25 resolve when this model carries force. Contact discretization also affects how that force relates to the physical assembly.

## Why the flush installation needs a separate check

The flush STEP has real interference. In the local model, a 2 mm search recovered the 1.178 mm crown overlap of the moving cell. A 1 mm search found only 1.000 mm, and a 0.6 mm search found only 0.598 mm. Tightening the search around the flush configuration can therefore hide real interference. Conversely, the default global search reported a peak overlap of 10.093 mm, versus 1.848 mm with a 2 mm search. These values identify a contact-model discrepancy. They do not establish an installation force. The flush comparison requires a verified path into contact, including the washers and strips, and force balance in the installed assembly before the vibration solve.

## Both raised cases restrain the edge cell after contact

The local fixture pulls the outer cell in positive X. Its other translations and rotations are free. With the case on a 3 mm stand-off, the last open state was at 0.196 mm and the first closed state at 0.221 mm for either screw material. Under the 8.54 N load that moved the bare cell 0.50 mm, both cased cells travelled about 0.20 mm. The interpolated estimates were 0.207 mm with nylon and 0.202 mm with aluminium. Both lie in the same 0.025 mm interval between native states.

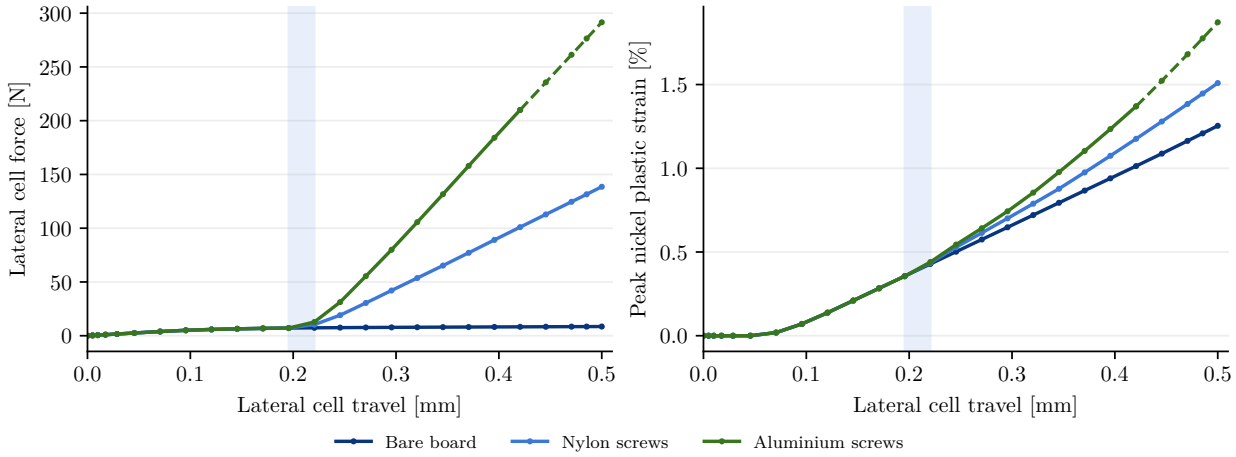



Figure A.5: All three native pull histories. The blue band brackets first contact. The dashed aluminium segment extends from the last state below both PLA tensile references to states above at least one reference.

| At 0.420625 mm prescribed cell travel | Bare | Raised nylon | Raised aluminium |
|---|---|---|---|
| Cell reaction in X [N] | 8.26 | 100.99 | 209.96 |
| Load on four case supports in X [N] | — | 80.00 | 174.54 |
| Peak equivalent plastic strain in nickel [%] | 1.013 | 1.176 | 1.370 |
| External work applied to cell [mJ] | 2.63 | 11.91 | 22.56 |
| Energy stored in printed case [mJ] | — | 5.41 | 12.93 |
| Energy stored in four screw shanks [mJ] | — | 2.05 | 2.61 |

At a fixed travel the stiffer aluminium attachment increased both the case load and the strain in the nickel. At a fixed 8.54 N, the native states bracketing that force instead gave 0.36–0.44% peak nickel plastic strain for the two cased curves, versus 1.25% bare. Integrating native nodal forces and moments over the displacements of the PLA and shanks independently reproduced their stored energies within 0.009% of each component's maximum change.

## Where the aluminium comparison leaves the coupon range

With aluminium screws, peak in-plane PLA tension crossed 52.3 MPa between 0.421 and 0.446 mm. Through-layer tension crossed 40.5 MPa between 0.446 and 0.471 mm. Both peaks were within 0.54 mm of the idealized screw bearing. At 0.50 mm the elastic continuation returned 291.5 N with aluminium, versus 138.5 N with nylon and 8.54 N bare. The nylon case stayed below both references, reaching 28.6 and 23.5 MPa. Its most tensile shank fibre reached 30.5 MPa, including 29.3 MPa from bending.

All fixtures saved 33 converged states. Peak penetration at a loaded cell/case corner was 0.0116 mm with nylon and 0.0198 mm with aluminium. The native cell contact forces and case-support reactions balanced to within 0.0019 N. These pulls assume fixed screw tops and omit direct PLA-to-nickel contact. At 0.421 mm cell travel, the axial-load resultants at the four tab bearings acted 5.41–6.54 mm from the screw axes with nylon and 5.41–5.49 mm with aluminium. All eight offsets exceed the modeled 3.75 mm bearing radius. A one-sided axial pressure across that footprint cannot carry the same force and moment. The computed pull stiffness therefore depends on how the installed joints transmit a couple. Page 17 gives the force check.

## Both raised cases reduce board deflection during tightening

The bare board and both 3 mm stand-off assemblies completed eight equilibrium states under the same six-bolt tightening sequence. Each original bolt carried 750 N before its length was locked and retained 749.996–750.000 N afterward. The two cased models differ only in the properties assigned to the four case screws. The load in this comparison comes from the six original bolts.

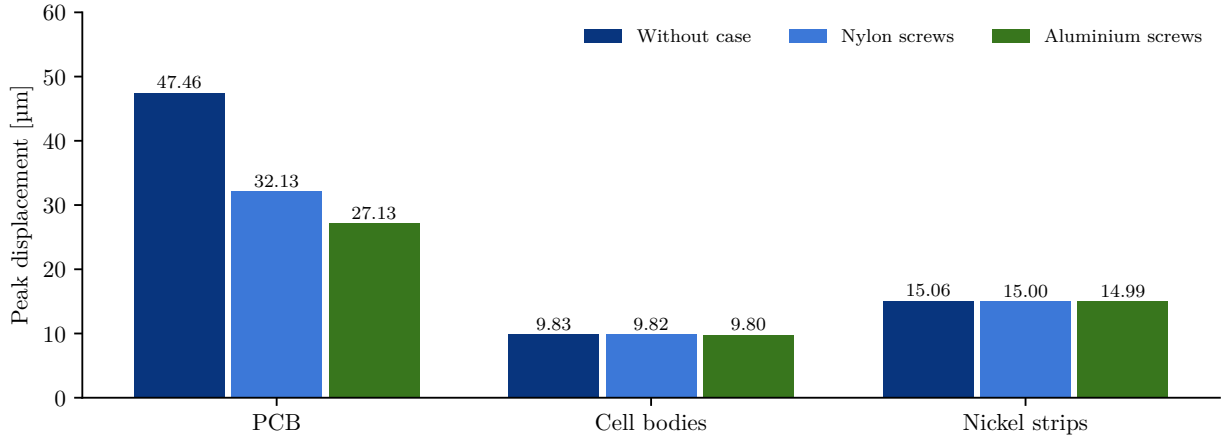


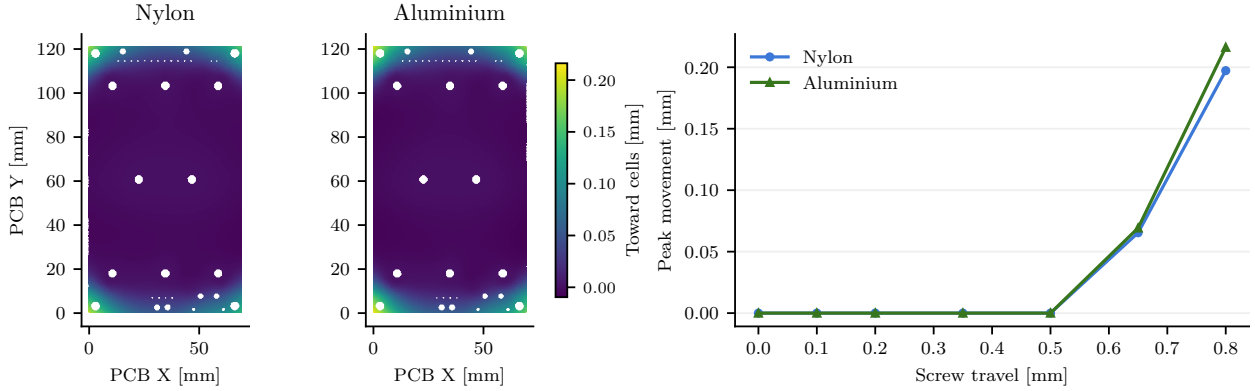

Figure A.6: Peak displacement magnitude relative to the undeformed assembly after bolt locking. Each comparison uses the same mesh nodes on each part. Micrometres are used here to show the small changes at the cells and nickel strips.

The peak PCB displacement occurred at node 207485 in all three models. It fell from 47.46 $\mu$m bare to 32.13 $\mu$m with nylon screws and 27.13 $\mu$m with aluminium screws, reductions of 32.3% and 42.8%. Changing the four case screws from nylon to aluminium reduced this peak by another 5.00 $\mu$m, or 15.6% relative to nylon. The cell-body peak changed by at most 0.30% from bare and the nickel-strip peak by 0.51%. All 8,038 case-to-assembly contact faces stayed open in both raised assemblies, with zero contact pressure at every stored state. The case altered board bending through its four corner attachments.

ANSYS DPF recovered translations on 217,704 common PCB nodes. On the bare control the two displacement readers agreed on every component. The six pretension adjustments agreed with native MAPDL readback within $10^{-11}$ m in all three assemblies.

## The PCB bends toward the cells in both flush installations

The PCB moved less than 1 $\mu$m through 0.50 mm screw travel. At 0.80 mm, after washer contact, the same corner at $(0, 0)$ had moved 0.197 mm toward the cells with nylon screws and 0.216 mm with aluminium. The aluminium screws raised the total drive force by 44% and peak board motion by 9.6%. The four PCB bearings moved 0.104–0.145 mm with nylon and 0.125–0.163 mm with aluminium.




Upper PCB face at 0.80 mm screw travel, with one displacement scale for both maps. Positive movement is toward the cells. The curves show the maximum across all 217,704 board nodes relative to six-bolt lock.

At 0.80 mm, the largest increase in upper-face principal tensile strain from six-bolt lock was 0.601% with nylon and 0.958% with aluminium. We compared the same element corner before and after driving. Excluding points within 2 mm of a tied PCB bearing node gave 0.601% and 0.670%, respectively. These local elastic strains describe the board laminate model.

## The raised enclosure adds resonances below the first bare-board mode

We extracted undamped natural frequencies about the verified bolt-locked states, using the same mounting constraints and a 1–4000 Hz search. The assembled masses were 251.82 g bare, 325.82 g with the raised nylon-screw case and 326.10 g with aluminium screws. The models returned 38, 49 and 48 modes, respectively. The lowest three frequencies were 253, 289 and 339 Hz with nylon and 320, 372 and 378 Hz with aluminium. All six fall below the first bare-board mode at 486 Hz.

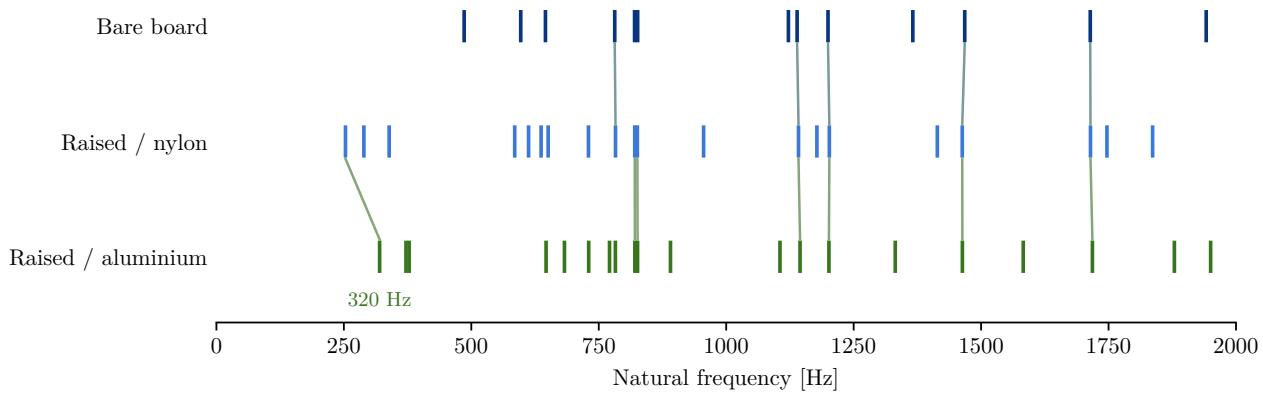



Figure A.7: Each tick marks one native frequency in the excitation band. Lines connect five bare-to-nylon matches above MAC 0.94, and nylon-to-aluminium matches above MAC 0.97 where the assigned match is also the best available match.

The 253 Hz mode carried 17.9% of the assembled mass in X translation. In this shape the largest cell displacement was 2.8% of the peak on the PCB, while the peak on the case was 126% of the PCB peak. Those are amplitude ratios *within a mode*; the actual vibration displacement requires a forcing spectrum and damping. The 253 Hz mode had no close counterpart in the bare set: its highest shape correlation was 0.56. Conversely, the bare 781.35 Hz mode matched cased mode 9 at 782.88 Hz (correlation 0.943). This battery-motion mode shifted by only 0.20% even though its rank changed from fourth to ninth.

| Effective translational mass, fraction of whole assembly | X | Y | Z |
|---|---|---|---|
| Bare, all 38 retained modes | 87.2% | 89.0% | 89.6% |
| Raised nylon, all 49 retained modes | 90.1% | 91.5% | 90.8% |
| Raised aluminium, all 48 retained modes | 90.1% | 91.4% | 90.3% |

## How the shape comparison was made

Both modal meshes contain their assembly's static deformation. Before comparing vectors, we checked every original solid node against its own preload displacement and matched the 369,584 node IDs and their undeformed coordinates. The modal assurance criterion (MAC) is the squared, unweighted correlation of the global X, Y and Z eigenvector components at these nodes. It ranges from zero to one and is independent of the sign and normalization of a mode. It excludes the new case nodes so that both vectors describe the same structure. Modes with weak or competing matches remain ambiguous. For the two screw materials we also matched all 399,194 solid nodes, including 29,610 PLA nodes. The first mode rose by 26.5% from 253.16 to 320.17 Hz with full-solid MAC 0.989 and PLA-only MAC 0.994. It is a closely corresponding shape. The next two matches had full-solid MACs of 0.739 and 0.785.

The October bolted movie shows mode 20 at 2364.18 Hz. The two raised assemblies have mode 20 at 1836.45 and 1950.27 Hz. Its MAC with the bare mode is below 0.00023 in both. Searching the full native bases instead places the closest shape at mode 27, at 2365.06 Hz with nylon (MAC 0.717) and 2368.47 Hz with aluminium (MAC 0.640). The movie preserves the original rank and identifies this match on screen.

## The linear model predicts more PCB motion with the raised case

We applied the October base acceleration spectrum separately along X, Y and Z to the three bolt-locked assemblies. For the 3 mm stand-off with nylon screws, the independent calculation predicts greater out-of-plane PCB motion for all three excitation directions. Under X excitation the peak PCB response in Z rises from 23.37 to 145.55 $\mu$m RMS, a factor of 6.23. In the same calculation the largest cell response in X falls from 61.43 to 53.79 $\mu$m RMS (12.4%). With aluminium screws, the peak PCB Z responses were 120.00, 82.33 and 100.55 $\mu$m RMS for base X, Y and Z. Compared with nylon, these fall by 17.6%, 5.0% and 23.6%. In particular, the 32% reduction in static board deflection during tightening did not carry over to this vibration metric.

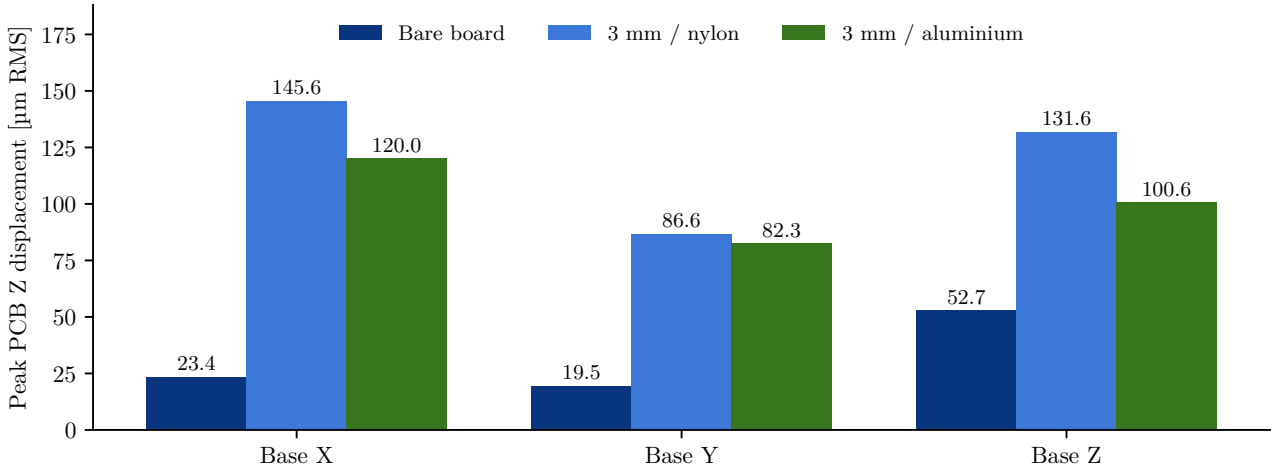



Figure A.8: Each group compares the largest board-normal RMS displacement anywhere on the PCB under the same single-axis base spectrum. Peaks can occur at different nodes. Values describe motion relative to the supports.

## How we checked the response

The independent calculation uses each native mode's signed displacement field and directional participation factor. We integrate the modal frequency responses against the one-sided acceleration spectrum (20–2000 Hz, 14.14 g RMS, 2% modal damping), including cross terms between modes driven by the same base input. This preserves cancellation and reinforcement. Halving the quadrature spacing from 0.25 to 0.125 Hz changed the modal covariance by less than $6 \times 10^{-8}$ in relative norm. For the bare assembly we compared all 18 peak directional displacements: three base directions, three response components, and both whole-assembly and PCB scopes. The worst difference from October's native ANSYS random-response output was 0.013%. The three native MAPDL repeats reproduced all 51 October Mechanical maxima within $4.75 \times 10^{-8}$ relative, including displacement, absolute acceleration, equivalent stress and the three normal elastic strains on the PCB. Peak PCB equivalent stresses under base X, Y and Z were 90.325, 17.850 and 32.482 MPa RMS.

We repeated the independent integration with modal cutoffs of 2000, 3000, 3500 and 4000 Hz, keeping the input spectrum fixed. For the raised nylon case under base X, both the 20-mode and 49-mode calculations gave a peak PCB Z displacement of 145.55 $\mu$m RMS. Across all three inputs, increasing the cutoff from 3500 to 4000 Hz changed the nine PCB directional peaks by at most 0.22% bare and 0.056% cased. The bare PCB's Y response to base Y differed by 8.32% with the 2000 Hz cutoff. These checks apply to displacement within the available basis.

The two native base-X solutions independently gave 120.00 $\mu$m for the raised aluminium case and 145.55 $\mu$m for the raised nylon assembly's peak PCB Z displacement. All 30 directional displacement peaks across five part selections per cased model agreed with the independent integrals within 0.0017%. Both calculations linearize about the tightened contact state.

## The 3 mm stand-off does not prevent dynamic cell contact in the model

The tab stand-off is not the air gap at the cells. The smallest sampled cell clearance on the tightened mesh was 0.143 mm. We paired nearby opposing surfaces and calculated their simultaneous relative motion along the local separation direction. Under base X the most exposed cell location had 0.145 mm clearance and 0.098 mm RMS relative motion. The gap there is only 1.47 times that RMS displacement. This ratio suggests contact is plausible, but it does not tell us when contact occurs.

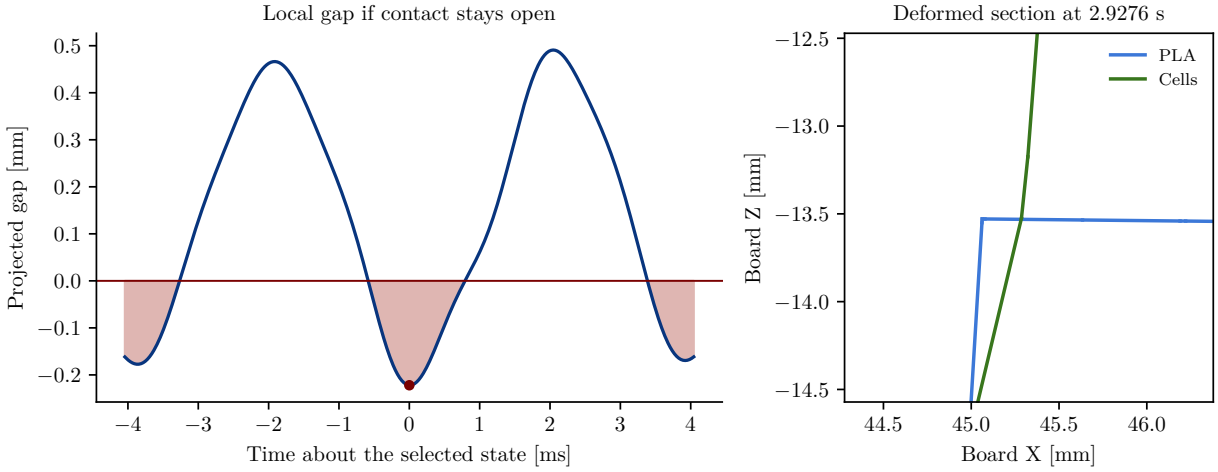



Figure A.9: Raised case, nylon screws, October X-axis waveform. Left: first-order gap of the selected pair. A negative value indicates that the linear solution continues past projected contact. Right: actual-scale section of both triangulated surfaces displaced by the same signed modes. The curves are a diagnostic of the fixed-open model.

We independently integrated the nine-second October acceleration history and reconstructed the two skins at the instant of greatest projected approach for this pair, 2.9276 s into the waveform. At the tightened rest position the cell and case meshes had no intersections. In the selected deformed state, 895 triangle pairs intersected, and a separate signed-distance calculation placed 24 cell vertices more than 1 $\mu$m inside the PLA skin. The deepest sampled vertex lay 0.214 mm inside the PLA. Subdividing the curved elements twice more gave 0.215 mm at the same instant. Both surfaces had positive clearance before excitation. Their intersection shows where the fixed-open vibration assumption breaks down.

| Excitation | Cell pair gap [mm] | Relative RMS [mm] | Intersecting cell/PLA pairs |
|---|---|---|---|
| Base X | 0.145 | 0.098 | 895 |
| Base Y | 0.143 | 0.019 | 0 |
| Base Z | 0.236 | 0.112 | 0 |

Table A.4: Each row selects the cell pair with the greatest RMS-to-gap ratio for that excitation, then checks the whole cell and case skins at one instant. A zero count at one instant does not establish separation throughout the record.

The surface search used both centroids and vertices of facing triangles within 1.2 mm in both directions, including the midside nodes of curved elements. Under base Z the nearest-line projection became negative, but the cell and case skins did not intersect at the selected instant. Tangential motion matters: a negative projected gap alone is insufficient evidence. Under base X both independent geometric checks found overlap. Until a nonlinear transient calculation or vibration test measures the response after closure, use the preceding fixed-contact RMS results as a comparison of the two linear models, not an installed flight stress, impact load or clearance margin.

## The local flush installation stopped before the tabs seated

Four nylon screws drew the printed case toward a rigid board datum from an initial 3 mm stand-off. The existing washer clamps were tightened first. ANSYS saved 88 converged equilibria before the third load step failed at the next increment. The final converged screw-head travel was 2.8685 mm, leaving 0.1315 mm of the commanded travel. The solver never reached the sideways pull.

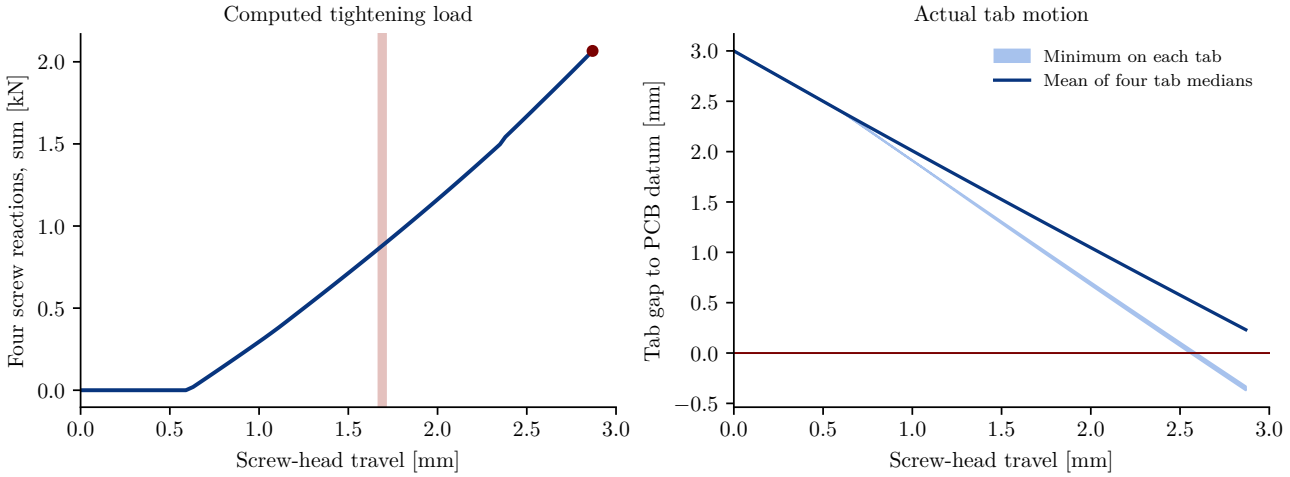
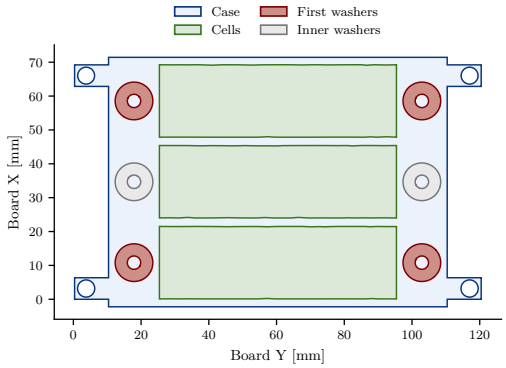



Figure A.10: Left, the computed total screw reaction rose to 2.066 kN. The shaded interval marks the first exceedance of the 0.05 mm numerical penetration screen. Right, the tab surfaces tilted during drawdown. At the last equilibrium every tab crossed the nominal PCB plane at some points and remained clear at others.




The native fixture first loaded the four outer washers at 0.59–0.63 mm of screw travel. This brackets the independent STEP result of 0.603 mm (page 2). At 0.63 mm, two closed contact facets at each washer carried 4.89–4.92 N in board Z, totalling 19.63 N. Each loaded node lay within 0.8 $\mu$m of its assigned washer. The next nearest candidate was over 8.6 mm farther away. These local reactions apply to the fixture with fixed neighboring hardware.

The moving edge cell and its two nickel strips stayed clear throughout the partial path. Loaded-corner penetration first exceeded 0.05 mm at 1.67–1.71 mm travel and reached 0.119 mm. The largest departure from force balance was 0.030 N.

| Final saved equilibrium | Tab 1 | Tab 2 | Tab 3 | Tab 4 |
|---|---|---|---|---|
| Screw-head reaction [N] | 484 | 517 | 530 | 536 |
| Minimum gap to board datum [mm] | −0.332 | −0.357 | −0.366 | −0.369 |
| Median sampled gap [mm] | 0.248 | 0.227 | 0.236 | 0.214 |

In this local fixture the seating springs moved with the screw grips, allowing a tilted tab to cross the board plane. The peak computed PLA equivalent stress reached 233 MPa under the elastic print model. The physical installation force, failure sequence, and full-flush side-pull response remain unresolved. The whole-board installation model instead places contact directly between each tab and the flexible PCB surface, so the model can develop a normal reaction at the bearing land.

## Contact deflects the raised case along the middle cell

The fixed-open calculation on page 10 placed parts of all three cell surfaces inside the enclosure at the selected X-waveform sample. Along the middle cell, the affected PLA facets extended from 25.2 to 95.6 mm in board Y. We therefore retained the whole length of this contact region in two native submodels. Their outer boundaries lay 9 or 15 mm from the middle-cell contact line. Each retained the original quadratic elements and directional PLA properties.

For each domain we imposed the tightened board displacements, then applied the signed motion at 2.9276 s to the two local cell bodies and the outer cut faces of the PLA. We increased the imposed motion from zero to its full value and unloaded in a series of static equilibria. With cell contact disabled, the free PLA nodes followed the parent solution to within 1 $\mu$m. With frictionless contact active, the largest change in their motion was 249 $\mu$m in the smaller domain and 244 $\mu$m in the larger one.

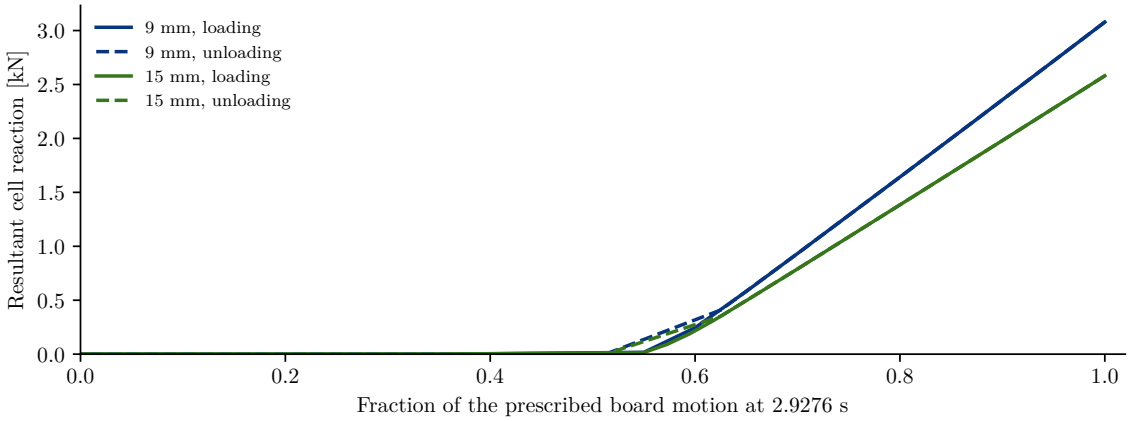



Figure A.11: Each curve gives the force needed to impose this particular displacement path. The larger domain leaves a wider section of PLA free to flex. The peak reaction fell by 16.2%. The local models hold the cell motion fixed as the case pushes back.

| At the fully imposed displacement | 9 mm domain | 15 mm domain |
|---|---|---|
| Resultant cell reaction [kN] | 3.079 | 2.581 |
| Active contact faces | 24 | 24 |
| Active faces touching the artificial cut | 0 | 0 |
| Maximum sampled skin overlap, contact disabled [mm] | 0.2165 | 0.2164 |
| Maximum sampled skin overlap, contact enabled [mm] | 0.0424 | 0.0445 |

The independent geometry check measured signed distances on vertices and facet centres inside a common 8 mm band. Some tessellated surfaces still intersected even though the largest loaded-corner penetration recorded by ANSYS was below 0.5 $\mu$m at this state. Changing the cut size changed the force appreciably despite whole-model force imbalances below 0.004 N. These results establish local deflection under imposed motion. A flight impact load requires the cells and the surrounding board to respond to contact together.

## Which cell first reaches the case in the transient window?

Using the native X history from rest, we checked every integration sample through 89.478 ms on all three curved cell skins. The centre cell was clear at 85.327 ms and intersected the case at 85.342 ms. The left cell first intersected at 85.388 ms. The right cell remained clear through the window. Both local domains captured the two intervals of centre-cell overlap. The left cell lies outside both domains, so the local reactions describe only part of the assembly load path.

At the prescribed time stations, the sealed transient inputs reproduced native translations at all driven nodes within 0.0022 $\mu$m. Between stations, linear interpolation in the 9 mm domain differed from the native motion by up to 1.04 $\mu$m with a 61.04 $\mu$s table interval and 0.26 $\mu$m with a 30.52 $\mu$s interval. These checks quantify the motion imposed on the submodels. The force curves above are from the static loading and unloading calculation.

## The raised case increases PCB stress under Y and Z excitation

The native MAPDL comparison used the same X-axis base spectrum and 2% damping on both assemblies. Each original board bolt had already been tightened and locked. With four nylon case screws at the 3 mm stand-off, the peak PCB equivalent stress fell by 1.6%, from 90.325 to 88.847 MPa RMS. Its location moved about 54 mm across the board. The PCB's maximum normal strains in X and Y both increased.

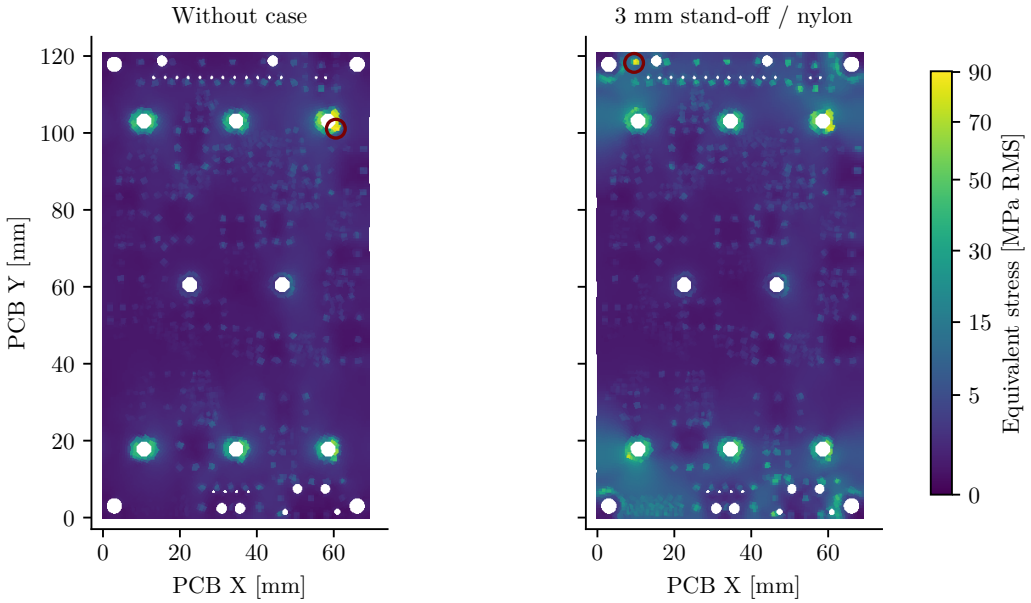



Figure A.12: PCB plan view in the original mesh coordinates. Each solid footprint shows its largest stored equivalent-stress RMS among its corners. Higher values are drawn last through the thickness. Red circles mark the maxima at $(60.55, 101.11)$ and $(9.54, 118.21)$ mm. Both maps use the same colour scale.

| Peak response, one sigma | Base axis | Bare | 3 mm nylon | 3 mm aluminium |
|---|---|---|---|---|
| PCB Z displacement [$\mu$m] | X | 23.37 | 145.55 | 120.00 |
| | Y | 19.45 | 86.65 | 82.33 |
| | Z | 52.67 | 131.64 | 100.56 |
| PCB equivalent stress [MPa] | X | 90.32 | 88.85 | 88.65 |
| | Y | 17.85 | 57.05 | 50.29 |
| | Z | 32.48 | 64.77 | 84.19 |
| Nickel equivalent stress [MPa] | X | 291.90 | 256.24 | 286.46 |
| | Y | 15.11 | 21.85 | 21.86 |
| | Z | 38.13 | 38.38 | 38.20 |

Under base Z, aluminium screws gave less board travel than nylon screws and higher peak PCB stress. Each entry is a maximum over the named part. The spectra hold the tightened contact state fixed (pages 10 and 12).

An independent sum of the signed modal stresses reproduced all nine native stress-peak locations across 312,112 PCB element corners. Reducing the modal cutoff from 4 to 3.5 kHz changed the nine peak values by at most 0.20%. With only the modes below 2 kHz, the bare base-Y peak was 34.65% lower. The excitation band remained 20–2000 Hz throughout.

## The printed tab carries a local load across its layers

The printed case and PCB react differently to the choice of screw. At the 3 mm stand-off, changing nylon screws to aluminium reduced peak PCB movement during tightening (page 7), while the greatest tensile stress normal to the layers anywhere in the PLA rose from 0.688 to 1.515 MPa. The independent orthotropic elastic-law check reproduced the stored tightening strains to within $4.0 \times 10^{-8}$ relative field norm across all 64,432 element corners.

For X-axis vibration with nylon screws, the native peak layer-normal stress was 17.16 MPa RMS, 0.53 mm from the nearest node tied to a screw bearing. Among corners at least 2 mm from those nodes it was 5.72 MPa RMS. The static tension at the corner of greatest RMS was 0.688 MPa. The six native component fields and stored equivalent-stress RMS agreed with integration of the signed modal stresses to within 0.001% relative field norm.

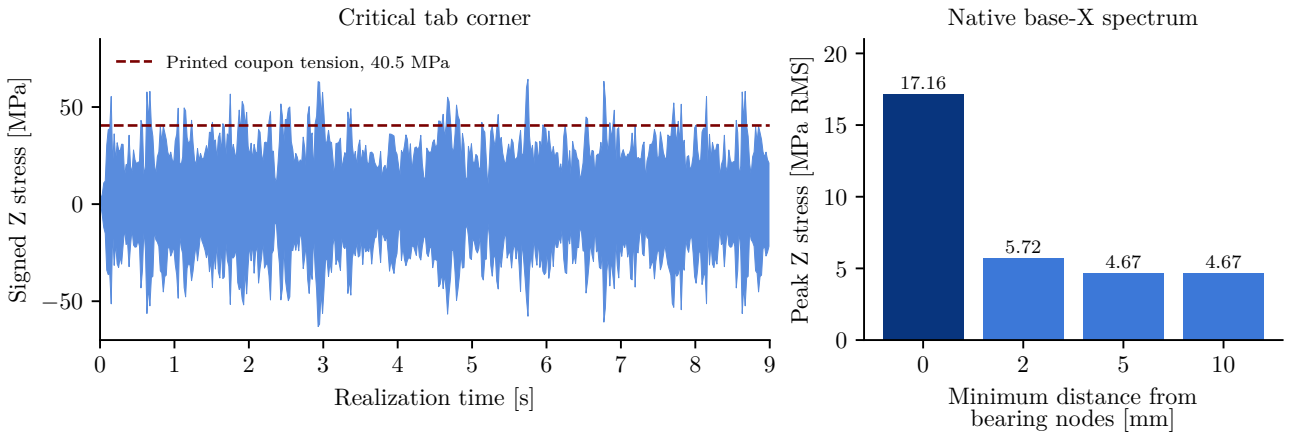



Figure A.13: Left, independent reconstruction at element 1215221, node 815413, with the static tension included. The band preserves sample maxima and minima in 20 ms windows. Right, maximum stored through-layer RMS over locations at or beyond each distance from a coupled screw-bearing node.

## Cell clearance at the peak printed-layer stress

The native nine-second histories gave peak layer-normal tension above the printed-coupon mean of 40.5 MPa under all three base inputs. The table uses the corner of highest layer-normal RMS for each input and includes the locked preload. All three cell skins intersected the case at the X stress peak. At the Y and Z stress peaks, the curved skins were disjoint. These selected instants describe a linear model with fixed contact.

| Base | Peak [MPa] | Coupon [%] | Time [s] | Left gap [mm] | Centre gap [mm] | Right gap [mm] |
|---|---|---|---|---|---|---|
| X | 64.3 | 159 | 5.74146 | −0.201 | −0.186 | −0.123 |
| Y | 44.0 | 109 | 5.18582 | 0.128 | 0.111 | 0.133 |
| Z | 56.2 | 139 | 7.76979 | 0.121 | 0.132 | 0.114 |

Raised nylon mount. Each gap is the minimum signed distance sampled on that cell's skin at the stress peak, with eight segments per quadratic edge. Negative gaps indicate overlap. The surface-intersection check agreed at two refinements.

Over 589,825 samples per axis, the native modal coordinates agreed with independent integration to within 0.0015% in relative norm. The attachment-force histories agreed within 0.0005 N over the X movie clip. The material reference comes from the PolyLite PLA coupon data. Actual print orientation, infill, bearing compliance and impact response remain unmeasured, so the table cannot establish whether the physical tabs crack.

## Aluminium screws reduce board motion and concentrate load in the tab

The two raised models differ only in the properties of the four case screws. Both native base-X spectra used the same excitation, printed case and board mesh. Relative to nylon, aluminium reduced the peak PCB Z displacement by 17.6% and the PCB Z acceleration by 18.6%. The two PCB equivalent-stress peaks were within 0.22% of each other. Changing to aluminium moved the peak back to the same node as the bare reference.

| Native base-X response, RMS | Bare | Raised nylon | Raised aluminium |
|---|---|---|---|
| PCB Z displacement [$\mu$m] | 23.37 | 145.55 | 120.00 |
| PCB Z acceleration [$g$] | 52.29 | 67.45 | 54.92 |
| Cell-body X acceleration [$g$] | 103.51 | 91.81 | 100.90 |
| PCB equivalent stress [MPa] | 90.325 | 88.847 | 88.655 |
| PLA layer-normal stress [MPa] | — | 17.16 | 24.78 |
| PLA layer-normal stress, at least 2 mm from bearings [MPa] | — | 5.72 | 5.41 |

The aluminium case had a peak layer-normal stress of 24.78 MPa RMS at a corner 0.50 mm from a coupled bearing node. That is 44.4% above the nylon peak. Among corners at least 2 mm away, aluminium gave a slightly lower peak. This dependence on the bearing region motivates the finer case mesh and inspection of how the actual screw head bears on the print.

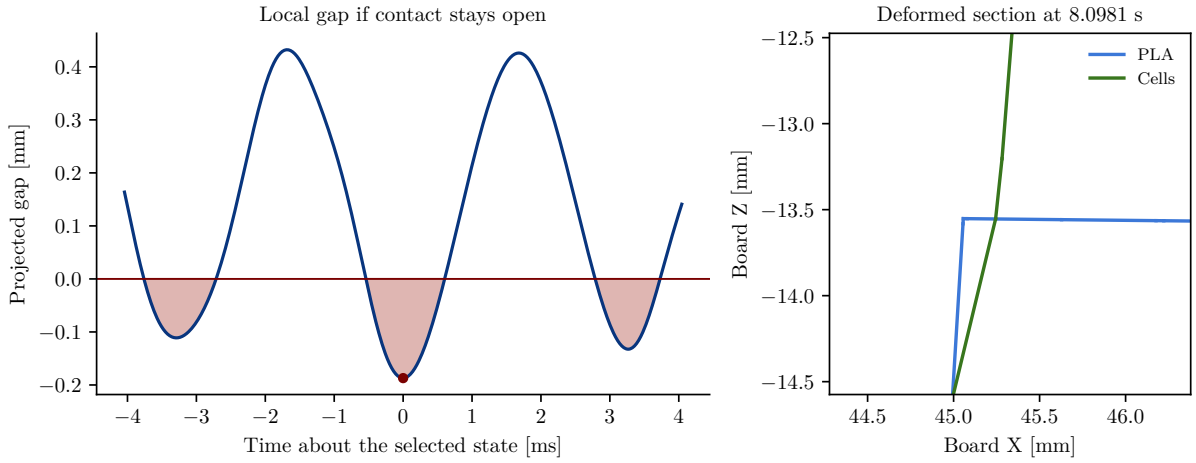



Figure A.14: Raised aluminium-screw case, independent response to the October X waveform. Near the selected point, the tightened gap was 0.145 mm and the normal relative motion was 0.070 mm RMS, compared with 0.098 mm for nylon. The section is plotted at physical scale.

## Cell clearance at the aluminium mount's peak layer stress

At the monitored X peak, all three cell skins crossed the PLA. The left cell had the greatest sampled overlap. The table compares the monitored stress maximum for each of the three independent base directions with the signed cell gaps from the same native time sample. Physical loads after impact and across the actual screw bearing remain unresolved.

| Base | Peak [MPa] | Coupon [%] | Time [s] | Left gap [mm] | Centre gap [mm] | Right gap [mm] |
|---|---|---|---|---|---|---|
| X | 110.1 | 272 | 8.09808 | −0.175 | −0.155 | −0.100 |
| Y | 91.6 | 226 | 5.14613 | 0.100 | 0.087 | 0.115 |
| Z | 104.2 | 257 | 6.32817 | 0.160 | 0.143 | 0.140 |

Raised aluminium mount under the original nine-second waveforms. The monitored corner maximizes layer-normal RMS for the given input axis. The signed distances and surface checks use the same definitions as on page 14.

## The raised case lowers the highest nickel-strip stress under base X

Each cell connects to the board through a nickel strip at each end. We compared the same 11,348 element corners across all three assemblies under the October X-axis vibration spectrum. With nylon case screws, peak equivalent stress fell in all six strips. The largest value occurred at the same corner on the high-Y strip of the right-hand cell in all three assemblies. It fell from 291.9 MPa RMS bare to 256.2 MPa with nylon (12.2%) and 286.5 MPa with aluminium (1.9%). Aluminium raised the middle cell's two strip peaks by 2.6% and 2.7% relative to bare.

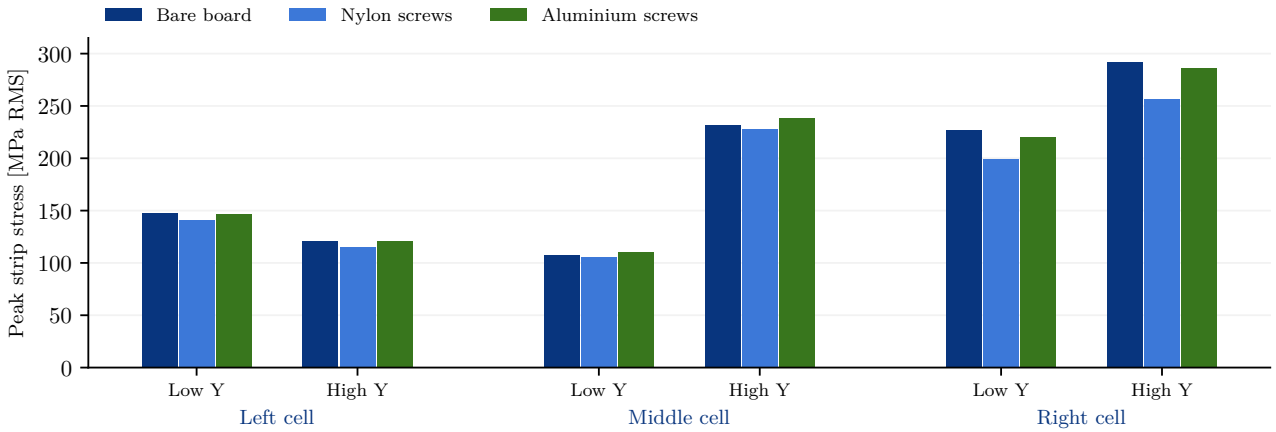



Figure A.15: Native one-sigma equivalent stress, maximized separately in each nickel strip. Low and high Y identify the strip ends near board coordinates 22 and 99 mm. Left, middle and right refer to increasing board X. For every strip, its maximum stayed at the same element corner across screw choices.

## The assumed nickel yield stress is reached while the cells still clear the case

The global model uses linear elastic nickel with modulus 220 GPa. The separate October nonlinear pull model assumes a 185 MPa initial yield stress for annealed nickel and a modulus of 205 GPa. To compare the two, we reconstructed the global *signed* stress history during the first 0.1 s of the seeded October base-X waveform. At each sampled instant we added the six stress components from the locked preload to their respective dynamic components, then calculated von Mises stress at each corner. Independent spectral integration also reproduced all six native RMS component fields and the stored equivalent-stress field to within 0.0033% relative field norm.

| First sampled crossing of 185 MPa | Bare | Raised / nylon | Raised / aluminium |
|---|---|---|---|
| Base X, time bracket [ms] | 38.116–38.132 | 38.956–38.971 | 38.116–38.132 |
| Peak after tightening alone [MPa] | 161.9 | 162.1 | 162.1 |
| Base X, minimum clearance bound [mm] | — | 0.038 | 0.068 |

Table A.5: Each time refers to this excitation with its prescribed 0.1 s ramp. The material threshold is a property assumed in the local model. Actual strip temper and weld properties would be needed to infer yielding or fatigue life.

The clearance check covers every quadratic exterior face on all three cells and the PLA. At the tightened position their separation exceeded 0.130 mm. We bounded the subsequent motion of each face by tracking its quadratic Bézier control points in a common translating frame at 65,536 samples/s. The two bodies' maximum excursions summed to 0.092 mm with nylon and 0.062 mm with aluminium through their respective threshold crossings. This places the early stress excursion before direct cell-to-case contact in both calculations. The air gap can close later in the same waveform (pages 10 and 15).

With base Z excitation, only the raised aluminium model crossed 185 MPa during the first 100 ms. The first crossing occurred at 77.911–77.927 ms on the middle cell attachment (STEP body 17). The same moving-surface bound gave at least 0.0256 mm of cell-to-case clearance through that sample. With base Y, all three models stayed below the reference throughout the 100 ms window.

## The local flush fixture exceeds the printed-PLA tensile references early in tightening

The four outer washers first loaded the nylon-screw fixture at 0.59–0.63 mm of screw-head travel. Between 0.91 and 0.95 mm, the greatest tensile stress in the assumed print plane crossed 52.3 MPa. Through-layer tension crossed 40.5 MPa between 0.99 and 1.03 mm. These are typical tensile strengths for the 100%-infill reference coupons. Comparing the native true stresses with these nominal coupon strengths gives an approximate check on the range of the elastic model. Polymaker coupon data.

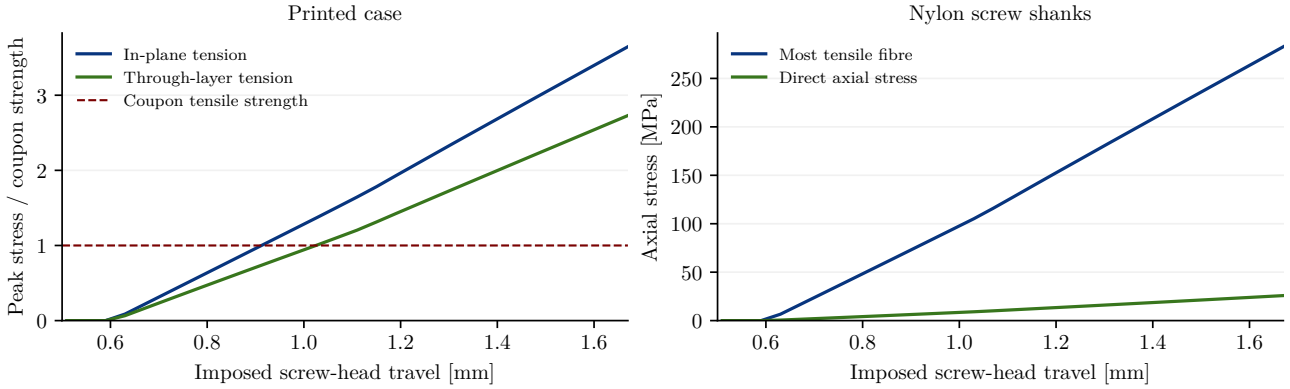



Figure A.16: Native converged local states with rigid neighboring hardware and prescribed screw-head travel. Left, maximum tensile stress over the 16,108 quadratic PLA elements. Right, section-node tension including bending and the direct axial component in the four idealized short screw shanks. Curves stop at the last sampled travel before loaded penetration first exceeded 0.05 mm.

| Local material reference | First corner crossing [mm] | First element with all four corners above [mm] |
|---|---|---|
| In-plane tension, 52.3 MPa | 0.91–0.95 | 1.55–1.59 |
| Through-layer tension, 40.5 MPa | 0.99–1.03 | 2.57–2.61 |
| Through-layer elongation, 1.8% | 1.23–1.27 | No sampled element |

Table A.6: The coupon elongation corresponds to logarithmic strain $\ln(1.018) = 0.01784$. The first in-plane and through-layer stress crossings occurred 3.72 and 3.78 mm from the nearest coupled bearing node. Contact penetration already exceeded the local screen by the time all four corners of an element crossed the through-layer stress reference.

At 0.95 mm travel each prescribed screw head carried 64.0–64.5 N along board Z. The largest direct axial screw stress was 7.37 MPa. Including bending, the most tensile modelled screw fibre reached 85.34 MPa. A rigid connection between the screw rotation and its bearing footprint strongly influences that value. The McMaster-Carr listing for the representative nylon screw (93640A145) gives its dimensions and lists tensile strength as unrated. Its installed strength, head rocking and thread engagement therefore remain unverified. McMaster-Carr screw data.

## How much moment each tab bearing carries

At 0.91 mm of screw travel, before the first coupon crossing, the four local tab bearings carried 56.5–56.9 N each along board Z. For each bearing, we summed the axial nodal forces $F_{z,i}$ and their first moment about the screw axis. The equivalent offset $e = \|\sum_i \mathbf{r}_{xy,i} F_{z,i}\| / |\sum_i F_{z,i}|$ was 4.44–4.46 mm. Pressure acting in one direction inside a circular footprint of radius $R$ gives $e \leq R$. For the modeled $R = 3.75$ mm, reproducing the observed moment using axial forces alone would require at least 5.2–5.4 N in the opposing direction per bearing. Forces from the printed solid and the screw/seat assembly agreed within 0.000003 N and moments within 0.00001 N mm. An actual joint can also transfer moment through the shank, thread or head restraint.

For comparison, tightening the six original bolts in the raised full-board models produced a maximum case-screw fibre tension of 1.15 MPa with nylon and 1.63 MPa with aluminium. Neither raised case reached a printed-PLA coupon reference during tightening. The local flush trial subsequently crossed the board datum (page 11). Resolving the force required for an actual flush seat depends on the full-board contact path and the as-built material and joint behaviour.

## The left and centre cells reach the raised nylon case in the X movie

The open-gap model puts the cell and PLA skins into intersection in frames 253–270 of the 480-frame X-axis movie. That is 4.20–4.48 s into playback, at 1.015442–1.016479 s in the load record. Checking all 1,917 integration samples in the clip identified two cells. The centre cell crossed first, then the left cell one sample later. The right cell stayed clear. All three stayed clear throughout the film interval for the other five mount and excitation combinations.

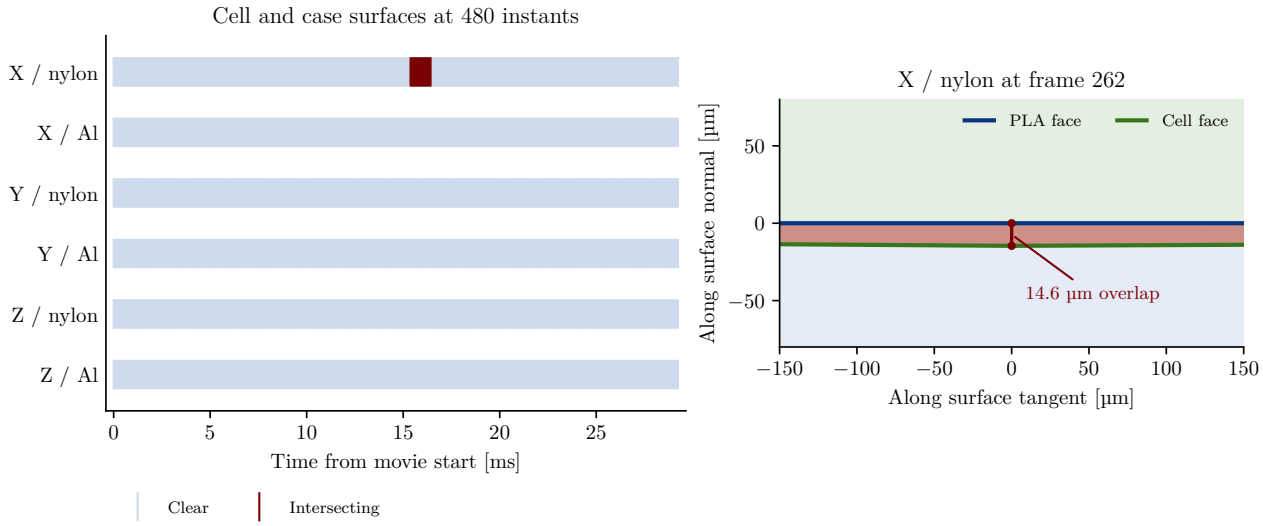



Figure A.17: Left, the cell and PLA surface check repeated at every displayed instant for each 3 mm mount and input axis. Right, the exterior surfaces at frame 262, cut in a plane through the closest PLA point and the most deeply embedded sampled cell vertex. Both section axes use the same scale.

| Cell | Entry bracket [ms] | Exit bracket [ms] | Overlap at 15.869 ms [$\mu$m] |
|---|---|---|---|
| Left | 15.335–15.350 | 16.388–16.403 | 12.1 |
| Centre | 15.320–15.335 | 16.418–16.434 | 11.3 |
| Right | Clear at all 1,917 samples | | — |

Table A.7: Physical time from the first movie sample. The 15.26 $\mu$s integration step brackets entry and exit. Overlap is the sampled cell-vertex depth inside the curved PLA skin at one common instant. The nine-second MAPDL runs reproduced the independent surface classifications at all 1,917 movie-interval samples for both raised mounts and all three input axes.

At each mount's own peak dynamic PCB stress in the X clip, the smallest sampled cell-vertex gap was 0.149 mm at frame 228 for nylon and 0.146 mm at frame 290 for aluminium. The fixed-contact motion calculation omits any change in stress or trajectory caused by earlier contact.

## Sensitivity to integration step and modal cutoff

Independent integration displaced all 8,038 exterior quadratic PLA faces and all 938 cell faces. Subdividing their convex hulls established separation at clear samples. Two successive triangle refinements confirmed each crossing. The full closed skins gave 15.0 $\mu$m of sampled overlap with four segments per edge and 14.6 $\mu$m with eight at frame 262. The movie magnifies movement 50 times. The section above uses physical displacements.

For the nylon X history we halved and quartered the integration step, holding the piecewise-linear acceleration input fixed. Relative to the exact response of the modal oscillators to that input, the modal-coordinate differences fell from 0.401% to 0.100% and 0.025%. All three time steps and the exact integration gave the same entry and exit brackets at the original sampling times. Retaining 35 modes through 2.95 kHz or 39 through 3.44 kHz gave the same brackets as the full 49-mode basis. Impact forces still require a contact-response calculation. Raised cases under X-axis excitation.

## The flush tabs make edge contact while most of each face stays open

With nylon screws all four tabs first carried pressure between 2.60 and 2.675 mm travel. With aluminium all four were still clear at 2.90 mm and carried pressure at the final 3.00 mm equilibrium. Even there only 18.3–18.9% of each aluminium-mounted tab lay within 0.05 mm of the board. The far sides retained 2.43–2.44 mm gaps. Both models exceeded the printed-PLA tensile references before tab contact. The resulting shapes describe an elastic continuation beyond the material data.

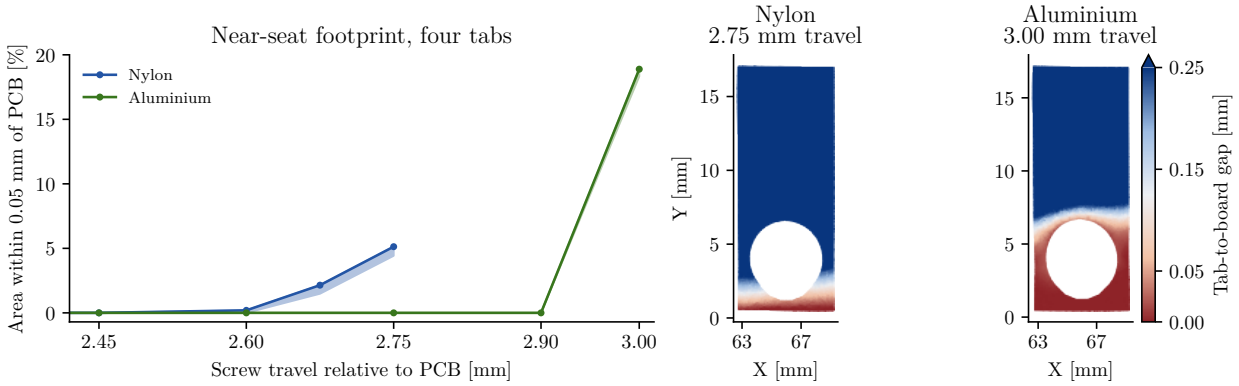



Figure A.18: Near-seat area throughout the last increments. Lines mark the largest fraction among the four feet and shading spans their range. The maps show the high-X, low-Y tab at each trial's last equilibrium. White marks the bore, and the darkest blue denotes gaps of 0.25 mm or greater.

| Tab position | Nylon near-seat area [%] | Nylon largest gap [mm] | Al smallest gap [mm] |
|---|---|---|---|
| Low X, low Y | 4.4 | 2.425 | 0.218 |
| Low X, high Y | 4.7 | 2.422 | 0.217 |
| High X, low Y | 5.1 | 2.417 | 0.204 |
| High X, high Y | 4.9 | 2.418 | 0.210 |

Table A.8: Near-seat area is the fraction of each overlapping tab footprint within 0.05 mm of the board. It was zero at every aluminium-mounted tab. All quantities are at 2.75 mm screw travel.

With nylon, the first through-layer tensile stress crossed 40.5 MPa between 0.95 and 1.10 mm, at a node tied to the idealized screw bearing. In-plane tension crossed 52.3 MPa at 1.10–1.25 mm. When the search excluded points within 2 mm of tied bearing nodes, both references were first crossed at 1.25–1.40 mm. The latter interval follows the first crossing of the 15 $\mu$m hardware-penetration screen (1.10–1.25 mm). At 2.75 mm the contact field reached 71 $\mu$m penetration. The nylon trial stopped at this equilibrium after three cutbacks. An installation-load rating requires measured print and contact behaviour.

The combined nylon result reproduced all 28 equilibria. The hardware contact audit included all 8,038 PLA contact faces. At 0.95 mm travel, before the first coupon crossing, the four original outer battery bolts gained 10.3–14.0 N above their 750 N locked preload. The two centre bolts changed by −0.9 and +1.4 N. At 0.95 mm the four drives supplied 25.59 mJ of work (28.41 mJ using alternate saved states). The PCB gained 8.81 mJ of stiffness energy, the PLA 8.01 mJ and the four screws 6.59 mJ. All element groups together accounted for 24.59 mJ.

## Aluminium screws increase the load at the same drawdown

In the two whole-board trials the screw shank geometry was identical. Only modulus, Poisson ratio and density differed between the decks. The first loaded hardware contact fell between 0.50 and 0.65 mm of relative travel for both mounts. At 0.80 mm, the four drives together carried 80.0 N with nylon and 115.6 N with aluminium. Peak through-layer PLA tension was 18.2 and 37.2 MPa respectively, against the 40.5 MPa coupon reference. With aluminium, both the in-plane and through-layer tensile references were first exceeded between 0.80 and 0.95 mm, while all four tab contacts remained open. In the 0.80 mm aluminium snapshot only the four outer washers intersected the PLA. Refining the curved surfaces gave up to 24.0 $\mu$m of sampled washer overlap, compared with 7.9 $\mu$m at loaded native contact corners. The smallest sampled cell clearance was 0.066 mm. At 3.00 mm travel, only the four outer washers intersected the PLA. Maximum sampled washer overlap was 0.136 mm and loaded native contact penetration was 0.109 mm, above the 0.015 mm screen. The three cells remained clear. A physically supported flush vibration state remains unresolved.

## The raised nylon case lowers the transverse load on all three cells

Under base-X vibration, the X component of the right-hand cell's net dynamic force fell from 30.51 N RMS bare to 26.67 N with nylon screws and 29.73 N with aluminium. These are reductions of 12.6% and 2.6%. Nylon also reduced this load on the left and centre cells. Aluminium raised the centre-cell load from 25.60 to 26.28 N RMS. The PSD calculation keeps the tightened cell-to-case gap open. The two nickel attachments on each cell therefore carry its net dynamic load until additional contact occurs.

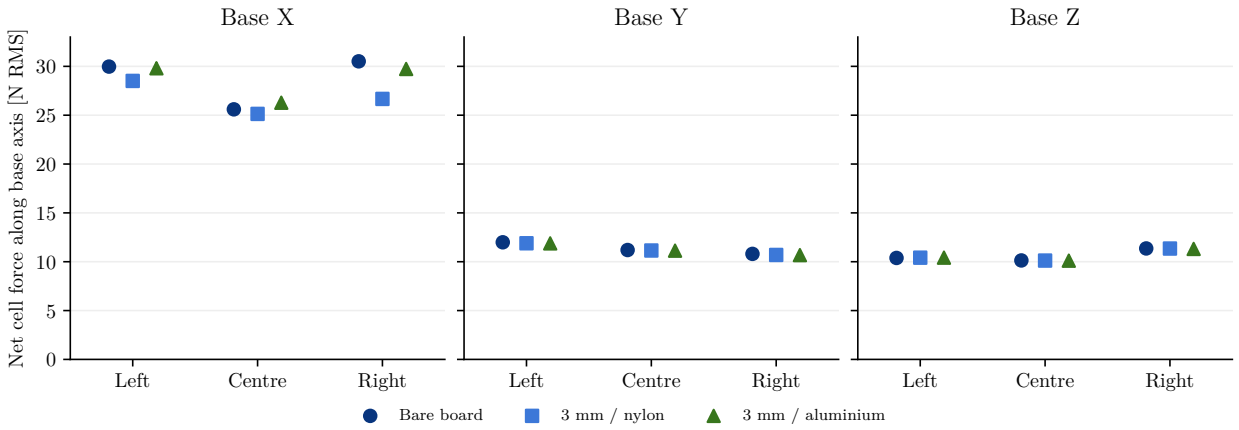



Figure A.19: Net force component parallel to each of three separate base excitations. Markers within each group correspond to the same physical cell. The ordinate is shared across panels. The material model gives each cell a mass of 55.38 g.

| Base X, cell force [N RMS] | Bare board | 3 mm / nylon | 3 mm / aluminium |
|---|---|---|---|
| Left | 29.97 | 28.51 | 29.80 |
| Centre | 25.60 | 25.13 | 26.28 |
| Right | 30.51 | 26.67 | 29.73 |

Along the excited direction, the Y and Z cases changed by less than 0.13 N on each cell. A transverse component can still grow. With Y input the X force on the left cell was 0.12 N RMS bare, 1.75 N with nylon and 1.01 N with aluminium. Page 22 compares the signed forces at both cell-to-nickel attachments.

## Mass averaging and verification

For each native mode we integrated the signed translations over each curved ten-node cell tetrahedron with its material density. The relative acceleration transfer function follows from differentiating modal displacement twice. We added the common base acceleration before integrating over the 20–2000 Hz spectrum. Combining the signed quantities in this order accounts for phase cancellation within the cell. Multiplying the resulting centre-of-mass acceleration by cell mass gives the force.

The four- and five-point-per-coordinate volume rules agreed within $10^{-9}$ in relative norm. Reducing the upper modal cutoff from 4 to 3.5 kHz changed the driven-axis forces by at most 0.30%. Independent nodal acceleration RMS fields matched all nine native PSD fields within 0.044% in relative norm over cell nodes. Separately, the original bare-board native time histories and original signed shapes reproduced the independently integrated seeded cell accelerations within 0.0075%.

For the eight stationary seconds of the historical forcing, the driven-axis RMS loads differed from the continuous-spectrum estimates by at most 0.88%. The largest discrepancy over all components was 10.4% and occurred in a smaller transverse component. Evaluating the same sampled waveform with the discrete Newmark transfer function reproduced the time integration. At one-quarter of the historical time step, its largest discrepancy from the continuous-frequency response at the same Fourier bins fell to 0.59%.

## The raised case preserves the first unpreloaded mode

The first default-contact mode remained at 66.61 Hz for the bare board and both raised cases. The signed displacements of the 368,754 original solid nodes give a modal assurance criterion (MAC) above 0.999999 for both screw materials. The 20th modes tell a different story. Although each is the 20th frequency in its own assembly, the three shapes differ substantially.

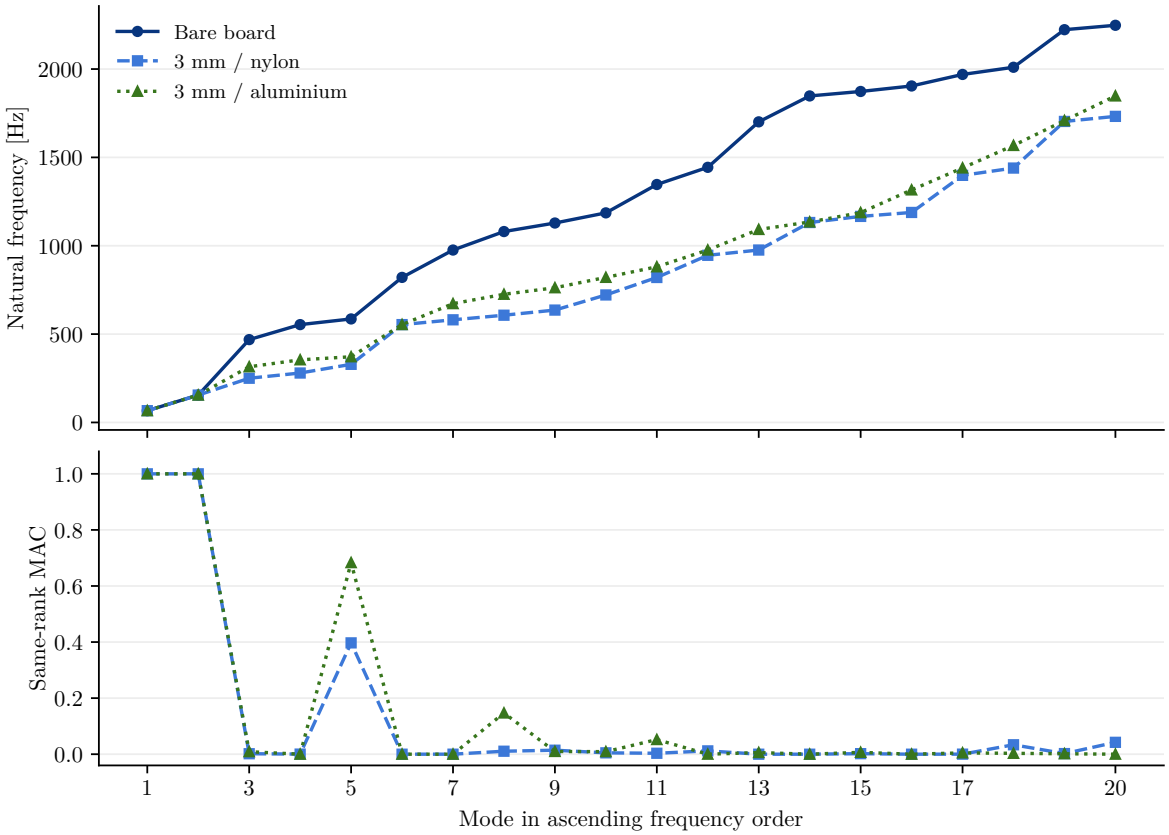



Figure A.20: The upper panel orders each assembly's 20 native frequencies. The lower panel compares the bare and cased displacement vectors at equal rank. A value near one means that the original solid nodes moved in the same relative pattern.

| Mounting | Mode 1 [Hz] | Mode 20 [Hz] | Mode 20 MAC to bare |
|---|---|---|---|
| Bare | 66.611 | 2247.59 | 1.000 |
| 3 mm, nylon screws | 66.614 | 1732.16 | 0.042 |
| 3 mm, aluminium screws | 66.615 | 1847.75 | $< 0.001$ |

The closest shape to bare mode 20 within either raised set was mode 19. Even those correlations were weak (0.129 for nylon and 0.137 for aluminium). The displayed aluminium mode 20 instead matches bare mode 14 at 1847.75 Hz (MAC 1.000). A matching shape for bare mode 20 may lie above the extracted range.

## Reading the mode movies

Both default-contact movie pairs preserve the October mode ranks, a common colour range, and a 6 mm peak display amplitude. The raised panels identify the screw material, native frequency and shape correlation. These runs retain the original unpreloaded contacts. The tightened assembly has different equilibrium contact conditions and prestressed frequencies (page 8). Across the full catalogue, all 22 subjects now have a bare view and a comparison of both raised configurations. The flush assemblies still need an admissible installed state before their vibration fields can be shown.

## The two cell attachments carry opposing lengthwise forces

The raised nylon case reduced the X force at both ends of all three cells under X excitation. Under Z excitation, however, both raised cases increased the force along the cell axis (Y) at all six attachments. The largest change occurred on the right-hand cell. The two ends carried 2.07 and 1.54 N RMS bare, 4.14 and 3.91 N with nylon screws, and 3.55 and 3.33 N with aluminium screws.

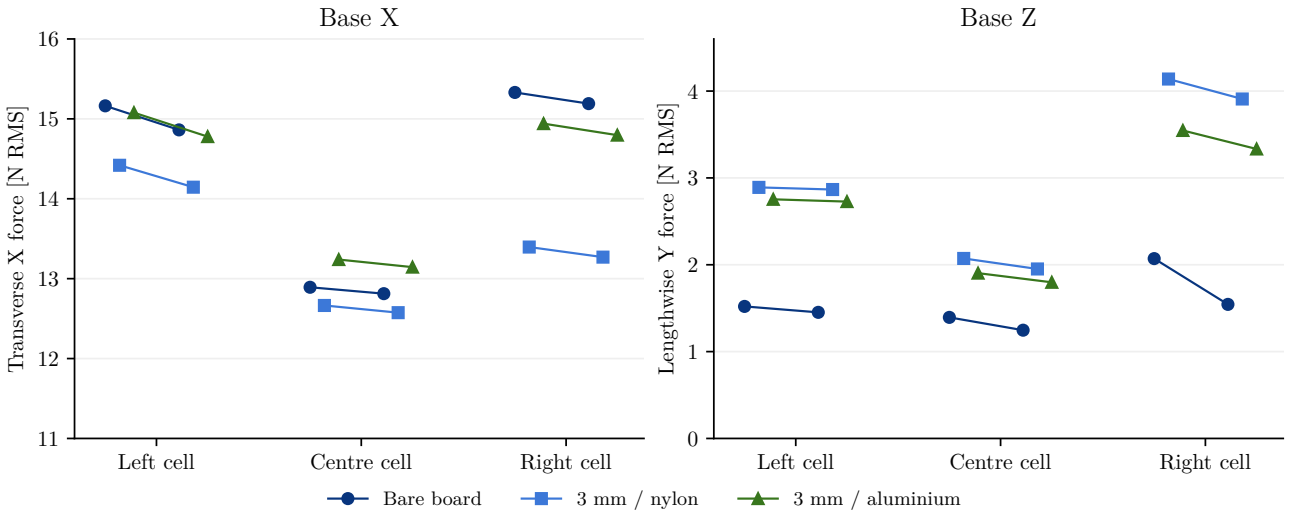



Figure A.21: Each line joins the low-Y and high-Y ends of one cell. Left, the X force for base X. Right, the lengthwise Y force for base Z. Each marker gives the one-sigma force integrated over a complete cell-to-nickel attachment.

## Why the whole-cell force can stay small

For the right-hand cell under base Z, the two lengthwise force histories had correlations of −0.88 bare, −0.97 with nylon and −0.97 with aluminium in the spectrum calculation. When the signed forces were added before taking RMS, the corresponding net loads were only 1.03, 0.99 and 0.90 N. The opposing forces matter to the individual attachments even when they nearly cancel across the cell.

| Right cell, base X | Bare | Raised nylon | Raised aluminium |
|---|---|---|---|
| Low-Y end, X force [N RMS] | 15.33 | 13.40 | 14.94 |
| High-Y end, X force [N RMS] | 15.19 | 13.27 | 14.80 |
| Combined X force [N RMS] | 30.52 | 26.66 | 29.73 |

## Force recovery and checks

Both sides of each bond were identified from the native contact and target connectivity. For each tightened mode, the assembled solid-element stiffness and inertia forces cancel at free cell nodes. Summing the residuals on a bonded face gives its signed modal force. The adjacent nickel strip reproduced the opposite resultant to within 0.00032% in vector norm for every mode. Independent frequency integration then combined the six bond resultants over the original base spectrum with 2% modal damping.

For 90 sampled cell and nickel elements, the independently combined stiffness-force components agreed with all nine native PSD cases within 0.0056% in relative norm. Under base X the summed X loads agreed with the whole-cell inertia calculation (page 20) within 0.42%. Across Y and Z the corresponding differences reached 4.73% and 2.16%. Removing modes above 3.5 kHz changed any component by at most 0.362 N. The original October mode forces and native coordinates reproduced all 480 bare-board movie frames for all three drives within 0.00043 N per component.

The bonds cover the areas joined in the model. Actual weld sizes, local separation and the response after cell-to-case impact remain unmeasured.

## Case support shifts the peak dynamic load at the nickel attachments

For each of the three original nine-second acceleration records, we recovered all three force components at both nickel attachments on each cell. We took the vector magnitude at each of the 589,825 saved instants, then found the largest magnitude across the six ends. These forces describe the extra load from vibration above the locked preload in the linear model.

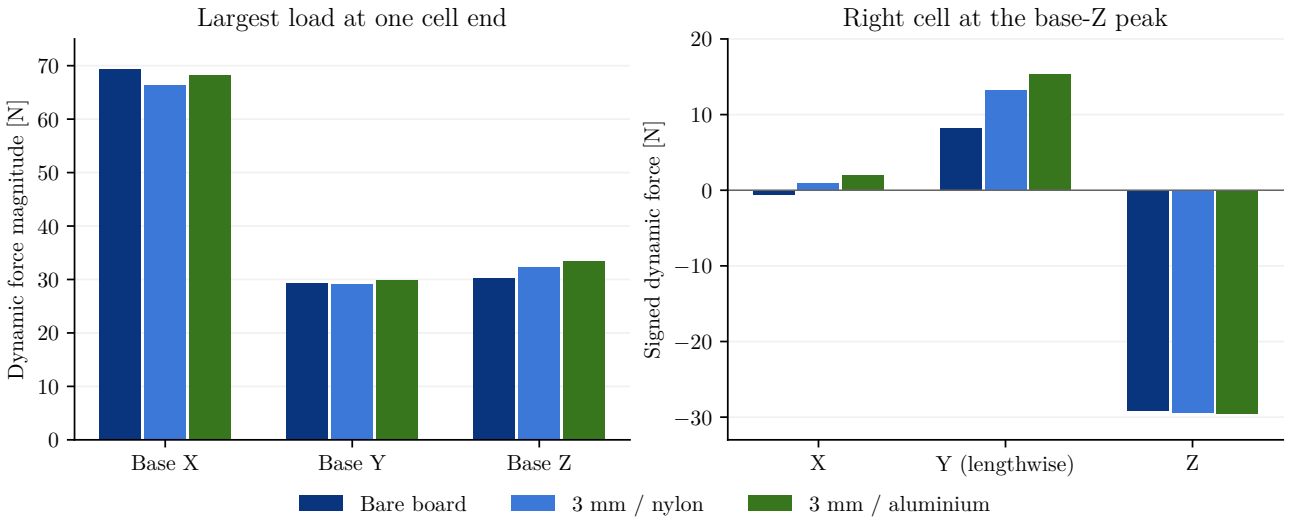



Figure A.22: Left, the maximum at a single cell end under each base direction. Right, the simultaneous signed components at the low-Y end of the right-hand cell. All three assemblies reached their base-Z maximum there at 1.86743 s. The positive lengthwise force increased with both raised mounts.

| Drive | Bare [N] | 3 mm nylon [N] | 3 mm aluminium [N] | Governing attachment |
|---|---|---|---|---|
| Base X | 69.39 | 66.39 | 68.23 | Left cell, low-Y end |
| Base Y | 29.25 | 29.17 | 29.79 | Left cell, end varies |
| Base Z | 30.22 | 32.21 | 33.30 | Right cell, low-Y end |

Table A.9: Largest dynamic force magnitude at one cell-to-nickel bond over the full record. Under base Y the bare and nylon peaks occur at the high-Y end and the aluminium peak occurs at the low-Y end.

Under base X the peak end load fell by 4.3% with nylon and 1.7% with aluminium. Under base Z it rose by 6.6% and 10.2%. At the base-Z peak the Z component stayed close to −29 N in all three assemblies. The lengthwise component rose from 8.17 N bare to 13.21 N with nylon and 15.32 N with aluminium. The force on an individual attachment can therefore grow even while the two lengthwise end loads partly cancel across the cell (page 22).

## Response before the nickel yield reference is reached

The first 38.116 ms of base X precede the first sampled exceedance of the local pull model's 185 MPa nickel reference in any assembly (page 16). Over this common interval, the right cell's low-Y end carried the largest force in all three assemblies. Its magnitude reached 9.57 N bare, 8.92 N with nylon and 9.50 N with aluminium. The bounds on the raised assemblies' relative surface motion leave the cells clear of the PLA through this interval. The full-record peaks above include later excursions beyond the elastic and fixed-contact assumptions.

## Numerical checks and physical scope

The main results use signed native MAPDL time histories and bonded-interface forces from the prestressed modes. A separate integration of the same base motion reproduced the complete force fields within 0.002% in relative norm. Restricting the modes to 3.5 kHz changed the nine peak magnitudes by at most 1.09%. The bonded end area idealizes the battery-to-nickel attachment. The model supplies a dynamic resultant over that area. Predicting a weld failure load requires the actual weld geometry and material response.

## During startup, screw material changes which limit appears first

In the raised aluminium mount under base X, tension across the assumed print layers reached the 40.5 MPa coupon mean at 63.54 ms. The first sampled cell-to-case intersection followed at 69.78 ms. With nylon screws the middle cell first intersected the case at 85.34 ms, while both PLA tensile stresses stayed below their respective coupon means throughout the 100 ms ramp. The nickel model had already exceeded its separate trial yield reference under X in all three assemblies (page 16).

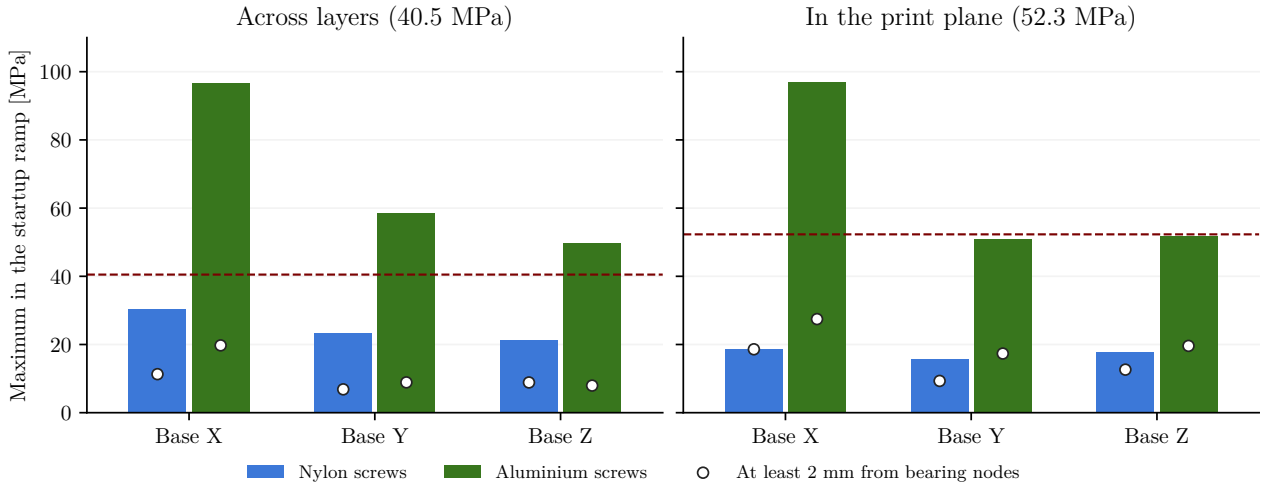



Figure A.23: Peak tensile stress over 6,555 samples of each native waveform, including the locked preload. Bars include all 64,432 PLA element corners. White markers give the maxima among corners at least 2 mm from any node tied to a screw bearing. Dashed lines mark the coupon means.

| | | First sampled event [ms] | | | Cell gap at layer |
|---|---|---|---|---|---|
| Screws | Base | Layer tension | In-plane tension | Cell intersection | crossing [mm] |
| Nylon | X | — | — | 85.34 | — |
| Nylon | Y | — | — | — | — |
| Nylon | Z | — | — | — | — |
| Aluminium | X | 63.54 | 85.21 | 69.78 | 0.035 |
| Aluminium | Y | 77.45 | — | — | 0.135 |
| Aluminium | Z | 81.39 | — | — | 0.132 |

Table A.10: Times identify the first integration station meeting each criterion. The previous station was 15.26 $\mu$s earlier. Dashes in the three event columns mean the criterion was absent through 100.006 ms. Gap is the minimum sampled cell-vertex distance from the case across all three cells at the layer crossing.

The three aluminium layer crossings occurred within 0.54 mm of a node tied to a screw bearing. Outside the 2 mm band, both stresses remained below the coupon means for every input and screw material. On the aluminium X path, a cell surface was already inside the PLA when in-plane tension reached 52.3 MPa. The local bearing representation matters to the stress maximum. The comparison identifies where the fixed-contact, elastic calculation first reaches these particular screens. Failure of an actual printed tab depends on its print and bearing conditions.

## Checks on the order of events

Signed modal stresses were added to the locked stress tensor before evaluating layer-normal tension and the greatest in-plane principal stress. The search covered both screw mounts, all three base inputs and every stored station. The exact response of each modal oscillator to the piecewise-linear input reproduced the native tensile crossing brackets. Omitting modes above 3.5 kHz preserved those brackets and the first base-X cell crossing for both mounts. Independent integration shifted the first aluminium X intersection by one sample (15.26 $\mu$s) and reproduced the nylon onset. The cell check used the complete curved exterior surfaces. A separate intersection test at two surface refinements confirmed the cell states at the four aluminium tensile crossings.

## With nylon screws, the raised PLA case carries most of the pull at 0.50 mm

The bare fixture and the raised fixture with nylon screws used the same nominal nickel damage law, 2,304 shell elements and two 750 N washer clamps. At a prescribed 0.500 mm of sideways travel, the bare edge cell required 11.33 N. The raised case required 141.82 N, of which the four case screws carried 112.62 N (79.4%). At this displacement the neighbouring-cell support carried no net X load. The largest tensile stresses in the PLA were 28.75 MPa in the print plane and 23.60 MPa across the layers.

Local sliding at the washer–nickel interfaces began in the same 0.0500–0.0875 mm travel interval in both fixtures, before the cell touched the case. At 0.500 mm the peak element-averaged frictional stress rose from 0.801 MPa bare to 1.144 MPa cased, and peak accumulated tangential motion rose from 1.29 to 1.77 $\mu$m. Ten bare and eleven cased contact corners registered sliding. Every face with a sliding corner also retained a sticking corner.

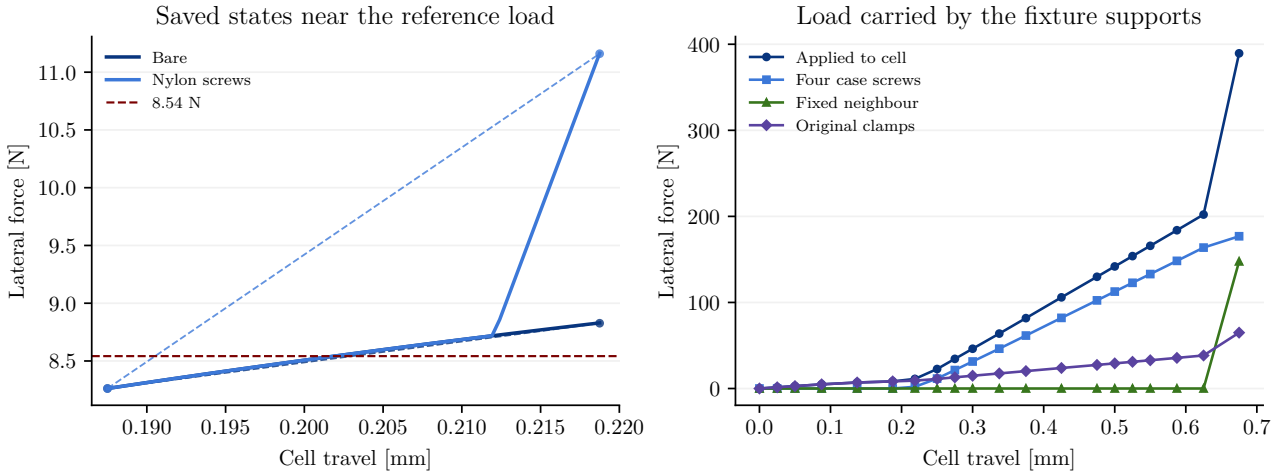



Figure A.24: Left, dashed lines join the original interval endpoints. The solid lines use 64 refined increments and coincide at 8.54 N. With nylon the native contact pair first carried load between 0.21191 and 0.21240 mm. Right, the load paths through the last station below the PLA and contact limits. The local fixture holds the other cells and the case screw tops fixed.

| Nickel model | Bare travel at 8.54 N | Raised travel at 8.54 N | Force at 0.50 mm, bare / cased |
|---|---|---|---|
| Bilinear | 0.500 mm | 0.207 mm | 8.54 / 138.53 N |
| Multilinear with damage | 0.20190 mm | 0.20190 mm | 11.33 / 141.82 N |

Table A.11: At 8.54205 N the refined bare, nylon and aluminium multilinear fixtures all gave 0.201904 mm between the same two saved stations. Both cases carried zero contact force there. At 0.212402 mm the first loaded face carried 0.114 N with nylon and 0.210 N with aluminium. The forces at 0.50 mm are saved equilibria for bare and nylon.

## Printed-material scope and the last equilibrium

The fixed neighbour began sharing the load between 0.625 and 0.675 mm of cell travel. At 0.675 mm the applied pull was 389.45 N. The case screws carried 176.89 N, the fixed neighbour carried 147.66 N and the original clamps carried 64.90 N. Between 0.675 and 0.725 mm the peak in-plane tension first exceeded the 52.3 MPa printed-coupon mean. The crossing occurred 4.36 mm from the nearest coupled bearing node. The first through-layer stress crossing followed at 0.800–0.825 mm. These local results depend on the actual stiffness of the mounts and neighbouring cells.

ANSYS retained 59 converged states, including 50 pull states, through 0.923275 mm. It terminated when the next increment failed to converge after cutbacks. None of the three nickel section summaries reached the $10^{-4}$ damage reporting threshold at any converged state. The two strips retained a graph path from the modelled battery attachment to the clamp at every station under the original October 0.99 section-damage diagnostic. The later equilibria extend beyond the onset of the printed-material limit. The contact-position comparison on page 5 also limits the interpretation of the low-load response. The physical rupture load remains unresolved.

## How the raised case changes strain in the nickel strips

The three local fixtures used the same 2,304 nickel shell elements, multilinear plasticity and damage law, and 750 N washer preloads. In positive X, both original raised contact pairs first carried load between 0.1875 and 0.21875 mm. The STEP surfaces already intersected at the lower station (page 5). The last travel saved at exactly the same value in all three models before either raised fixture exceeded its material limits was 0.3375 mm.

| Fixture | Cell pull [N] | Case screws [N] | Strip clamps [N] | Peak nickel plastic strain |
|---|---|---|---|---|
| Bare | 10.15 | — | 10.15 | 0.673% |
| 3 mm, nylon | 63.82 | 46.30 | 17.52 | 0.727% |
| 3 mm, aluminium | 125.98 | 100.18 | 25.81 | 0.783% |

Table A.12: Native reactions at 0.3375 mm. The fixed neighbour carried no lateral reaction at this travel. The maximum strain occurred in the same flat nickel shell element in all three models.

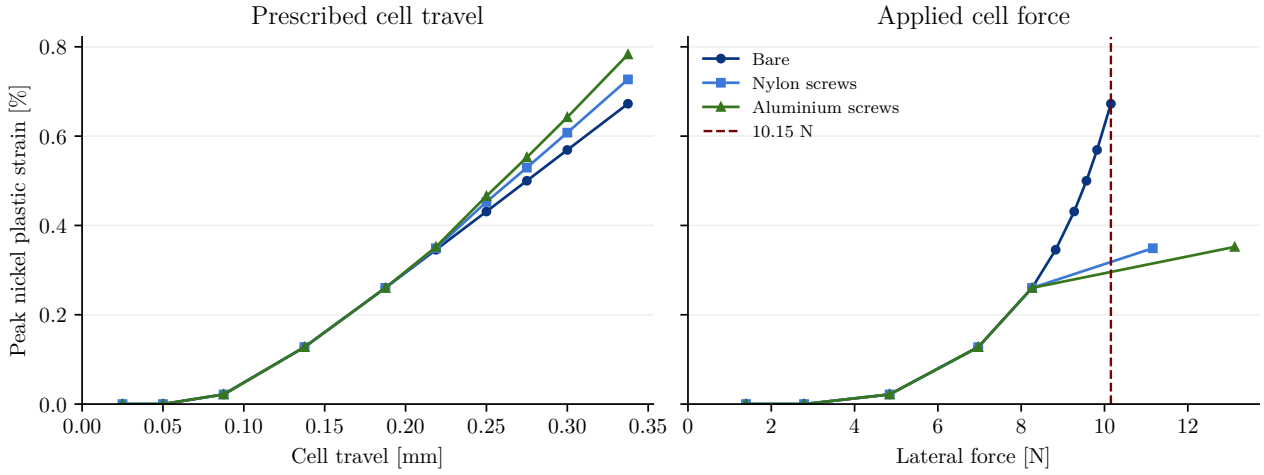



Figure A.25: Each marker is a converged native state. The vertical line marks the 10.15 N reaction of the bare fixture at 0.3375 mm. Curves connect neighbouring states. Strain is the maximum element-averaged equivalent plastic strain across the three saved shell sections.

At equal travel, the aluminium fixture took almost twice the force of the nylon fixture. Both cased cells also strained the nickel more than the bare cell. At equal force, the comparison reverses. The cased stations bracketing 10.15 N had peak strains of 0.260–0.349% with nylon screws and 0.260–0.352% with aluminium, compared with 0.673% at the bare station. The load bracket spans first contact in both cased fixtures. The difference between the screw materials at that force remains unresolved.

## Where the aluminium result reaches its material limit

At 0.425 mm the aluminium fixture carried 218.57 N, including 179.80 N through its four case screws. At the next saved equilibrium, 0.475 mm, in-plane PLA tension reached 60.16 MPa and through-layer tension reached 44.10 MPa. Both exceeded the printed-coupon means (52.3 and 40.5 MPa), within 0.54 mm of a tied screw-bearing node. With nylon screws the first tensile crossing occurred later, between 0.675 and 0.725 mm.

The aluminium solve retained 37 lateral equilibria through 0.87435 mm and stopped on the next increment. All 37 passed the original force-balance checks. The largest loaded-corner penetration was 0.0439 mm against the 0.05 mm local contact screen. Both nickel strips retained their modeled load paths and all three section damage summaries remained zero. The late equilibria exceed the PLA reference range, and the physical rupture load remains undetermined.

## Nickel material-law check

Three single-element tensile coupon reruns reproduced the October reaction, transverse displacement, axial stress, equivalent plastic strain and damage at all 1,074 shared stations. The peak reactions lay between 66.87 and 66.89 N. All three paths softened and stopped before the requested 1 mm extension. These numerical coupons verify the implemented material law. Predicting separation of the assembled strips also requires material and attachment measurements.

## How the vibration loads compare with component strength

The high through-layer tension occurs in the **printed PLA tab beside the screw bearing**. The four case screws change how force enters the tab. For the assumed print, the manufacturer's tensile specimens reached a mean of 40.5 MPa across the layers and 52.3 MPa in the print plane. Each percentage below is the largest tensile stress in the stated record divided by its corresponding coupon mean. These ratios include bolt preload.

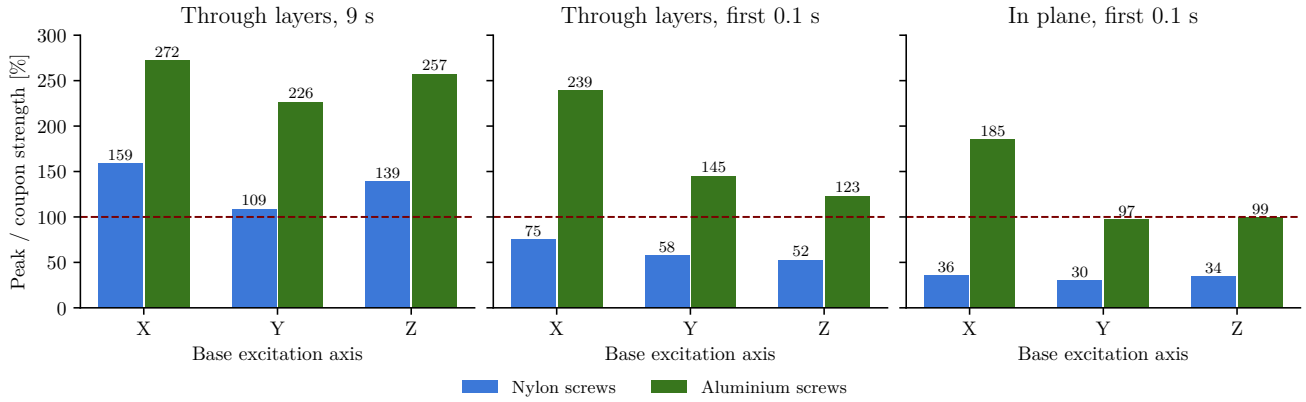



Figure A.26: Dashed lines mark 100% of mean coupon strength. Left: peak signed through-layer stress at the bearing corner with the highest layer-normal RMS for each nine-second test. Centre and right: maxima over all PLA corners during the first 0.1 s. X, Y and Z were separate base excitations.

For board-normal vibration the monitored PLA tab reached 56.2 MPa with nylon screws and 104.2 MPa with aluminium screws. Both peaks occurred with the sampled cell and case skins clear at that instant. The linear model continues to store elastic stress beyond the coupon strength. At these peaks, actual load redistribution or cracking would require the print properties and screw-bearing compliance of the installed case. The earlier aluminium startup crossings occurred within 0.54 mm of a node tied to a screw bearing.

## Other components

**Nickel strips.** The nickel damage model assumes 185 MPa initial yield. Bolt tightening alone reached about 162 MPa (88% of this reference). Under base X the signed, preload-inclusive stress crossed 185 MPa in all three assemblies within 39 ms (page 16). The actual temper and weld geometry are needed to assess strip rupture or fatigue life.

**Board and soldered parts.** Board-normal base vibration produced 32.5, 64.8 and 84.2 MPa RMS in the equivalent PCB material for bare, nylon and aluminium, respectively. These statistical stresses show how the board response changes. A strength margin for a populated assembly requires the actual laminate allowables and package-specific limits on local board strain or solder fatigue.

**Case screws and cell attachments.** At 0.3375 mm lateral travel the modeled peak tensile screw-fibre stresses were 13.1 MPa (nylon) and 34.2 MPa (aluminium). The nylon reference screw has no rated tensile strength in its listing, and the aluminium alloy and temper are unspecified. The attachment calculations give force over the idealized bonded footprint. A tested weld strength is needed to turn that force into a margin.

Printed PLA tensile data | IPC/JEDEC guidance on measuring board strain near components

## A 2.5 mm stand-off leaves clearance at the lips and under the cells

For the supplied Balius STEP at its nominal hole alignment, four outer washers set the closest rigid mounting position. They first touch at a 2.397 mm tab-to-PCB spacing. At 2.500 mm they clear by 0.103 mm. The cells approach the sides of their circular pockets much more closely than the floors.

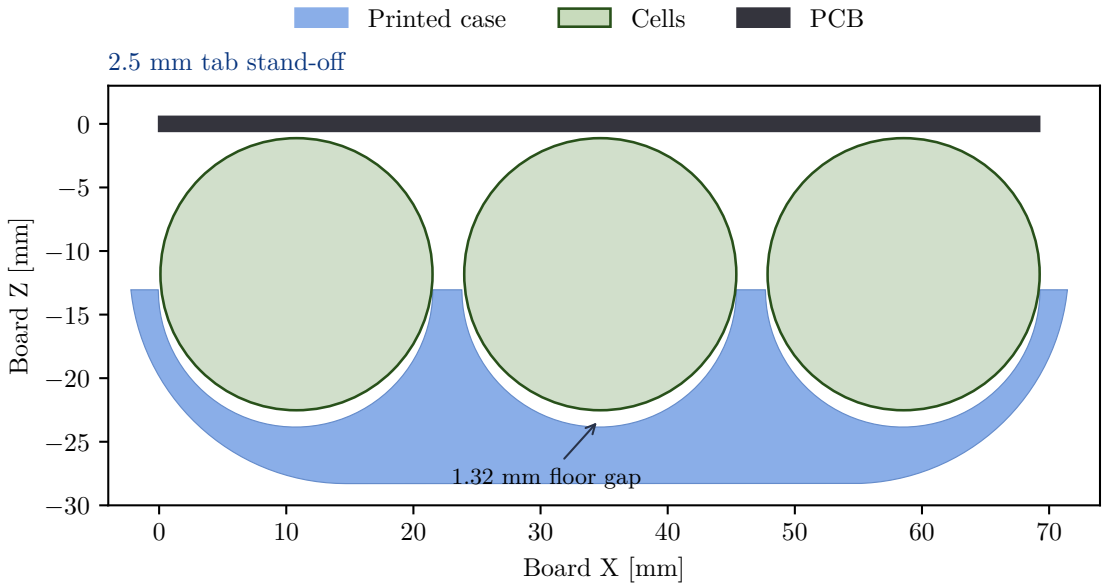



Figure A.27: Section at the case midspan using the STEP profile and 10.700 mm cell radii. The 1.32 mm dimension is vertical beneath the middle cell. Washers and mounting tabs sit outside this plane. The drawing uses the same scale for both axes.

| Tab-to-PCB spacing | Left cell minimum | Middle cell minimum | Right cell minimum |
|---|---|---|---|
| 2.400 mm | 0.082 mm | 0.065 mm | 0.082 mm |
| 2.500 mm | 0.093 mm | 0.076 mm | 0.093 mm |
| 3.000 mm | 0.161 mm | 0.144 mm | 0.161 mm |

Shortest three-dimensional surface distances between each undeformed cell and the case. The 2.400 mm setting leaves just 0.003 mm between the case and the first four washers.

Holding the board and cells fixed, the middle cell would first touch as spacing fell to about 1.240 mm, then the outer cells near 1.210 mm. Reaching either position requires interference with the existing washers. A 2.5 mm spacer can therefore give a close nominal side fit, while simultaneous supported contact of all three rigid cells requires a change to the installation geometry or compliance. Actual print dimensions, washer stack and cell positions determine the as-built clearances.

## Which battery case fits the archived board?

The Balius SolidWorks part gives the closest nominal fit of the three files examined. The supplied Balius STEP and the fresh native export share the same three cell-pocket axes and radii. The native part provides additional space around the washers. SolidWorks 2026 SP4.1 exported the default configuration of each native part to AP203 STEP with zero reported errors or warnings. Each export reopened as one solid.

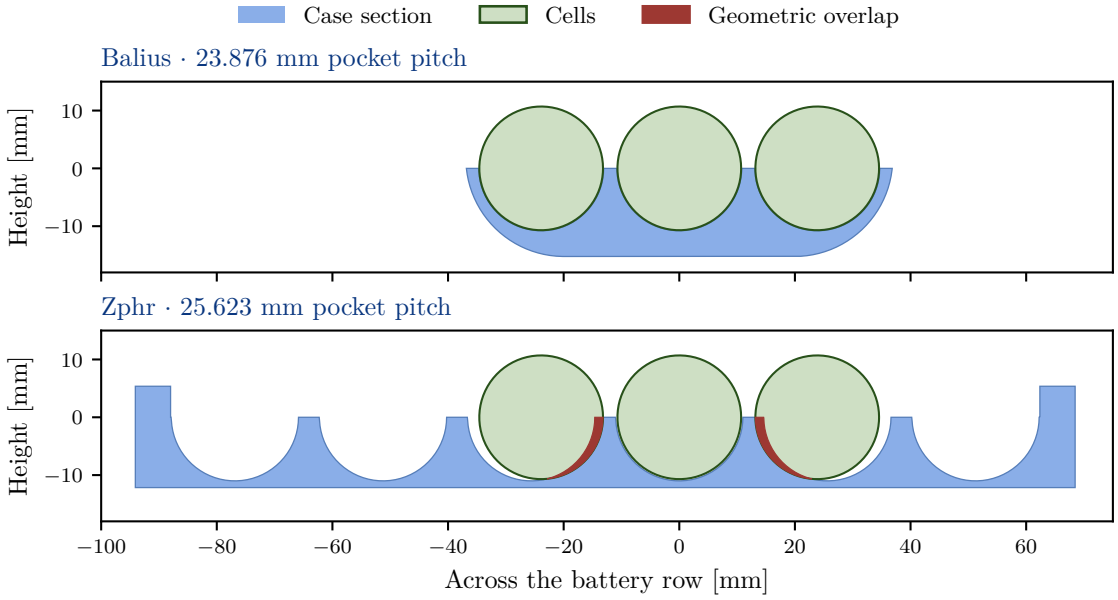



Figure A.28: Midspan sections at equal scale. For this pocket comparison, the middle cell axis coincides with one pocket axis in each direction. Red marks where the outer cells intersect the Zphr case in this pose. The mounting holes are checked separately.

## Balius at 1.25 mm tab-to-board spacing

A rigid sweep over the 35 archived hardware solids found that the native part reaches the middle cell first at 1.240 mm stand-off. The outer cells reach their pockets at 1.210 mm if the geometry is allowed to continue through first contact. The earliest outer-washer contact in that continuation is at 0.733 mm. With the original supplied STEP, the washers touch at 2.397 mm, before any cell can reach its pocket.

| Case and tab spacing | Middle cell gap | Outer cell gaps | Smallest outer-washer gap |
|---|---|---|---|
| Supplied Balius STEP, 2.50 mm | 0.076 mm | 0.093 mm | 0.103 mm |
| Balius native export, 2.50 mm | 0.076 mm | 0.093 mm | 1.250 mm |
| Balius native export, 1.25 mm | 0.003 mm | 0.020 mm | 0.328 mm |

At 1.25 mm all three cells sit close to the native pockets. The small clearances depend on the actual print, battery sleeves, washer stack and board alignment. The reported structural and vibration results use the originally supplied STEP.

## Zphr pocket and screw alignment

Zphr has six pockets of radius 11.000 mm and pitch 25.623 mm. The three 10.700 mm-radius cells have adjacent pitches of 23.893 and 23.859 mm. Centring the middle cell in a Zphr pocket puts the outer cell axes 1.729 and 1.764 mm off-centre in the neighbouring pockets. The available concentric radial clearance is only 0.300 mm.

The 14 Zphr bores have 3.400 mm through diameter. Comparing inter-hole distances over every assignment of four bores to the board corners proves that at least one centre is displaced by at least 10.525 mm under any rigid planar alignment. A nominal M3 screw can accommodate only 0.800 mm of centre offset across the case and PCB bores. Changing stand-off alone cannot register this pattern.

## The raised case engages later when the pull reverses

We pulled the same outer cell in both directions along PCB X. For each mounting configuration, the two solved input decks had the same geometry, clamp preload, bilinear nickel law and load increments. The only numerical change was the sign of the prescribed cell travel. Each native result contains 33 converged equilibria.

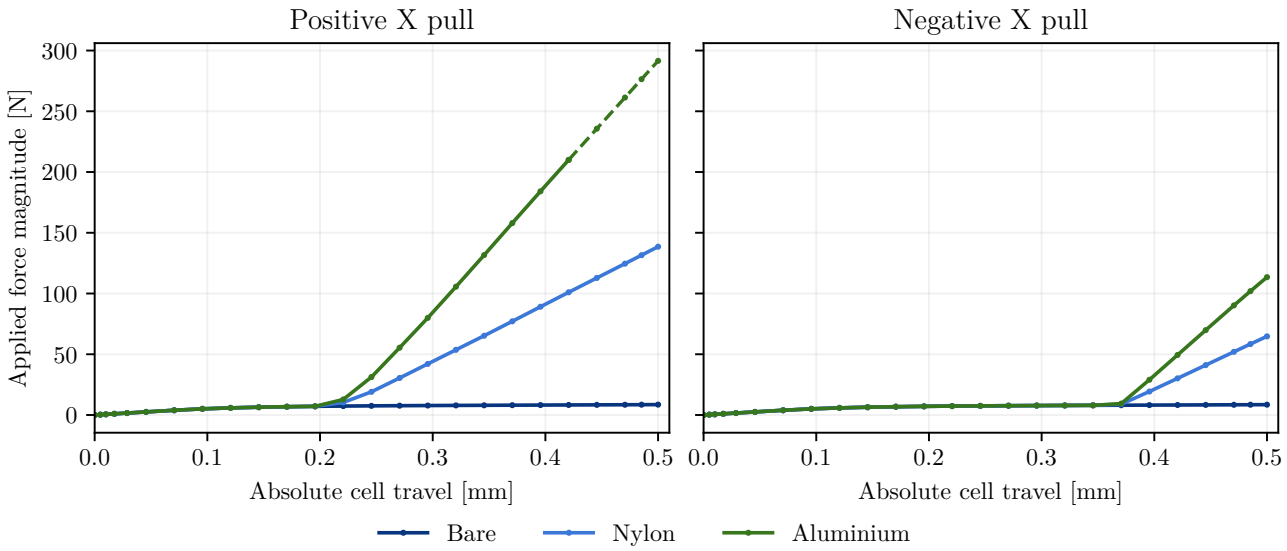



Figure A.29: Applied cell force at equal travel magnitudes for the bare board and the two 3 mm mounts. Dashed segments follow the linear PLA model beyond its first printed-material reference. All six curves use the October bilinear nickel law. The multilinear nickel results are on pages 25–26.

## Force at a common displacement

At 0.420625 mm, all six fixtures remained within the printed-material and contact-penetration screens where applicable. The bare forces in opposite directions differed by only 0.027 N. Reversing the pull reduced force by 70% with nylon case screws and 76% with aluminium screws.

| | Applied cell force [N] | | Via four case screws [N] | |
|---|---|---|---|---|
| Mount | $+X$ | $-X$ | $+X$ | $-X$ |
| Bare | 8.26 | 8.24 | — | — |
| Nylon | 100.99 | 30.17 | 80.00 | 18.91 |
| Aluminium | 209.96 | 49.42 | 174.54 | 35.58 |

Force magnitudes at 0.420625 mm of cell travel. Each value comes from a saved native equilibrium. The original nickel clamps carry the remainder in these fixtures.

## Clearance and first load

Under a pure X translation of the undeformed STEP cell, the gap first closes after 0.163 mm toward positive X and 0.308 mm toward negative X. In the native fixtures the cell can also rotate. The moving-cell contact pair first registered closure between 0.196–0.221 mm in positive X and 0.346–0.371 mm in negative X for both screw materials. The later engagement leaves less reverse travel over which the case can take load.

At 0.500 mm in negative X, the applied force was 8.51, 64.73 and 113.46 N for the bare board, nylon and aluminium respectively. The corresponding maximum nickel plastic strains were 1.252, 1.346 and 1.434%. Peak in-plane PLA tension at that reverse endpoint reached 23.58 MPa with nylon and 43.07 MPa with aluminium (45% and 82% of the 52.3 MPa coupon mean). Both reverse paths stayed within the recorded contact and PLA screens. The fixed screw tops represent a local fixture. These loads describe a prescribed displacement and do not establish a rupture load.